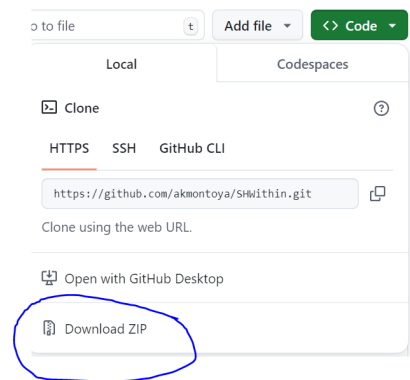


Mediation and Moderation Analysis for Simple Within-Subject Designs

Amanda K. Montoya
University of California, Los Angeles

November 20 & 21, 2025

Please go to <https://github.com/akmontoya/SHWithin.git>
Download the whole folder.



General Outline

Day 1

Introduction

- Within- and between-subject designs
- Overview of between-subject mediation

Within-Subject Mediation

- Estimation & Inference
- MEMORE Macro
- Issues of Causality
- Power Analysis
- Preregistration and writing guides
- Visuals for Publication
- Multiple Mediator Models

Wrap Up: Extensions and Next Steps

General Outline

Day 2

Between-subject moderation (review)

Within-Subject Moderation

- Estimation & Inference
- Visualizations
- Probing interactions
- MEMORE Macro
- Power Analysis
- Preregistration and writing guides

(If time) Within-Subject Moderated Mediation

- Estimation & Inference
- Visualizations
- Probing Conditional Effects
- MEMORE Macro
- Preregistration and Writing Guides

Wrap Up

Workshop Procedures

Assuming some familiarity with:

- Linear Regression
- (Some) Mediation & Moderation in BS-Design
- SPSS or SAS

Class time:

10:30am – 12:30pm EST

BREAK/Activity

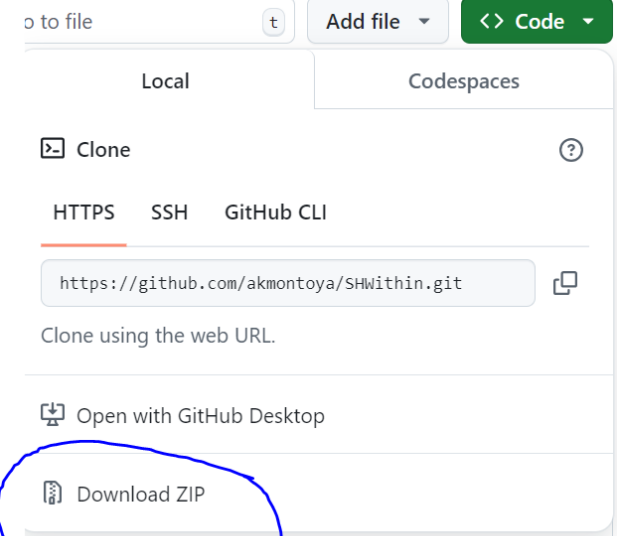
1:30pm – 3:30pm EST

How we will learn:

- Combination of theory and practice
- Follow along with the analysis as we go
 - Use syntax!
 - Ask questions about concepts or anything that is confusing

Download files at

<https://github.com/akmontoya/SHWithin.git>



What You'll Need

- ❑ This course is hands-on. Hopefully you brought a laptop with R 3.6.1, SPSS 19+ or SAS 9.1+ with PROC IML. If not, that is ok. You'll still benefit.

SPSS Code

SAS Code

R Code

- ❑ Various files available on GitHub.
 - ❑ SPSS, SAS, and R data folders. SPSS data files are ready to go. SAS files are programs thus must be executed to make them “work” files. R files are CSV and need to be read into R.
 - ❑ Macro files. This contains the MEMORE macro we'll heavily rely on, and some documents related to it.
 - ❑ Miscellaneous folder. Various files, including some PDFs and other miscellaneous things of relevance to this course.
- ❑ A lot of stamina.

Setting Up in R

A few packages I'm going to use throughout the workshop:

```
install.packages(c("mosaic", "ggformula", "jtools", "dplyr"))  
library(mosaic)  
library(ggformula)  
library(jtools)  
library(dplyr)
```

Loading in data:

```
filelocation <- "C:\\Your\\Path\\Here\\harass.csv"  
harass <- read.csv(filelocation, header = TRUE)
```

You are welcome to deviate from the way I do things, if you have a preferred package for certain types of operations, feel free to do so.

Experimental Designs

This course focuses on two types of designs:

- Between-subject
- Within-subject

Whether we are doing mediation or moderation, there will always be a focal predictor (X) and an outcome (Y).

To determine the type of design, we only need to focus on X and Y .

Between-Subject Design

- X is observed only once per subject
 - Randomly assigned (e.g., condition in a study)
 - Observed (e.g., handedness)
- Y is observed only once per subject
 - Only observed (e.g., typing speed)

Examples of Between-Subject Designs

Human participants are randomly assigned to complete a math test in either a hot or cold (X = temperature) room. Their performance on that math test (Y) is then measured.

Assigned

Researchers want to compare start-ups in finance to those in health (X = type of company) on their number of employees in their first two years (Y)

Observed

In this course X will typically be categorical, but it can also be continuous.

Within-Subject Design

- X is observed or assigned multiple times per subject
 - Assigned (e.g., participate in both conditions)
 - Every subject is observed under the same values of X
 - Observed (e.g., measured across time)
 - Values of X may differ across participants
- Y is observed under each instance of X
 - Only observed (e.g., typing speed)

Examples of Within-Subject Designs

Human participants are complete a math test both a hot and a cold (X = temperature) room. Their performance on that math test (Y) is then measured.

Assigned

Number of employees (Y) is measured at the end of the first year and second year (X = age of startup) across many start-ups.

Assigned

Number of employees (Y) is measured one year apart for each start-up (X = age of startup)

Observed

Repeated-Measures vs Within-Subjects

Used very similarly

Repeated Measures

Within-Subjects

Types of Repeated-Measurements

- Each person *over time*
- *Nested/Multilevel* data (individuals within schools, cohorts, etc)
- *Dyadic* data (twins, couples, labmates, roommates)
- Each person in a *variety of circumstances*
- and many more...

Repeated Measures Data

It's important to think about how/when/how many times the variables in your mediation model are measured

- Within-Subject: X and Y are assigned/observed repeatedly
- *Multilevel* has a nice system referring to levels (1-1-1 mediation, 2-2-1, mediation etc.
- Is your focal predictor measured repeatedly?
- Is your focal predictor what differentiates your repeated measurements?

Our Focus: Two Instance Repeated-Measures

The focal predictor is the factor which differs by repeated measures.

X : is assigned twice

M : measured in each of the two instances

Y : measured in each of the two instances

Examples:

- Participants read two scenarios. Interested in how scenario influences Y through M . Measure M and Y in each scenario.
- Pre-post test: Therapist measures certain symptoms and various outcomes before administering some intervention, and after administering the intervention.
- Researcher interested in if male partners in heterosexual relationships believe fights are less severe because they are less perceptive of small “squabbles”. Measure both male and female partners in relationships, self report number of small “squabbles” and severity of last fight.

Counter Examples

- Does calorie consumption (X) impact body image (Y) through weight gain (M) over time?
 - X is observed (not assigned), so different subjects have different X s. Can be dealt with in Multilevel or Latent Growth approaches.
- Any instance where repeated-measure factor is a “nuisance” (e.g. studying schools, but not interested in comparing schools directly).
 - X is observed and doesn’t vary within subjects of study

Mediation

- Between Subjects
 - Path analytic approach
 - Interpretation
 - Estimation
 - Inference
- Two-Instance Repeated-Measures
 - Judd Kenny and McClelland (2001)
 - Path analytic approach
 - Estimation of Indirect Effects
 - MEMORE
 - Reporting (Writing and Figures)
 - Causal inference
 - Study Planning (Power Analysis & Preregistration)
 - Multiple Mediator Models

Running Example: Group Work in Computer Science (BS)

Montoya, A. K. (2013) Increasing Interest in Computer Science through Group Work: A Goal Congruity Approach. <https://osf.io/preprints/psyarxiv/ahgfy>

Between-Subjects Version (CompSci_BS.sav, CompSci_BS.sas) :

Female participants (N = 107) read *one of two* syllabi for a computer science class. One of the syllabi reported the class would have group projects throughout (cond = 1), and the other syllabi stated that there would be individual projects (cond = 0) throughout the class.

Measured Variables:

- Interest in the class ($\alpha = .89$)
 - How interested are you in taking the class you read about?
 - How much would you want to take the class you read about?
 - How likely would you be to choose the class you read about?
 - How interested are you in majoring in computer science?
 - 1 Not at All – 7 Very much

University of Washington

Computer Science & Engineering 142:

Introduction to Programming I

Course Syllabus

Instructor

name: John Johnson
email: j.johnson@uw.edu
office: CSE 800
office phone: (206)555-1234
office hours: see course website

Course Overview

This course provides an introduction to computer science using the Java programming language. CSE 142 is primarily a programming course that focuses on common computational problem solving techniques. No prior programming experience is assumed, although students should know the basics of using a computer (e.g., using a web browser and word processing program) and should be competent with math through Algebra 1. The information, concepts, and analytical thinking introduced in lecture provide a unifying framework for the topics covered in CSE 143.

Lecture Time

MWF 12:00 PM - 1:00 PM, Classroom TBA

Discussion Sections

You will be expected to participate in a weekly discussion section, held on Thursdays (see course website for details). The TA who runs your section will grade your homework assignments. In section, we will answer questions, go over common errors in homework, and discuss sample problems in more detail than lecture.

Course Web Site

- <http://www.cs.washington.edu/142/>

Textbook

- Reges/Stepp, *Building Java Programs: A Back to Basics Approach (2nd Edition)*.

Grading

The primary assessment for your success in this class is exams. There will be 2 midterms and 1 final, and together they make up 85% of your grade. The homework assignments are designed to prepare you for your exams. The exams are designed to assess your ability to utilize the concepts you've learned from your homework and in lecture in new contexts.

5% participation
10% weekly homework assignments
25% midterm 1
25% midterm 2
35% final exam

Exams

Our exams are closed-book and closed-notes, although each student will be allowed to bring a single index card with hand-written notes (no larger than 5" by 8"). No electronic devices may be used, including calculators. Make-up exams will not be given except in case of a serious emergency.

Homework

Homework consists of weekly assignments done in optional groups and submitted electronically on the course web site. Disputes about homework grading must be made within 2 weeks of receiving the grade. If you don't make an honest effort on the homework, your exam score will reflect it.

Academic Integrity and Collaboration

Computer Science is best learned through interacting with your fellow students to ensure that you thoroughly understand each concept. Homework assignments may be completed with other students. You are strongly encouraged to discuss general ideas of how to approach an assignment with other students, and may discuss specific details about what to write with other students. Any help you receive from or provide to classmates should be cited in your assignment. You may seek help from University of Washington CSE 142 TAs, professors, and classmates.

You must abide by the following rules:

- You are highly encouraged to work with another student on homework assignments.
- You may not show another student outside of your class your solution to an assignment, nor look at his/her solution.
- You may not have anyone outside of your class describe in detail how to solve an assignment or sit with you as you write it.
- You may not post online about your homework, other than on the class discussion board, to ask others for help.

University of Washington

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Homework consists of weekly assignments done individually and submitted

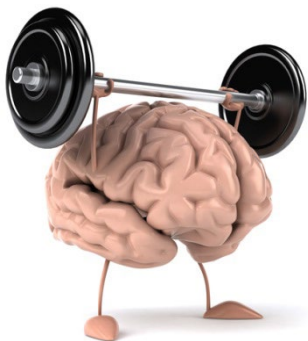
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Academic Integrity and Collaboration

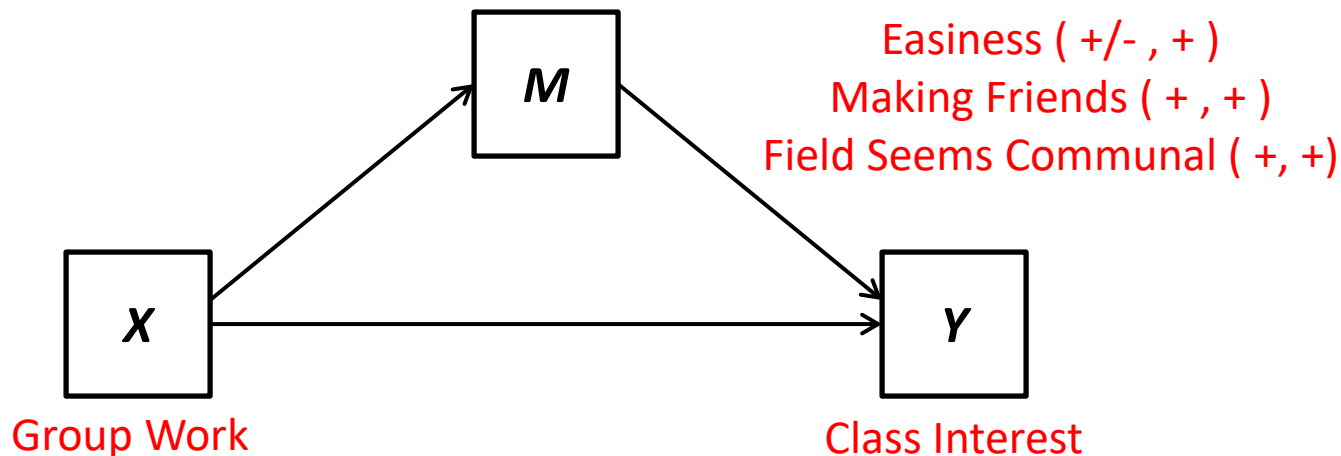
Computer Science is best learned through interacting with the material to ensure that you thoroughly understand each concept. Homework assignments must be completed individually. You may not discuss general ideas of how to approach an assignment with other students or discuss specific details about what to write with other students. Any help you receive from or provide to classmates should be limited. You may seek help from University of Washington CSE 142 TAs and professors.

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- You may not have anyone describe in detail how to solve an assignment or sit with you as you write it.
- You may not post online about your homework to ask others for help.



Mediation



A simple mediation model connects an **assumed** causal variable (X) to an **assumed** outcome variable (Y), through some mechanism (M).

M is frequently referred to as a *mediator*, *intermediary variable*, or *surrogate variable*.

Many different kind of variables may act as mediators. Emotional variables, situational, individual level variables, cognitive variables, environmental variables, etc.

Mediation can be found throughout the psychology literature and is particularly common in social psychology

A quick example: *Name some possible mediators!*

Running Example: Group Work in Computer Science (BS)

Montoya, A. K. (2013) Increasing Interest in Computer Science through Group Work: A Goal Congruity Approach.

Between-Subjects Version (CompSci_BS.sav, CompSci_BS.sas) :

Female participants (N = 107) read *one of two* syllabi for a computer science class. One of the syllabi reported the class would have group projects throughout (cond = 1), and the other syllabi stated that there would be individual projects (cond = 0) throughout the class.

Measured Variables:

- Interest in the class ($\alpha = .89$)
 - How interested are you in taking the class you read about?
 - How much would you want to take the class you read about?
 - How likely would you be to choose the class you read about?
 - How interested are you in majoring in computer science?
 - 1 Not at All – 7 Very much
- CSCComm: Perceptions that computer science is communal ($\alpha = .90$)
 - Computer science would assist me in _____.
 - Helping others, serving the community, working with others, connecting with others, caring for others.
 - 1 Strongly Disagree – 7 Strongly Agree

Mediation: Path Analysis

Consider a , b , c , and c' to be measures of the effect of the variables in the mediation model.

These could be measured using regression coefficients from OLS or path estimates in a structural equation model using maximum likelihood estimation.

Indirect effect of X on Y (through M) = $a \times b$

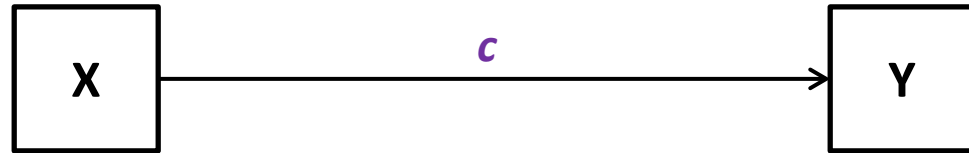
Direct effect of X on Y (not through M) = c'

Indirect effect = total effect - direct effect

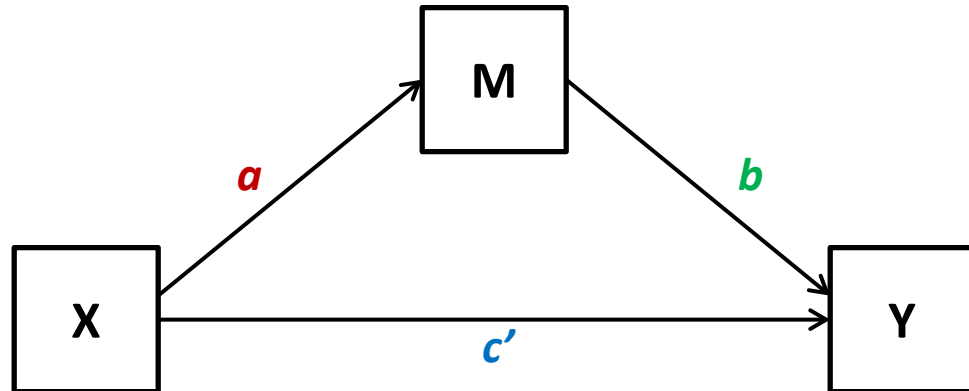
$$a \times b = c - c'$$

Total effect = direct effect + indirect effect

$$c = c' + a \times b$$



$$Y_i = i_{Y^*} + cX_i + e_{Y_i^*}$$

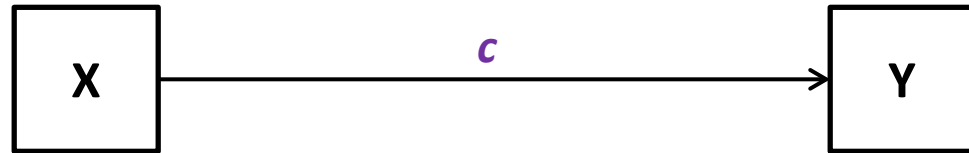


$$M_i = i_M + aX_i + e_{M_i}$$

$$Y_i = i_Y + c'X_i + bM_i + e_{Y_i}$$

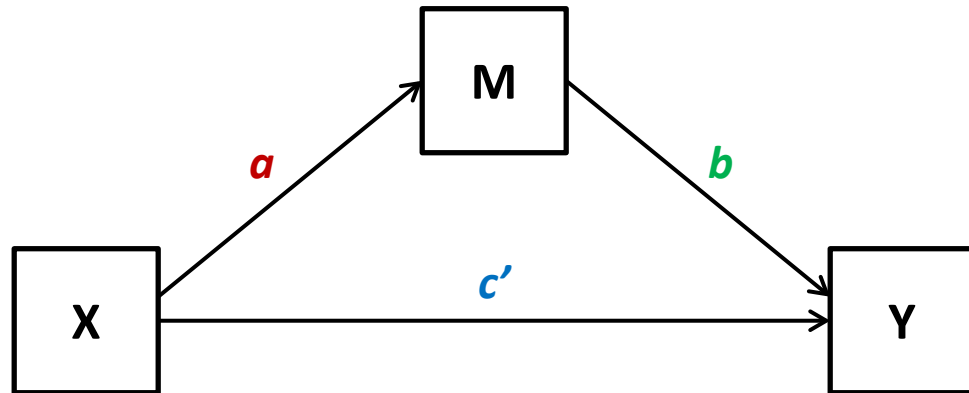
Interpreting the Coefficients

Total Effect (c): The effect of our presumed cause (X) on our outcome (Y), without controlling for any other variables.



a -path: The effect of our presumed cause (X) on our mediator (M).

b -path: The effect of our mediator (M) on the outcome (Y) while controlling for X . (i.e. predicted difference in Y for two people with the same score on X but who differ on M by one unit).



Direct effect (c'): The effect of our presumed cause (X) on Y while controlling for M . (i.e. predicted difference in Y for two people who differ by one unit on X but with the same score on M)

Indirect Effect (ab): Product of effect of X on M , and effect of M on Y controlling for X . The effect of X on Y through M .

The Data: CompSci_BS

SPSS

CompSci_BS.sav [DataSet5] - IBM SPSS Statistics Data Editor

	Subject	Cond	Interest
1	101	1.00	4.00
2	102	1.00	4.00
3	108	1.00	4.67
4	111	1.00	1.67
5	117	.00	4.00
6	118	.00	2.33
7	119	1.00	4.67
8	120	1.00	1.67
9	122	.00	1.00
10	123	.00	1.00
11	125	.00	6.00
12	126	1.00	3.33

The SPSS file is ready for analysis.

SAS

```
data CompSci_BS;
input Subject Cond Interest CSComm grppref
datalines;
101 1 4 4.2 6 6
102 1 4 5.2 6.33333333333333 6.3333
108 1 4.66666666666667 5.6 5.33333333
111 1 1.66666666666667 2 6.33333333
117 0 4 1.8 3.66666666666667 0
118 0 2.33333333333333 4.8 3 0
119 1 4.66666666666667 5 3.66666666
120 1 1.66666666666667 4 4 4
122 0 1 3 5 0
123 0 1 1 3.66666666666667 0
125 0 6 4.2 3.33333333333333 0
126 1 3.33333333333333 3.6 1.66666666
127 1 3 4.2 4.33333333333333 4.3333
128 0 4 3.6 2 0
129 1 6.66666666666667 4.8 3 3
130 0 1 1 5.33333333333333 0
133 1 2.33333333333333 1.8 5.66666666
134 1 2 4 5 5
```

The SAS version is a SAS program that must be executed to produce a temporary work data file.

R

Subject	Cond	Interest	CSComm	grppref	condxgrppref
101	1	4	4.2	6	6
102	1	4	5.2	6.333333	6.333333
108	1	4.666667	5.6	5.333333	5.333333
111	1	1.666667	2	6.333333	6.333333
117	0	4	1.8	3.666667	0
118	0	2.333333	4.8	3	0
119	1	4.666667	5	3.666667	3.666667
120	1	1.666667	4	4	4
122	0	1	3	5	0
123	0	1	1	3.666667	0
125	0	6	4.2	3.333333	0
126	1	3.333333	3.6	1.666667	1.666667
127	1	3	4.2	4.333333	4.333333
128	0	4	3.6	2	0
129	1	6.666667	4.8	3	3

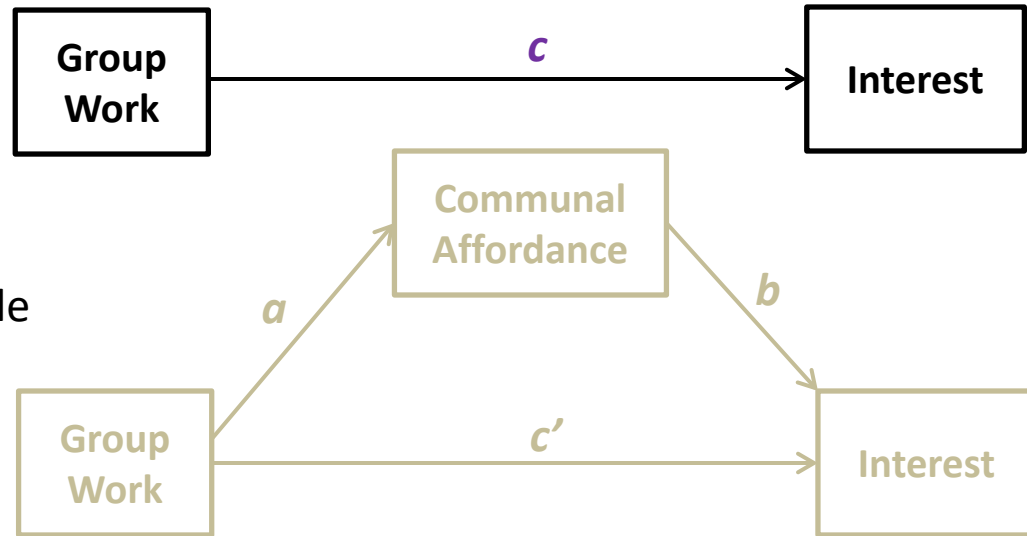
```
filelocation <-
"C:\\Data\\CompSci_BS.csv"
CompSci_BS <- read.csv(filelocation,
header = TRUE)
```

Estimation with CompSci_BS Data

Research Question: Can group work in computer science classes increase women's interest by increasing their perception that computer science is communal?

The c-path can be estimated in a sample using the regression equation below.

$$Y_i = c_0 + cX_i + e_{Y_i^*}$$



```
regression /dep = interest /method = enter cond.
```

```
proc reg data=CompSci_BS;model interest = cond/stb clb;run;
```

```
summary(lm(interest~cond, data = CompSci_BS))
```

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1	(Constant)	2.701	.193	14.002	.000
	Cond	.462	.285	.156	1.621

a. Dependent Variable: Interest

Overall women were .462 units more interested in the class with group work.

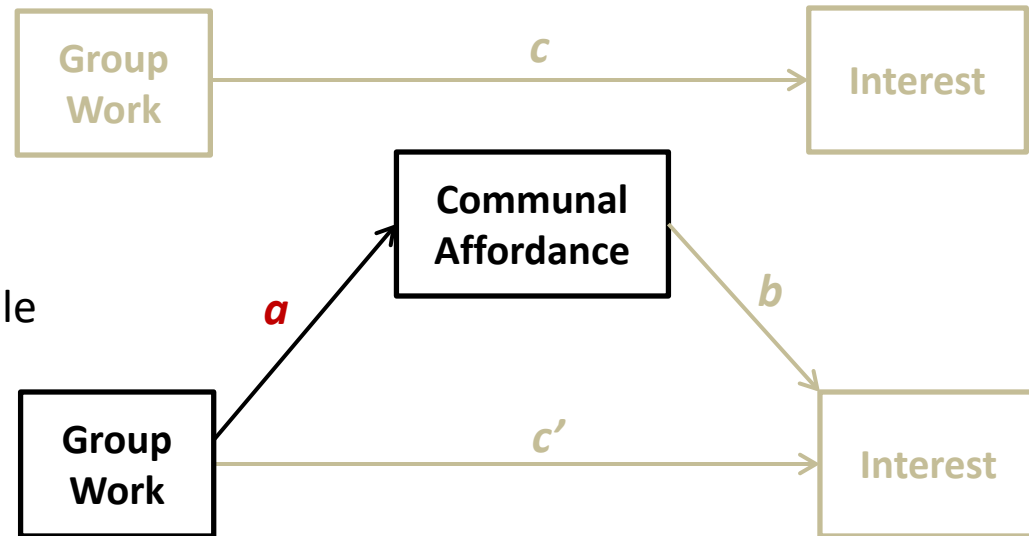
$c = .462$

Estimation with CompSci_BS Data

Research Question: Can group work in computer science classes increase women's interest by increasing their perception that computer science is communal?

The a -path can be estimated in a sample using the regression equation below.

$$M_i = a_0 + aX_i + e_{M_i}$$



```
regression /dep = CScomm /method = enter cond.
```

```
proc reg data=CompSci_BS;model CScomm = cond/stb clb;run;
```

```
summary(lm(CScomm~cond, data = CompSci_BS))
```

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	3.421	.159		21.472	.000
	Cond	.488	.237	.198	2.060	.042

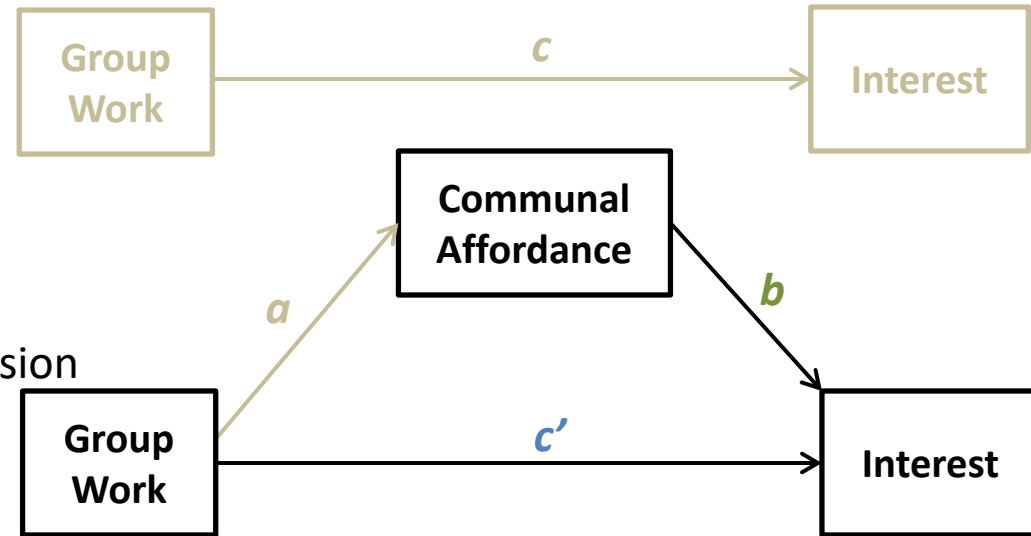
a. Dependent Variable: CSComm

Women saw computer science as .488 units more communal after reading a syllabus with group work.

$a = .488$

Estimation with CompSci_BS Data

Research Question: Can group work in computer science classes increase women's interest by increasing their perception that computer science is communal?



The b -path and direct effect can be estimated in a sample using the regression equation below.

$$Y_i = i_Y + c'X_i + bM_i + e_{Y_i}$$

```
regression /dep = interest /method = enter cond CScomm.
```

```
proc reg data=CompSci_BS;model interest = cond CScomm/stb clb;run;
```

```
summary(lm(interest~cond+CScomm, data = CompSci_BS))
```

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.964	.413		2.336	.021
	Cond	.218	.268	.073	.812	.419
	CSComm	.508	.109	.421	4.663	.000

a. Dependent Variable: Interest

Controlling for communal affordance, women in the group work condition were .218 units more interested in the class with group work.

$c' = 0.218$

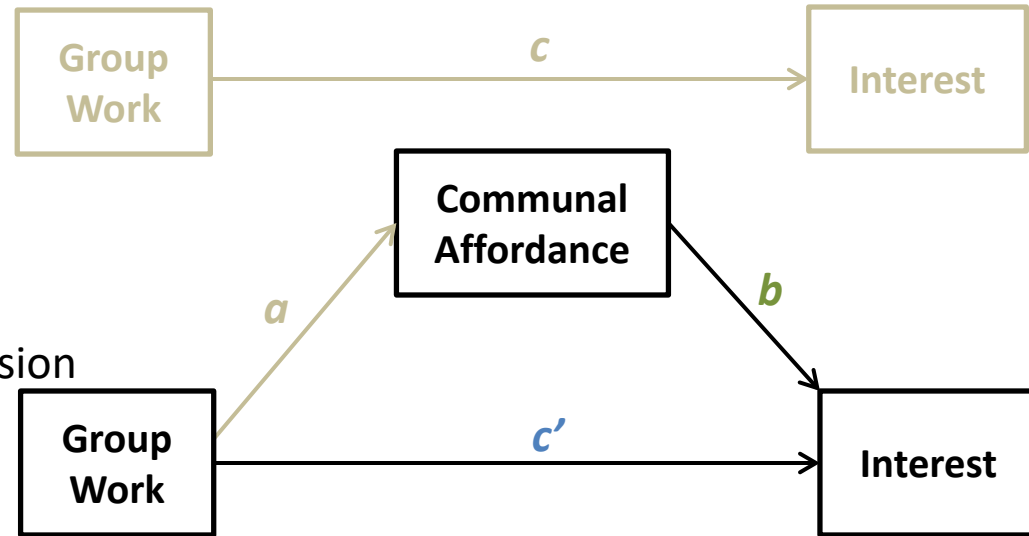
$b = .508$

Estimation with CompSci_BS Data

Research Question: Can group work in computer science classes increase women's interest by increasing their perception that computer science is communal?

The b -path and direct effect can be estimated in a sample using the regression equation below.

$$Y_i = i_Y + c'X_i + bM_i + e_{Y_i}$$



```
regression /dep = interest /method = enter cond CScomm.
```

```
proc reg data=CompSci_BS;model interest = cond CScomm/stb clb;run;
```

```
summary(lm(interest~cond+CScomm, data = CompSci_BS))
```

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.964	.413		2.336	.021
	Cond	.218	.268	.073	.812	.419
	CSComm	.508	.109	.421	4.663	.000

a. Dependent Variable: Interest

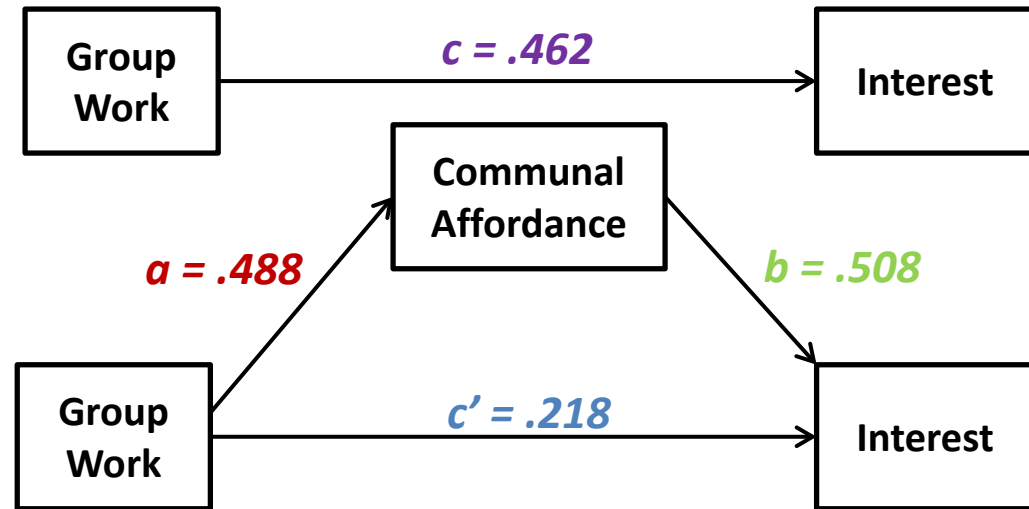
$c' = 0.218$

For two people in the same condition, a one unit difference in communal goals results in a 0.51 unit difference in interest, on average.

Interpreting the Coefficients

Research Question: Can group work in computer science classes increase women's interest by increasing their perception that computer science is communal?

On average, women were .46 units more interested in the class with group work ($p = .108$). Similarly, computer science was perceived as .49 units more communal after reading a syllabus with group work ($p = .042$). Controlling for condition, a one unit increase in communal affordance resulted in a .508 unit increase in interest ($p < .001$). Controlling for communal affordance, group work did not predict additional interest ($c' = .22$, $p = .42$).



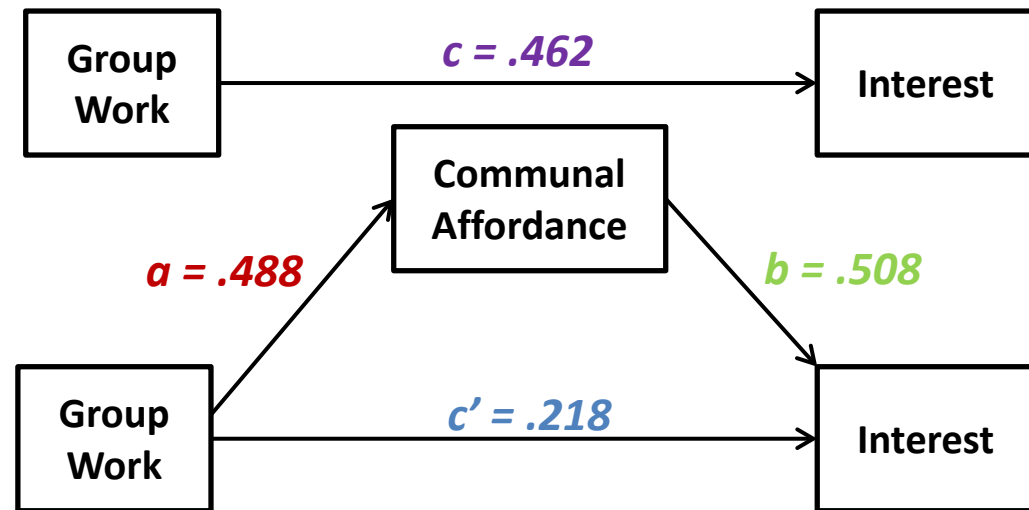
But what about the indirect effect?

Interpreting Indirect, Direct, and Total Effects

Indirect Effect

$$a \times b = .488 \times .508 = .249$$

Group work increased interest by .249 units indirectly through communal affordance. Where group work increased perceptions of communal affordance by .488 units, and a one unit increase in communal affordance resulted in a .508 unit increase in interest.



Direct Effect

$$c' = .218$$

Group work increased interest by .218 units directly (not through communal affordance).

Total Effect

$$c = .462$$

Group work increased interest by .462 units in total.

Inference for the direct and total effects can be drawn from the regression results because these are based on a single regression parameter.

$$p = .419$$

$$p = .108$$

Inference about the Indirect Effect

- How to make proper inference about the indirect effect may be the most active area of research in mediation analysis
- Some methods you may have heard of
 - Causal Steps / Baron and Kenny Method / Baron and Kenny Steps
 - Test of Joint Significance
 - Sobel Test / Multivariate Delta Method
 - Monte Carlo Confidence Intervals
 - Distribution of the Product Method
 - Bootstrap Confidence Intervals
 - Percentile Bootstrap
 - Bias-Corrected Bootstrap
 - Bias Corrected and Accelerated Bootstrap
- Why is this so hard?
 - The product of two normal distributions is not necessarily normal. The shape of the distribution of the indirect effect depends on the true indirect effect.
 - There are many instances where the indirect effect could be zero (either a or b could be zero, or both could be zero).

Causal Steps Method

Method

1. Test if there is a significant total effect ($c \neq 0$).
2. Test if there is a significant effect of X on M ($a \neq 0$).
3. Test if there is a significant effect of M on Y controlling for X ($b \neq 0$).
4. If all three steps are confirmed, test for partial vs. complete mediation.
 1. If X still has an effect on Y controlling for M ($c' \neq 0$), this is partial mediation
 2. If X does not have a significant effect on Y controlling for M , complete mediation

Appeal

- Easy to do, just need regression
- Intuitive

What's wrong with it?

- No estimate of the indirect effect
- No quantification of uncertainty about conclusion
 - p -value
 - Confidence Interval
- Requirement that the total effect is significant before looking for indirect effect
- Issues with *complete* and *partial* mediation

Joint Significance

Method

1. Test if there is a significant effect of X on M ($a \neq 0$).
2. Test if there is a significant effect of M on Y controlling for X ($b \neq 0$).

Appeal

- Easy to do, just need regression
- Intuitive
- Solves issues of requirement of significant total effect to claim an indirect effect.
- Good method balance Type I Error and Power

What's wrong with it?

- No estimate of the indirect effect
- No quantification of uncertainty about conclusion
 - p -value
 - Confidence Interval

Sobel Test / Normal Theory / Delta Method

Method

1. Calculate a Z-statistic

$$Z = \frac{ab}{\sqrt{b^2 s_a^2 + a^2 s_b^2 + s_a^2 s_b^2}}$$

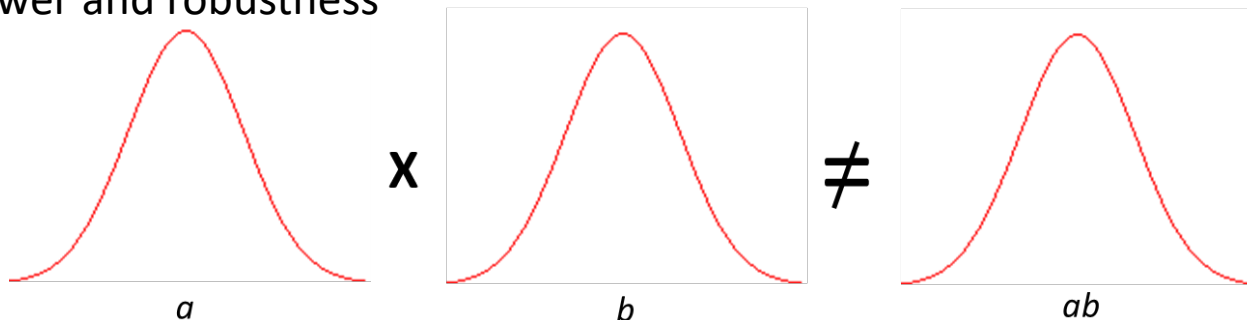
2. Calculate a p-value from Z-statistic

Appeal

- Very similar to process from regression
- Single test of the indirect effect
- Follows from asymptotic theory

What's wrong with it?

- Assumes indirect effect is normally distributed, which is not the case at finite sample sizes
- Poor power and robustness



Bootstrap Confidence Intervals (Percentile)

Empirically estimate sampling distribution of the indirect effect. From this distribution compute confidence intervals which can be used for estimation and hypothesis testing.

Method

1. Randomly sample n cases from your dataset with replacement.
2. Estimate the indirect effect using resampled dataset, call this $ab^{(1)}$
3. Repeat steps 1 and 2 a total of K times where K is many (10,000 recommended), each time calculated $ab^{(k)}$.
4. The sampling distribution of the $ab^{(i)}$'s can be used as an estimate of the sampling distribution of the indirect effect.
5. For a 95% confidence interval the lower and upper bounds will be the 2.5th and 97.5th percentiles of the K estimates of the indirect effect.

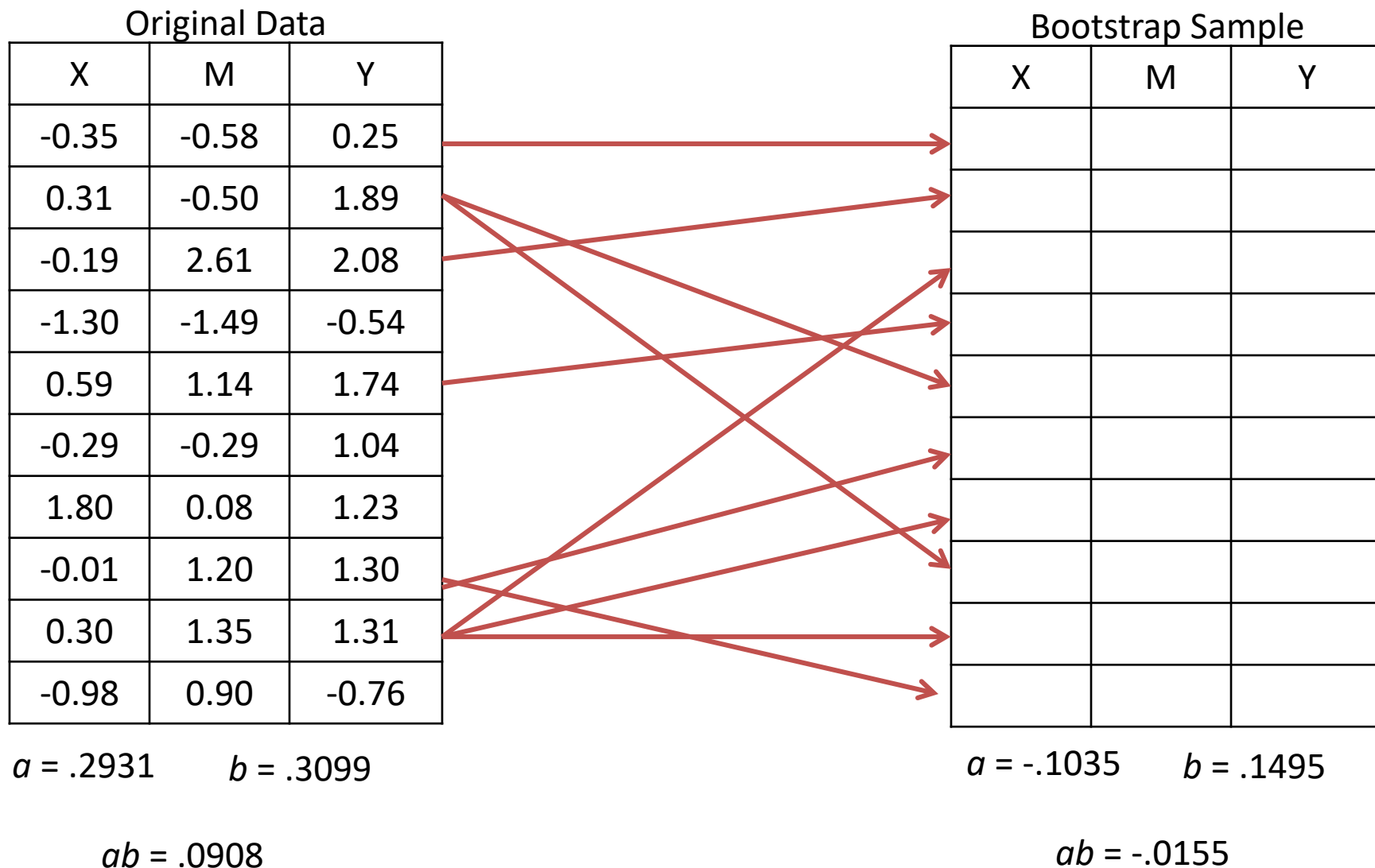
Appeal

- No assumptions about the sampling distribution of the indirect effect
- Provides point estimate of indirect effect
- Can calculate confidence intervals
- Good method balance Type I Error and Power

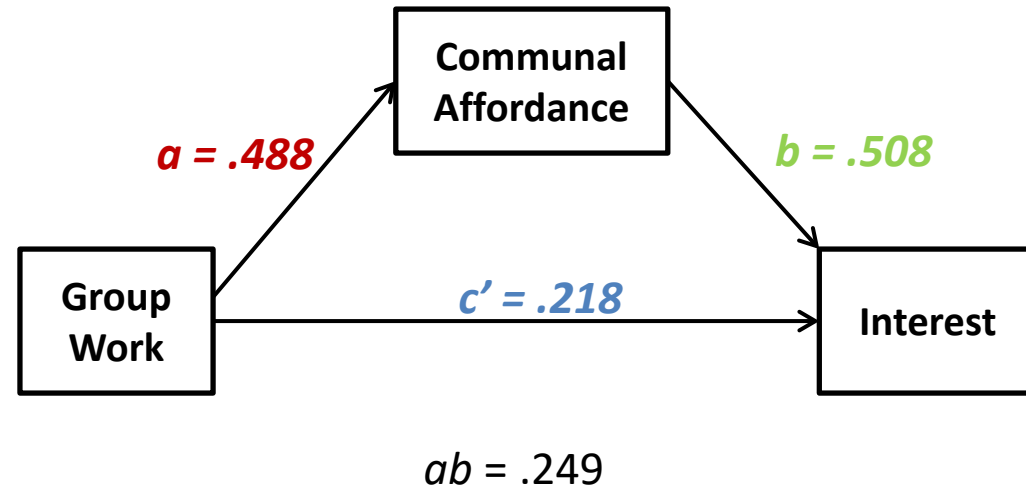
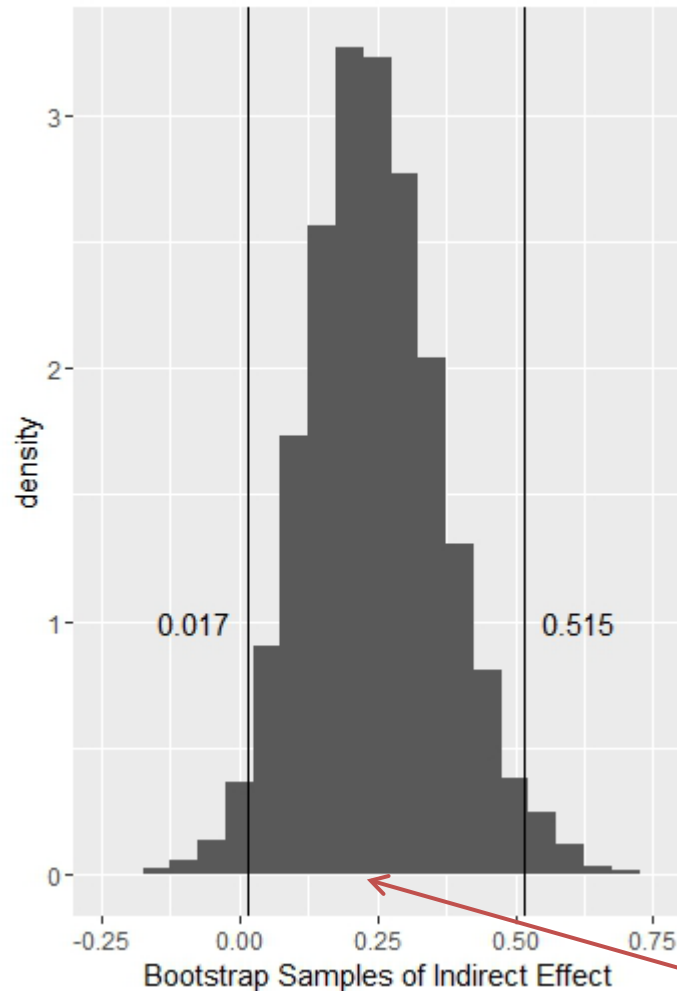
What's wrong with it?

- Not all software has this built in
- Requires original data

Bootstrap Confidence Intervals



Bootstrap Confidence Intervals (CompSci Data)



Zero is not contained in the confidence interval $[0.017, 0.515]$ so we conclude the indirect effect is different from zero with 95% confidence. This is similar to rejecting the null hypothesis at $\alpha = .05$.

$ab = .249$

Other Kinds of Bootstrap Confidence Intervals

All bootstrap confidence intervals use the same basic sampling technique, just use different methods for choosing the end points of the confidence intervals

Bias-Corrected Confidence Interval

- Percentile bootstrapping assumes that your sample estimate (ab) is unbiased in estimating the population indirect effect
- Bias-corrected reduces this assumption to assuming that the bias of ab is a constant (i.e. as N goes to infinity ab will go to the population indirect effect plus some constant)
- Bias-corrected confidence intervals estimate the bias of ab then adjust edges of confidence interval to be “bias-corrected” (i.e. centered not around your original estimate of ab), but around the point based on the bias estimation.

Bias-Corrected and Accelerated

- Same principles as BC regarding bias correction
- Acceleration allows for the assumption that the standard error of the indirect effect depends on the population value of the indirect effect
- Acceleration parameter, which is used to adjust the ends of the confidence interval is estimated using leave-one-out estimates of skew of the estimates of the indirect effect.

BC and BCA Bootstrap Confidence Intervals have been shown to have inflated Type I Error rates compared to other methods and are **not recommended**

The Monte Carlo Interval

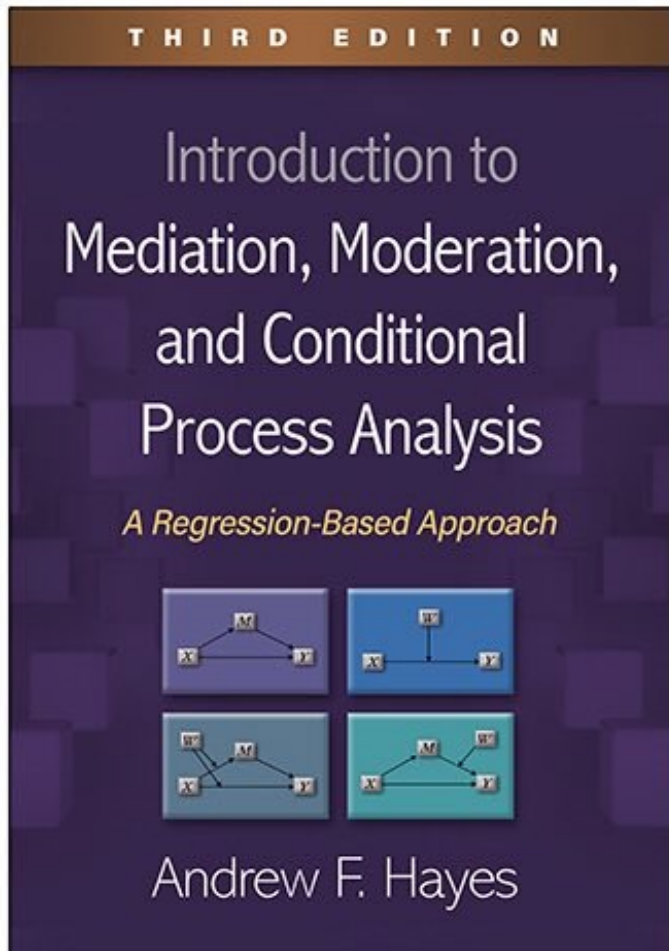
Monte Carlo empirically estimate the sampling distribution of the indirect effect and generate a confidence interval (CI) for estimation and hypothesis testing. This simulation based method assumes each individual path (a and b) are normally distributed.

- (1) Generate k samples from a normal distribution with mean a and standard deviation s_a
- (2) Generate k samples from a normal distribution with mean b and standard deviation s_b
- (3) Multiply samples together to get a distribution of k estimates of ab .
- (4) Rank order estimates and select estimates which define the lower percentile of sorted k estimates and upper percentile of sorted estimates which define CI of interest.
- (5) For 95% CI lower and upper bounds are 2.5th and 97.5th percentile in k bootstrap estimates of the indirect effect.

This method performs well (similarly to bootstrapping) in a variety of simulation studies, but is still less popular.

This method makes stronger assumptions than bootstrapping, but does not seem to result in greater power.

PROCESS Macro



Published in January 2022 and available through The Guilford Press, Amazon.com, and elsewhere.

- First released in beta form in March of 2012 and later documented in Hayes (2013, IMMCPA, published by The Guilford Press).
- Available for both SPSS (in macro and “custom dialog” form), SAS, and R.
- An integration of functions available in my other published macros for mediation and moderation analysis (SOBEL, INDIRECT, MODMED, MODPROBE, MED3C) and a whole lot more, all in one command.
- A handy tool for both “confirmatory” and “exploratory” approaches to data analysis.
- Freely available at www.processmacro.org. The current release is v4.3

For more on the use of PROCESS take my Statistical Horizons Course

THINKING DEEPLY ABOUT CAUSALITY

Understanding Cause and Effect

As scientists we're often looking to support a claim that "X causes Y." Many of us are familiar with the phrase "correlation is not causation." But what then is needed to support a claim of cause?

Often we rely on experimentation to help us support the claim of cause. But what happens when we cannot (ethically or practically) manipulate our causal variable?

Consider the claim "Smoking tobacco causes lung cancer." Is it unethical to randomize people to smoke or not smoke. How then do we know this claim is true?

Necessary Conditions for Cause:

1. Covariation
2. Temporal Ordering
3. Elimination of competing explanations

Research on tobacco use easily found evidence for 1 & 2, and slowly over time accumulation of evidence supported 3.

Understanding Cause and Effect

Necessary Conditions for Cause:

1. Covariation
2. Temporal Ordering
3. Elimination of competing explanations

The following methods are frequently used to support the claim of cause. Which conditions for cause are supported by each method?

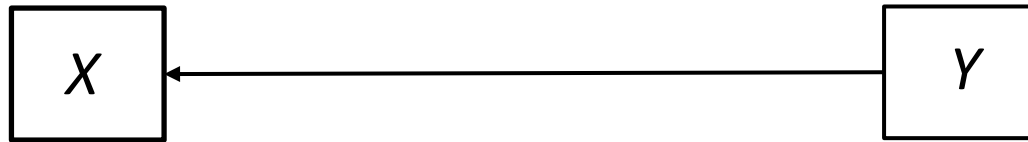
Experimental Manipulation:

Longitudinal Studies:

Cross-sectional Data Collection analyzed using Linear Regression:

If it's not a cause, what is it?

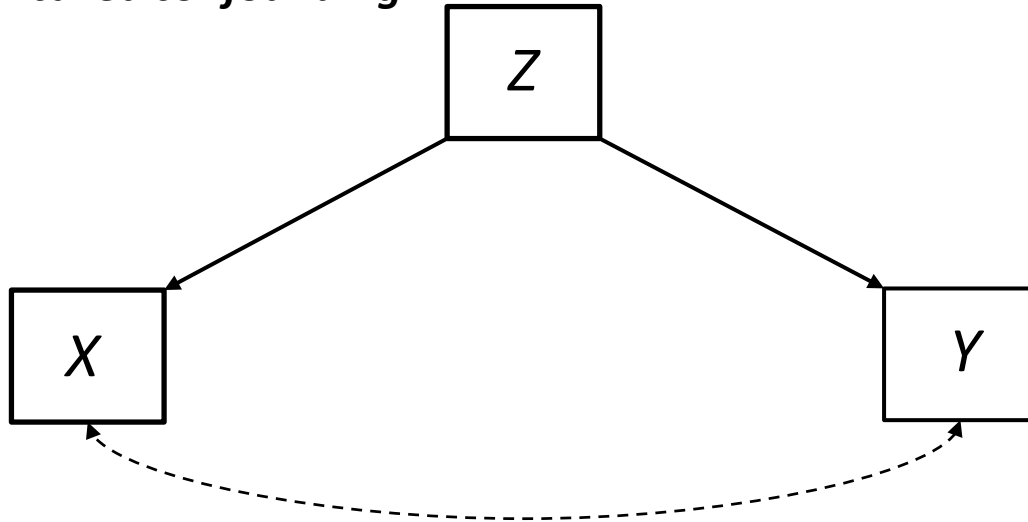
Effect/Reverse Causation: It's possible that X is an effect of Y , rather than Y being an effect of X



If it's not a cause, what is it?

Spurious association: when relationship between X and Y is induced by a shared cause.

This is often called *confounding*

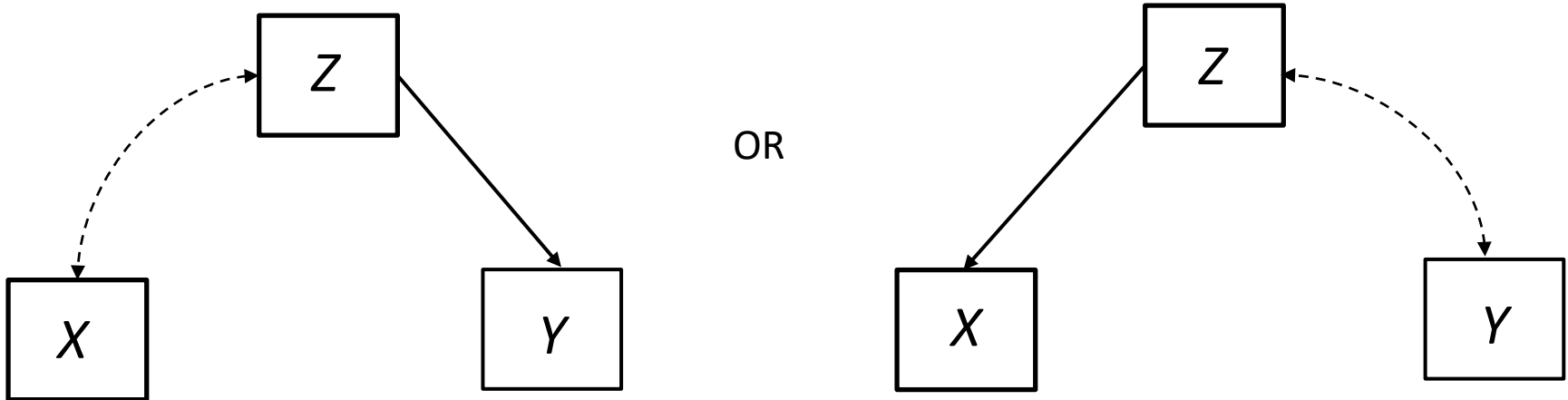


Example: [Skirt length theory](#) is one that suggests that skirt lengths predict the stock market (short skirt → market going up). Likely this is not a causal relationship but rather both skirt length and market trends are influenced by other larger cultural/economic trends.

Controlling for Z helps us eliminate spurious explanations.

If it's not a cause, what is it?

Epiphenomenality: X and Y are related because X is correlated with a cause of Y

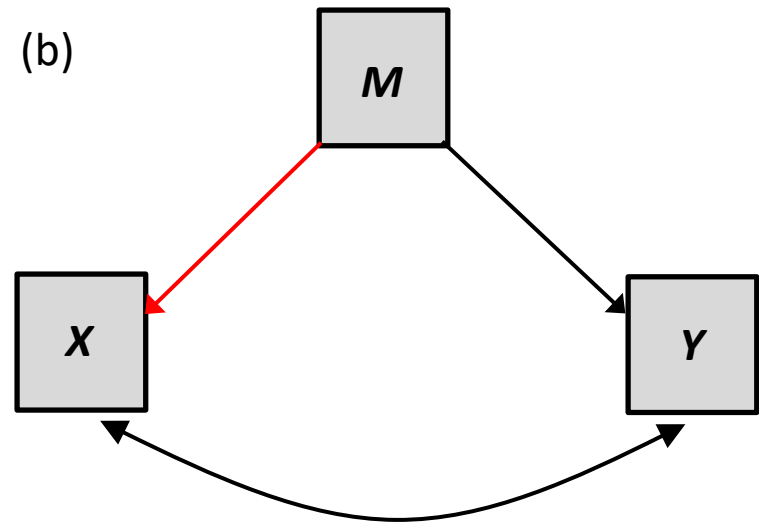
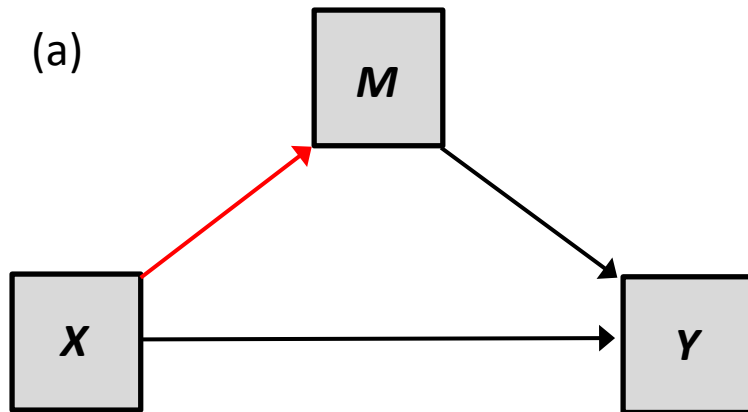


Example: Having an increased risk of breast cancer (Y) concurrent with taking an antibiotic is an **epiphenomenon**. It is not the antibiotic that is causing the increased risk, but the increased inflammation associated with the bacterial infection (Z) that prompted the taking of an antibiotic (X).

Controlling for Z helps us eliminate epiphenomenal explanations.

Mediation and spuriousness

Mediation analysis cannot distinguish between (a) mediation and (b) spuriousness. If (b) can be deemed plausible, that weakens the case for (a) regardless of what the data analysis tells you.

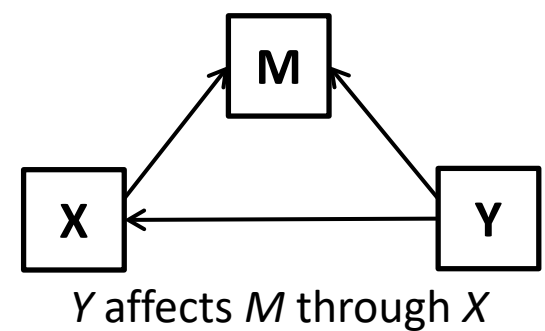
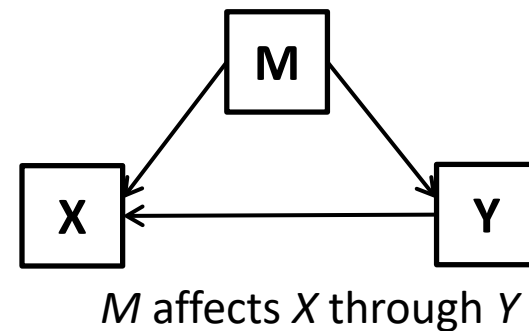
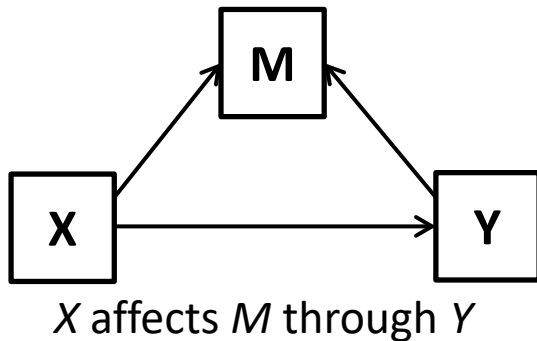
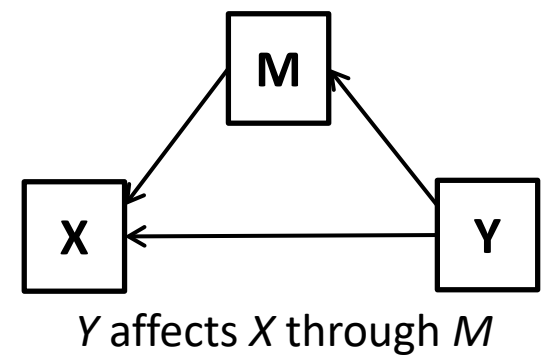
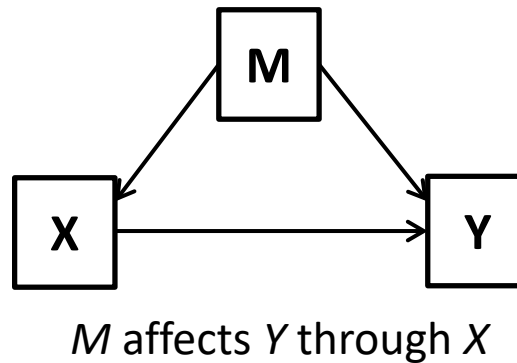
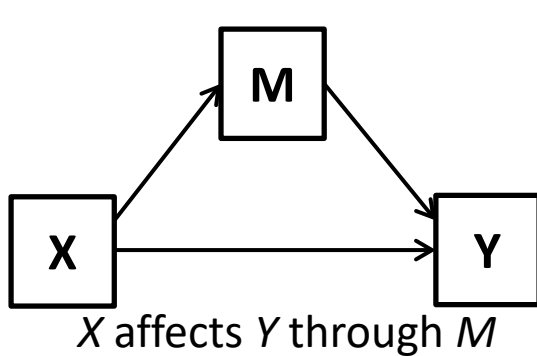


Inferences are always design-bound.

Mediation is a causal process, but causal claims are only justified if the design allows such claims, regardless of what the statistics say.

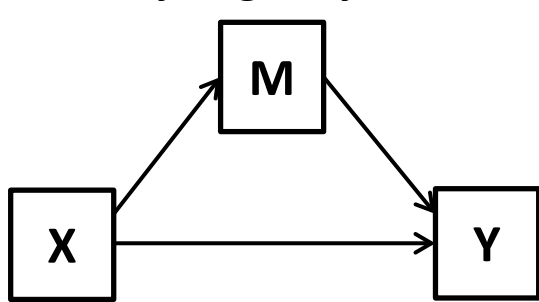
Mediation and Causality

There are a number of alternative causal processes that may be occurring when a *statistical indirect effect* is present:

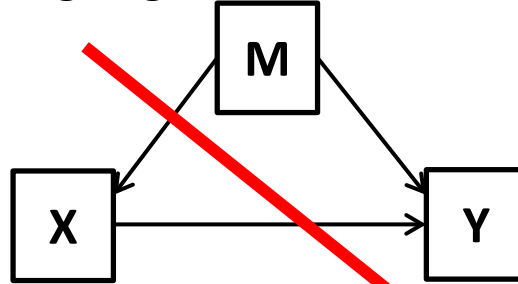


Mediation and Causality

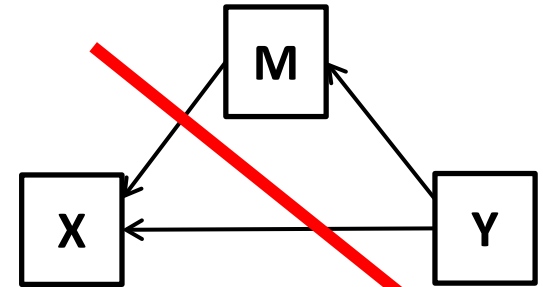
What you get by randomly assigning X .



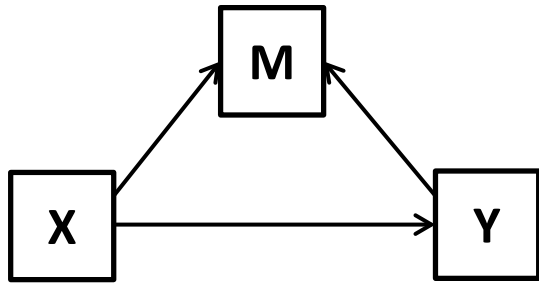
X affects Y through M



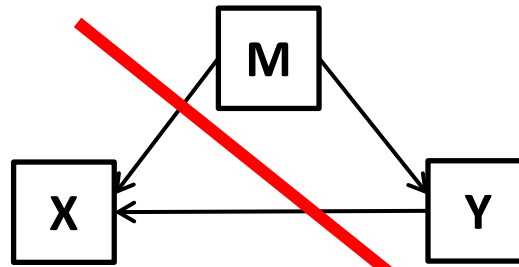
M affects Y through X



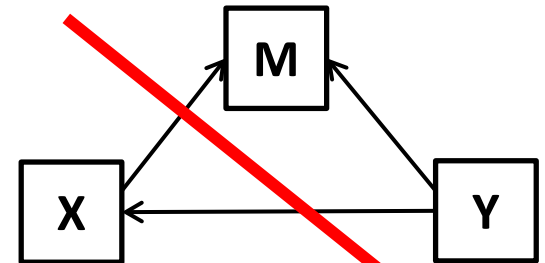
Y affects X through M



X affects M through Y



M affects X through Y



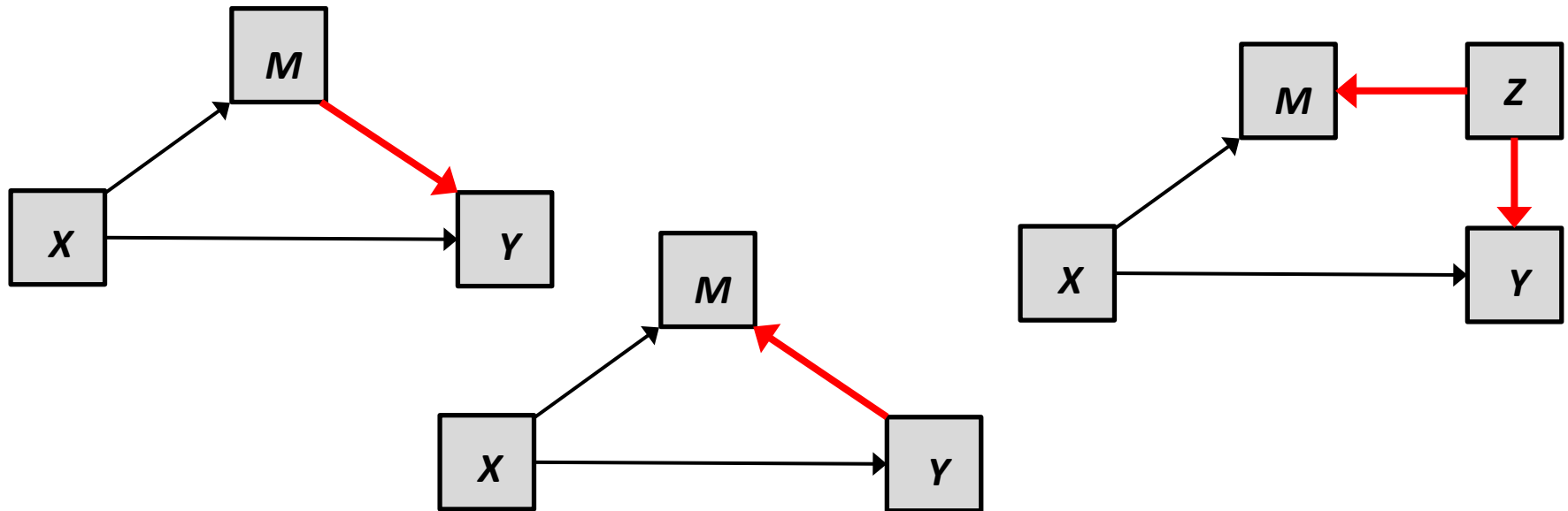
Y affects M through X

Even when X is randomly assigned, we can not provide evidence for the causal order between M and Y . This can only be supported using other experiments or previous research.

A statistically significant indirect effect does not lend credence to one model over another (Thoemmes, 2015, Basic and Applied Social Psychology).

Manipulation of X

Manipulation of and random assignment to X affords causal inference for the effect of X on M and Y , but not the effect of M on Y . We cannot establish causal order for the M - Y path using the methods that are the focus here. Theory is important. Multiple studies can help, one of which involves manipulation of M .



When X is not experimentally manipulated, all paths are subject to potential alternative causal orders or confounding.

Understanding Cause and Effect

Consider two male, Caucasian politicians. One of these two has been convicted of political corruption. Which one is it?



Understanding causal effects

Never Convicted



Convicted



Understanding causal effects

Lin, C., Adolphs, R., & Alvarez, R. M. (2018). Inferring whether officials are corruptible from looking at their faces. *Psychological Science*, 29(11), 1807-1823. 



Research Article

Inferring Whether Officials Are Corruptible From Looking at Their Faces



Chujun Lin, Ralph Adolphs, and R. Michael Alvarez

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aps
ASSOCIATION FOR
PSYCHOLOGICAL SCIENCE

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www.psychologicalscience.org/PS
SAGE

Abstract

While inferences of traits from unfamiliar faces prominently reveal stereotypes, some facial inferences also correlate with real-world outcomes. We investigated whether facial inferences are associated with an important real-world outcome closely linked to the face bearer's behavior: political corruption. In four preregistered studies ($N = 325$), participants made trait judgments of unfamiliar government officials on the basis of their photos. Relative to peers with clean records, federal and state officials convicted of political corruption (Study 1) and local officials who violated campaign finance laws (Study 2) were perceived as more corruptible, dishonest, selfish, and aggressive but similarly competent, ambitious, and masculine (Study 3). Mediation analyses and experiments in which the photos were digitally manipulated showed that participants' judgments of how corruptible an official looked were causally influenced by the face width of the stimuli (Study 4). The findings shed new light on the complex causal mechanisms linking facial appearances with social behavior.

Keywords

face perception, corruption, social attribution, stereotyping, political psychology, open data, open materials, preregistered

Received 9/25/17; Revision accepted 6/3/18

Table 1. Results for Correctly Categorized Officials Based on Aggregate-Level Trait Inferences and Individual-Level Trait Inferences From Study 1

Trait	Aggregate-level accuracy				Average individual-level accuracy ^a				
	Percentage of correctly categorized officials ($N = 72$)	Lower bound of 95% CI	$\chi^2(1)$	p	Mean accuracy ($N = 82$)	SD	Lower bound of 95% CI	$t(81)$	Cohen's d
Corruptibility	69.44%	59.22%	10.13	< .001	55.73%	6.95%	54.46%	7.47	0.82
Dishonesty	70.83%	60.67%	11.68	< .001	54.82%	6.41%	53.64%	6.81	0.75
Selfishness	66.67%	56.36%	7.35	.003	55.10%	6.76%	53.86%	6.83	0.75
Trustworthiness	68.06%	57.79%	8.68	.002	55.03%	6.41%	53.85%	7.10	0.78
Generosity	63.89%	53.53%	5.01	.013	54.97%	5.99%	53.87%	7.51	0.83

Note: CI = confidence interval.

^aAll p s for this variable are less than .001.

Faces are rich in information: They provide clues about gender, race, age, and trait attributes, which are inferred spontaneously and ubiquitously (Engell, Haxby, & Todorov, 2007; Todorov, 2017). Moreover, such inferences often guide our social behavior—for instance, we decide whom to trust on the basis of how trustworthy a face looks (Rezlescu, Duchaine, Olivola, & Chater, 2012; Van't Wout & Sanfey, 2008). Many trait judgments made by participants across generations and cultures show consensus (Cogsdill, Todorov, Spelke, & Banaji, 2014; Lin, Adolphs, & Alvarez, 2017; Rule et al., 2010). But are trait judgments from faces accurate?

Previous research has shown that trait judgments from faces can be associated with important real-world social outcomes, such as dating and mating (Olivola et al., 2014; Valentine, Li, Penke, & Perrett, 2014), earnings and fundraising (Genevsky & Knutson, 2015; Hamermesh, 2011; Ravina, 2012), science communication (Gheorghiu, Callan, & Skylark, 2017), sentencing decisions (Berry & Zebrowitz-McArthur, 1988; Blair, Judd, & Chapleau, 2004; Wilson & Rule, 2015; Zebrowitz

& McDonald, 1991), and leader selection (Todorov, Mandisodza, Goren, & Hall, 2005; for reviews, see Antonakis & Eubanks, 2017; Todorov, Olivola, Dotsch, & Mende-Siedlecki, 2015). Yet this prior research on the association between trait judgments from faces and real-world outcomes leaves open two important questions. First, most associations have focused on prosocial outcomes (e.g., correlations between competence judgments and election success; Todorov et al., 2005). Second, most associations are plausibly driven not by the behavior of the targets whose face is being judged but by the interests of the perceivers who are making the judgments (e.g., correlations between interesting-looking scientists and the perceiver's interest in their work). Here, we investigated an antisocial judgment that

Corresponding Author:

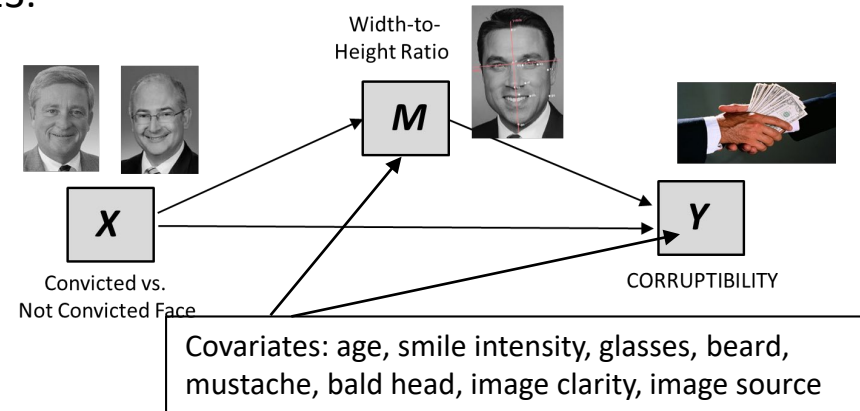
Chujun Lin, California Institute of Technology, Division of Humanities and Social Sciences, HSS 228-77, 1200 E. California Blvd., Pasadena, CA 91125
E-mail: cln7@caltech.edu

Understanding Cause and Effect

Lin, C., Adolphs, R., & Alvarez, R. M. (2018). Inferring whether officials are corruptible from looking at their faces. *Psychological Science*, 29(11), 1807-1823.

Necessary Conditions for Cause:

1. Covariation
2. Temporal Ordering
3. Elimination of competing explanations



1. Data from Studies 1 – 3 were combined to run the above mediation analysis. Have the necessary conditions for cause been met? Which necessary conditions does the mediation analysis support? Which require further support?
(Hint: Think about what would make you feel more confident in this causal order)

2. Study 4b experimentally manipulated the width-to-height ratios of faces to measure the impact on corruptibility. Which condition for cause does this help support? Are we now completely convinced about the causal claims of this mediation analysis?

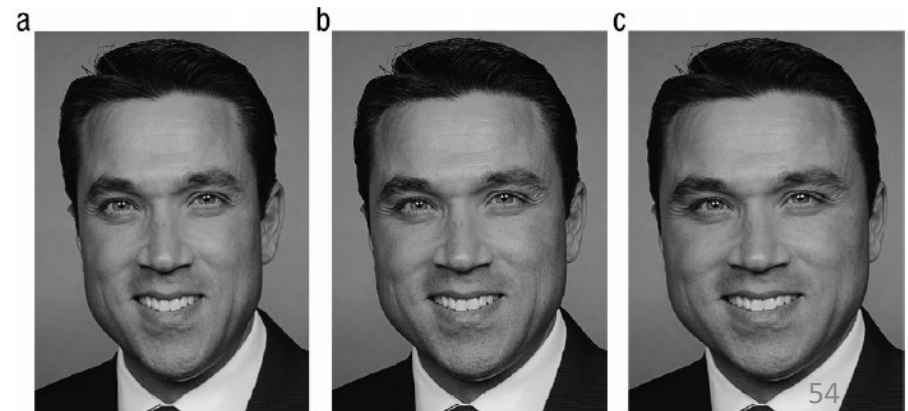


Fig. 6. Example of the same face in (a) slim, (b) original, and (c) fat versions.

MEDIATION IN WITHIN-SUBJECT DESIGNS

Running Example: Group Work in Computer Science (WS)

Montoya, A. K. (2013) Increasing Interest in Computer Science through Group Work: A Goal Congruity Approach (Undergraduate Honors Thesis).

Within-Subjects Version (CompSci_WS.sav, CompSci_WS.sas, CompSci_WS.csv) :

Female participants (N = 51) read two syllabi for a different computer science classes. One of the syllabi reported the class would have group projects throughout, and the other syllabi stated that individual project would be scheduled throughout.

- Syllabi also differed in professor's name (but not gender), and the primary programming language used in the class.

Measured Variables:

- Interest in each the class (same as BS version)
 - Two measures: `int_i` `int_g`
- Perceptions that the class has a communal environment.
 - Two measures: `comm_i` `comm_g`
 - Taking this class would assist me in _____.
 - Helping others, serving the community, working with others, connecting with others, caring for others.
- How difficult would you rate the class you read about?
 - Two measures: `diff_i` `diff_g`

Judd, Kenny, and McClelland (2001)

Judd, C. M., Kenny, D. A., & McClelland, G. H. (2001). Estimating and testing mediation and moderation in within-subject designs. *Psychological Methods*, 6, 115-134.



One of the few treatments of mediation analysis in this common research design.

A “causal steps”, Baron and Kenny type logic to determining whether M is functioning as a mediator of X 's effect on Y when both M and Y are measured twice in difference circumstances but on the same people.

1. On average, does Y differ by condition?
2. On average, does M differ by condition?
3. Does difference in M predict a difference in Y ?
4. Does the difference in M account for all the difference in Y ?

Computer Science Within-Subjects Data Example

Montoya, A. K. (2013) Increasing Interest in Computer Science through Group Work: A Goal Congruity Approach (Undergraduate Thesis).

Research Question: Can group work in computer science classes increase women's interest by increasing their perception that computer science is communal?

Data is in *wide form*: repeated measurements of the same variables are saved as separate variables (one row per participant). *Long form* is when there is a variable coding instance of repeated measurements (multiple rows per participant, one for each instance).

CompSci_WS.sav

int_I	int_G	comm_I	comm_G
1.50	4.00	1.00	6.80
2.75	3.25	2.00	5.40
5.75	2.50	3.20	3.60
3.50	5.75	1.60	5.20
2.25	2.00	4.40	4.60
1.50	1.75	3.00	5.00
2.50	4.25	4.20	4.40
6.00	1.75	4.80	2.40
3.00	2.00	2.60	5.80
4.00	5.25	1.60	5.00
5.00	5.00	4.60	6.20
2.00	1.75	3.80	4.20
1.00	1.75	2.60	3.20
1.25	4.50	1.00	6.00
5.75	4.50	2.60	6.00
3.25	4.75	3.00	6.20
2.75	2.25	4.80	4.60
5.50	2.00	4.00	7.00
1.75	5.25	1.60	5.60
4.00	5.50	1.80	5.40
2.25	4.00	2.20	4.80
4.00	6.50	2.00	6.80
5.00	4.50	3.20	6.00

Analysis using Judd et al. (2001)

1. On average, does Y differ by condition?

Setup a model of the outcome in each condition:

$$Y_{1i} = c_1 + e_{Y_{1i}^*}$$

$$Y_{2i} = c_2 + e_{Y_{2i}^*}$$

Is c_1 different from c_2 ?

Based on these models, setup a new model where you can directly estimate and conduct inference on what you are interested in (in this case $c_2 - c_1$):

$$Y_{2i} - Y_{1i} = (c_2 - c_1) + (e_{Y_{2i}^*} - e_{Y_{1i}^*}) = c + e_{Y_i^*}$$

Use intercept only regression analysis, or a paired sample t-test, or a one sample t-test on the differences to conduct inference on $c_2 - c_1$

With the data: On average, is class interest higher in the group work condition?

Paired Samples Test

	Paired Differences					t	df	Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
Pair 1 int_G - int_I	.37255	1.99585	.27948	-.18879	.93389	1.333	50	.189

```
T-TEST PAIRS = int_G with int_I (PAIRED) .
```

```
PROC TTEST DATA=CompSci_WS; PAIRED int_G*int_I; RUN;
```

```
t.test(Pair(int_G, int_I)~1, data = CompSci_WS)
```


Analysis using Judd et al. (2001)

2. On average, does M differ by condition?

Setup a model of the mediator in each condition:

$$M_{1i} = a_1 + e_{M_{1i}}$$

$$M_{2i} = a_2 + e_{M_{2i}}$$

Is a_1 different from a_2 ?

Based on these models, setup a new model where you can directly estimate and conduct inference on what you are interested in (in this case $a_2 - a_1$):

$$M_{2i} - M_{1i} = (a_2 - a_1) + (e_{M_{2i}} - e_{M_{1i}}) = a + e_{M_i}$$

Use intercept only regression analysis, or a paired sample t-test, or a one sample t-test on the differences to conduct inference on $a_2 - a_1$

With the data: On average, is communal goal affordance higher in the group work condition?

Paired Samples Test

	Paired Differences					t	df	 Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
Pair 1 comm_G - comm_I	2.29412	1.77870	.24907	1.79385	2.79438	9.211	50	.000



```
T-TEST PAIRS = comm_G with comm_I (PAIRED).
```

```
PROC TTEST DATA=CompSci_WS; PAIRED comm_G*comm_I; RUN;
```

```
t.test(Pair(comm_G, comm_I)~1, data = CompSci_WS)
```


Analysis using Judd et al. (2001)

3. Does difference in M predict a difference in Y ? / Does M predict Y controlling for condition?

Setup a model of the outcome in each condition:

$$Y_{1i} = g_{10} + g_{11}M_{1i} + e_{Y_{1i}}$$

$$Y_{2i} = g_{20} + g_{21}M_{2i} + e_{Y_{2i}}$$

Note that there are **two estimates** of the effect of M on Y . Let's average them to estimate an average effect of M on Y . Setup a new model where you can directly estimate and conduct inference on what you are interested in (in this case $\frac{1}{2}(g_{21} + g_{11})$):

$$Y_{2i} - Y_{1i} = (g_{20} - g_{10}) + g_{21}M_{2i} - g_{11}M_{1i} + (e_{Y_{2i}} - e_{Y_{1i}})$$

Optional
board work

$$Y_{2i} - Y_{1i} = (g_{20} - g_{10}) + \underbrace{\frac{g_{21} + g_{11}}{2}}_b (M_{2i} - M_{1i}) + \underbrace{\frac{(g_{21} - g_{11})}{2}}_d (M_{2i} + M_{1i}) + (e_{Y_{2i}} - e_{Y_{1i}})$$

Analysis using Judd et al. (2001)

3. Does *M* predict *Y* controlling for condition?

With the data: Does communal goal affordance predict interest in the class?

```
compute int_diff = int_G - int_I.  
compute comm_diff = comm_G - comm_I.  
compute comm_sum = comm_G+comm_I.  
EXECUTE.  
regression dep = int_diff /method = enter comm_diff comm_sum.
```

```
data CompSci_WS;set CompSci_WS;int_diff=int_g-int_I;  
comm_diff = comm_G - comm_I; comm_sum = comm_G+comm_I;run;  
proc reg data=CompSci_WS;model int_diff=comm_diff comm_sum;run;
```

```
CompSci_WS <- transform(CompSci_WS,  
                        int_diff = int_G - int_I,  
                        comm_diff = comm_G - comm_I,  
                        comm_sum = comm_G +comm_I)  
summary(lm(int_diff~comm_diff+comm_sum, data = CompSci_WS))
```

Analysis using Judd et al. (2001)

3. Does *M* predict *Y* controlling for condition?

With the data: Does communal goal affordance predict interest in the class?

```
compute int_diff = int_G - int_I.  
compute comm_diff = comm_G - comm_I.  
compute comm_sum = comm_G+comm_I.  
EXECUTE.  
regression dep = int_diff /method = enter comm_diff comm_sum.
```

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.310	1.877		.698	.489
	comm_diff	.590	.135	.526	4.385	.000
	comm_sum	-.275	.216	-.153	-1.272	.210



a. Dependent Variable: int_diff

Analysis using Judd et al. (2001)

4. Does the difference in communal goal affordance account for all the difference in interest?

$$Y_{2i} - Y_{1i} = (g_{20} - g_{10}) + \underbrace{\frac{g_{21} + g_{11}}{2}}_b (M_{2i} - M_{1i}) + \underbrace{\frac{(g_{21} - g_{11})}{2}}_d (M_{2i} + M_{1i}) + (e_{Y_{2i}} - e_{Y_{1i}})$$

Next we center the sum term, so the intercept has the interpretation of the predicted difference in Y for someone with no difference in M 's but is average on M 's.

$$Y_{2i} - Y_{1i} = c' + b(M_{2i} - M_{1i}) + d(M_{2i} + M_{1i} - (\overline{M_2 + M_1})) + (e_{Y_{2i}} - e_{Y_{1i}})$$

$$\text{where } c' = (g_{20} - g_{10} + d(\overline{M_2 + M_1}))$$

Intercept is predicted *outcome* when all regressors are zero. This means predicted difference in Y when there is no difference in M and a person is average on the sum of M .

Analysis using Judd et al. (2001)

4. Does the difference in communal goal affordance account for all the difference in interest?

With the data: Is there a significance difference in interest predicted when there is no difference in communal goals?

```
compute comm_sumc = comm_sum- 8.325490.  
EXECUTE.  
regression dep = int_diff /method = enter comm_diff comm_sumc.
```

```
data CompSci_WS;set CompSci_WS;comm_sumc = comm_sum-8.325490;run;  
proc reg data=CompSci_WS;model int_diff=comm_diff comm_sumc;run;
```

```
CompSci_WS <- transform(CompSci_WS,  
                        comm_sumc = comm_sum-8.325490)  
summary(lm(int_diff~comm_diff+comm_sumc, data = CompSci_WS))
```

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-.981	.388		-2.527	.015
	comm_diff	.590	.135	.526	4.385	.000
	comm_sum	-.275	.216	-.153	-1.272	.210

a. Dependent Variable: int_diff



Analysis using Judd et al. (2001)

- 
1. On average, is interest higher in the group work condition?
 2. On average, is communal goal affordance higher in the group work condition?
 3. Does difference in communal affordance predict a difference in interest?
 4. Does the difference in communal goal affordance account for all the difference in interest?

According to Judd, Kenny, and McClelland we do not have a mediated effect!

Because there is no evidence that interest is higher in the group work condition, the Judd et al. (2001) method would conclude there is not mediation.

Judd et al. Criticisms and Misuses

All criticisms of the causal steps approach apply to this approach:

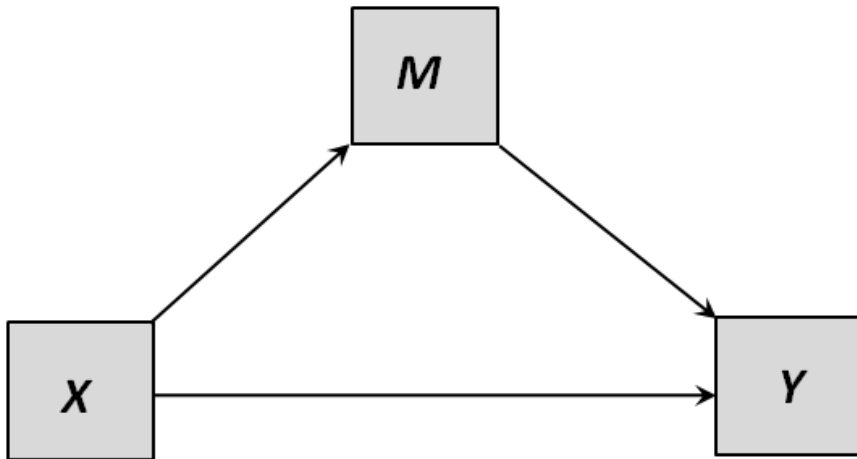
- There is no explicit quantification of the indirect effect
 - Inference about an indirect effect should be the result of a test on a *quantification* of the indirect effect
- Requiring that there must be a total effect is too restrictive
 - The direct and indirect effect could be of opposite sign
 - There is greater power to detect the indirect effect than total effect (*Judd, Kenny, 2014, Psych Science*)

This method has been used by a variety of researchers:

- Approximately 1150 citing papers, with around 1/3 using this method
- Many researchers do not report or estimate the partial regression coefficient for the sum of the mediators
- Because the estimate of the indirect effect is not made explicit, researchers often misinterpret the coefficients
 - *b* path is often interpreted as indirect effect
- Extensions to more complicated models have been poorly implemented until recently

Can we think about it like a path analysis?

Analytic Goal: Can group work in computer science classes increase women's interest by increasing their perception that computer science is communal?



Where is *X* in the data?

Y_1		Y_2	M_1		M_2
int_I		int_G	comm_I		comm_G
1.50		4.00	1.00		6.80
2.75		3.25	2.00		5.40
5.75		2.50	3.20		3.60
3.50		5.75	1.60		5.20
2.25		2.00	4.40		4.60
1.50		1.75	3.00		5.00
2.50		4.25	4.20		4.40
6.00		1.75	4.80		2.40
3.00		2.00	2.60		5.80
4.00		5.25	1.60		5.00
5.00		5.00	4.60		6.20
2.00		1.75	3.80		4.20
1.00		1.75	2.60		3.20
1.25		4.50	1.00		6.00
5.75		4.50	2.60		6.00
3.25		4.75	3.00		6.20
2.75		2.25	4.80		4.60
5.50		2.00	4.00		7.00
1.75		5.25	1.60		5.60
4.00		5.50	1.80		5.40
2.25		4.00	2.20		4.80
4.00		6.50	2.00		6.80
5.00		4.50	3.20		6.00
5.00		3.75	4.00		4.80
4.75		5.25	1.20		6.60

Advantages of a path analytic approach

Provides an estimate of the indirect, total, and direct effects

- Allows us to conduct inferential tests directly on an estimate of the indirect effect

Connects researchers understanding of between-subjects mediation to within-subjects mediation

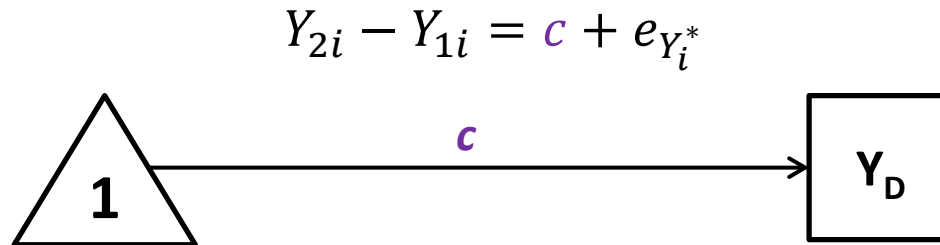
- Reduce misinterpretation of regression coefficients

Using a path analytic framework will help extend the simple mediation model to more complicated questions

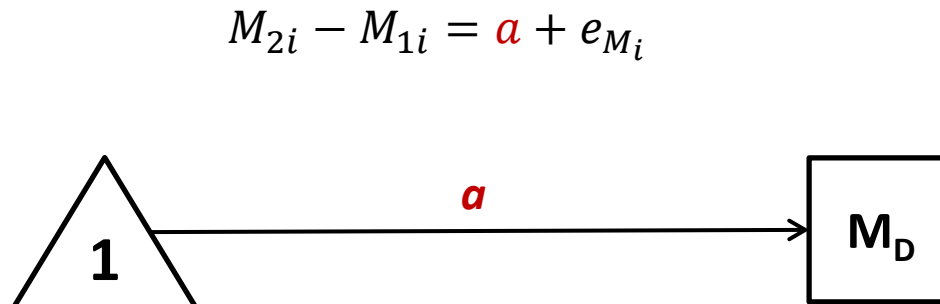
- Multiple mediators
- Moderated mediation
- Integration of between and within-subjects designs

Path-Analytic Approach

Total Effect (c): The effect of our presumed cause (X) on our outcome (Y), without controlling for any other variables. (i.e. mean difference in outcome between the two conditions).



a -path: The effect of our presumed cause (X) on our mediator (M). (i.e. mean difference in mediator between the two conditions).



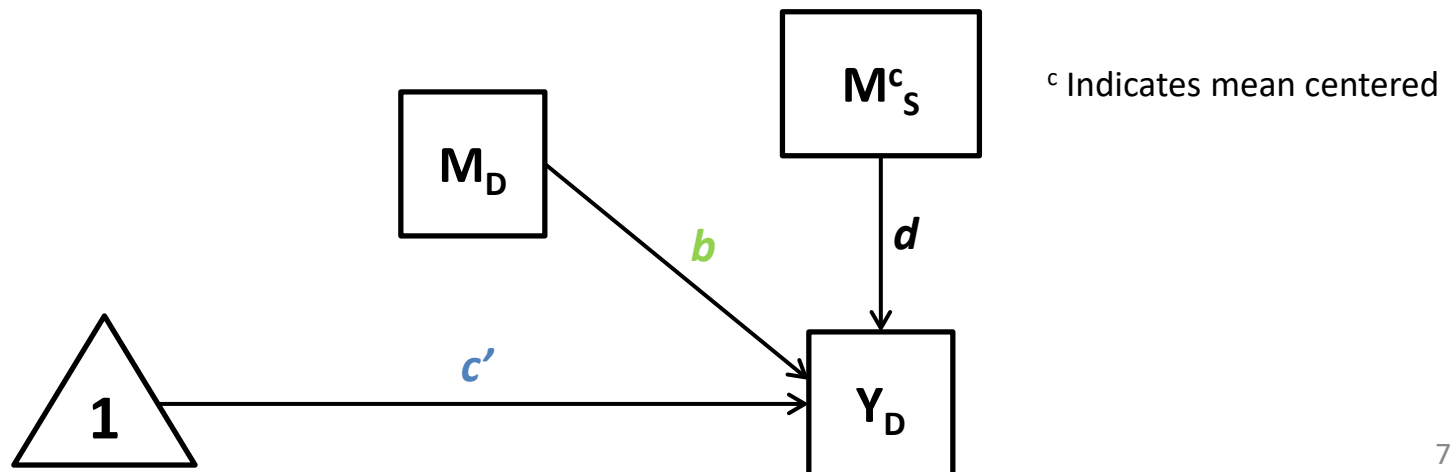
Path-Analytic Approach

b-path: The effect of our mediator (M) on the outcome (Y) while controlling for X . (i.e. predicted difference in Y for two people with the same score on X but who differ on M by one unit).

d-path: The difference in the effect of our mediator (M) on the outcome (Y) between the two levels of the presumed cause. (i.e. interaction between X and M in predicting Y).

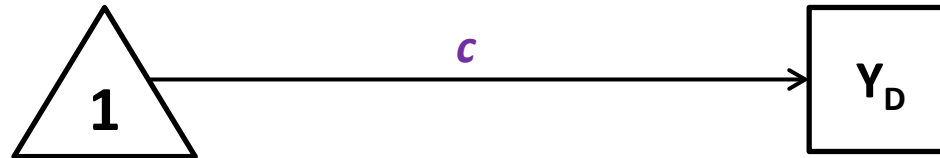
Direct effect (c'): The effect of our presumed cause (X) on Y while controlling for M . (i.e. predicted difference in Y for two people who differ by one unit on X but with the same score on M)

$$Y_{2i} - Y_{1i} = c' + b(M_{2i} - M_{1i}) + d(M_{2i} + M_{1i} - (\overline{M_2} + \overline{M_1})) + e_{Y_i}$$

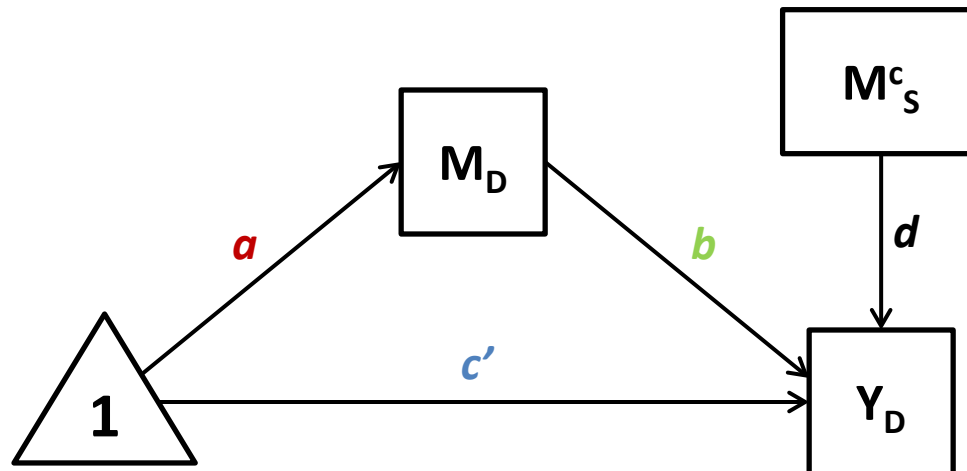


Path-Analytic Approach

Indirect Effect (ab): Product of effect of X on M , and effect of M on Y controlling for X . The effect of X on Y through M .



^c Indicates mean centered

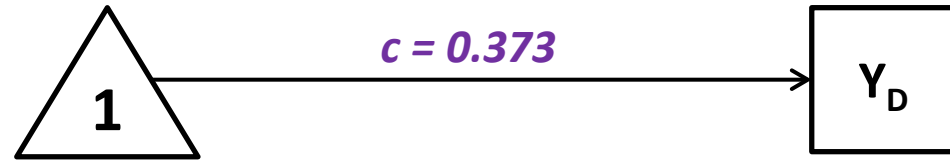


Within Subjects: Path Estimates

Total Effect c : (Regress Y_D on a constant)

$$\widehat{Y}_D = c$$

$$\widehat{Y}_D = .373$$



a path: (Regress M_D on a constant)

$$\widehat{M}_D = a$$

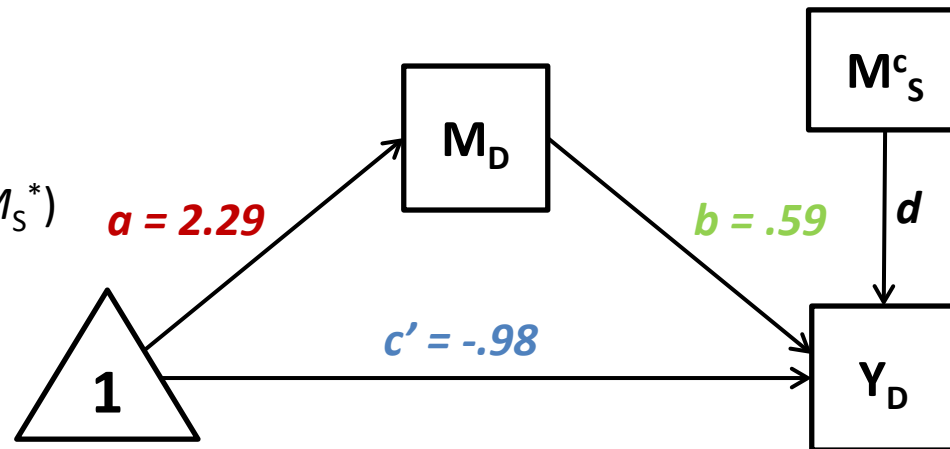
$$\widehat{M}_D = 2.29$$

^c Indicates mean centered

b path and c' path: (Regress Y_D on M_D and M_S^*)

$$\widehat{Y}_D = c' + b_1 M_D + d M_S^c$$

$$\widehat{Y}_D = -.98 + .59 M_D - .28 M_S^c$$



A one unit increase in the difference in communal goal affordance is expected to result in a .59 unit increase in the difference in interest.

People with no difference in communal goal affordance perceptions are expected to be .98 units more interested in the individual class than the group work class .

Note: M_S must be mean centered for c' to have intended interpretation

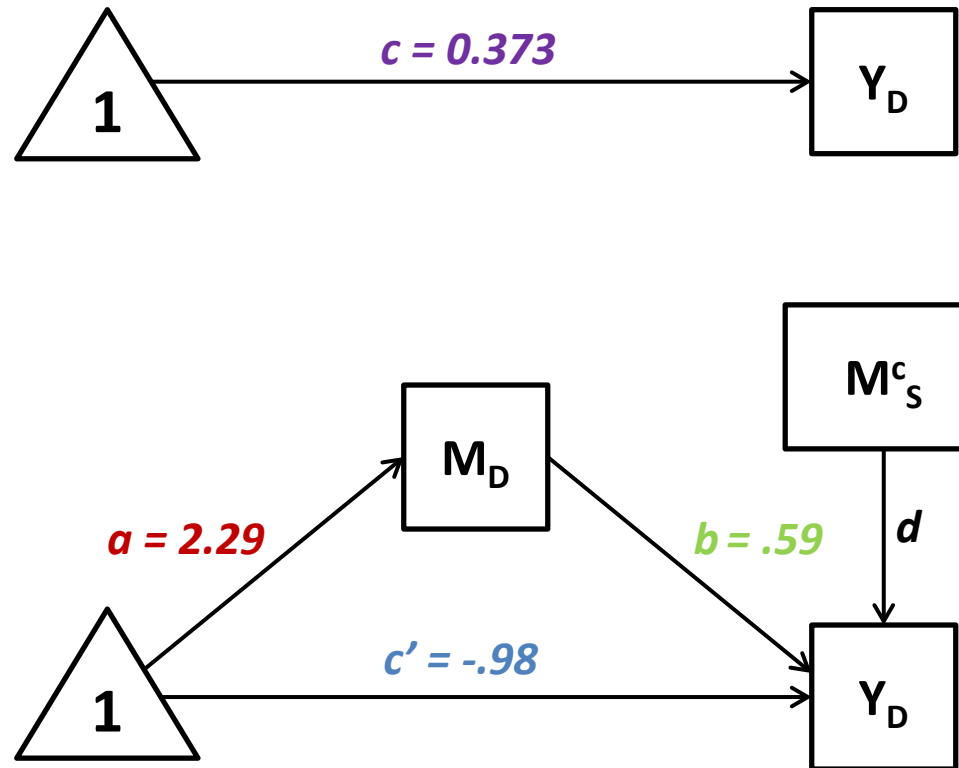
Data Example: Partitioning effect of X on Y

The effect of *X* on *Y* partitions into two components: direct and indirect, in the usual way.

$$c = c' + a \times b$$
$$.373 = -.98 + 2.29 \times .59$$
$$.373 = -.98 + 1.35$$

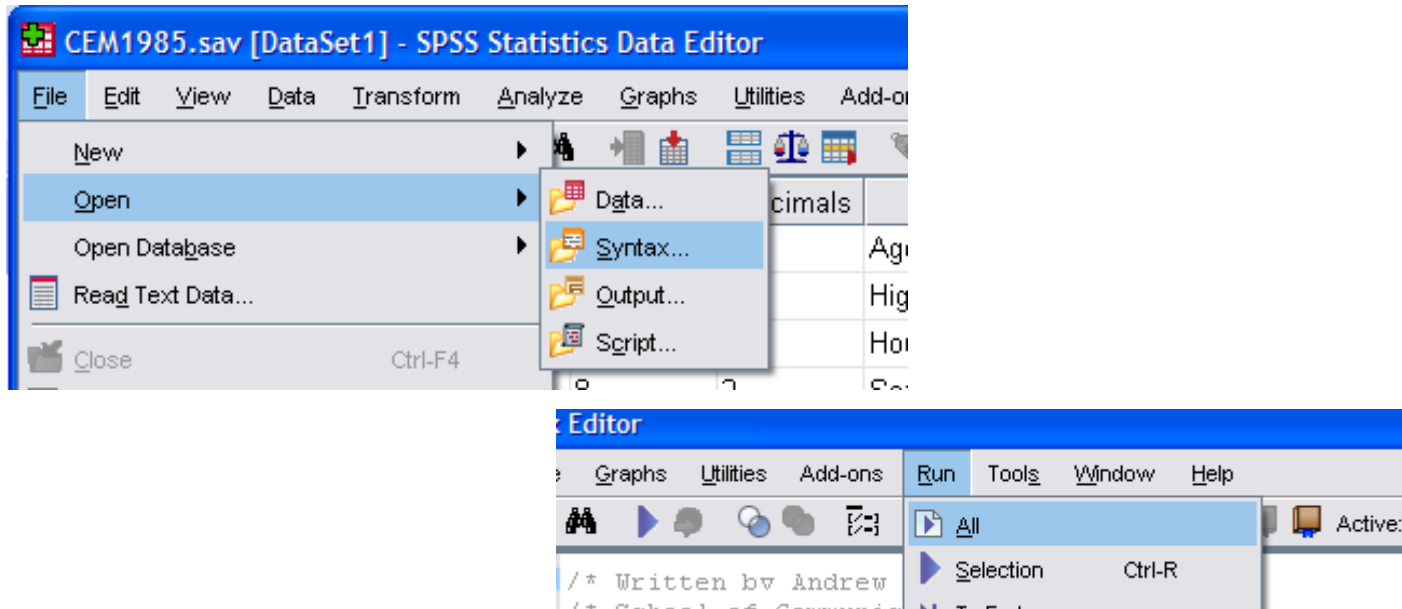
We can conduct inferential tests on the estimate of the indirect effect as in any other mediation analysis.

MEMORE has three methods of inference for the indirect effect available: bootstrapping, Monte Carlo confidence intervals, Sobel Tests



Teaching your package MEMORE (SPSS)

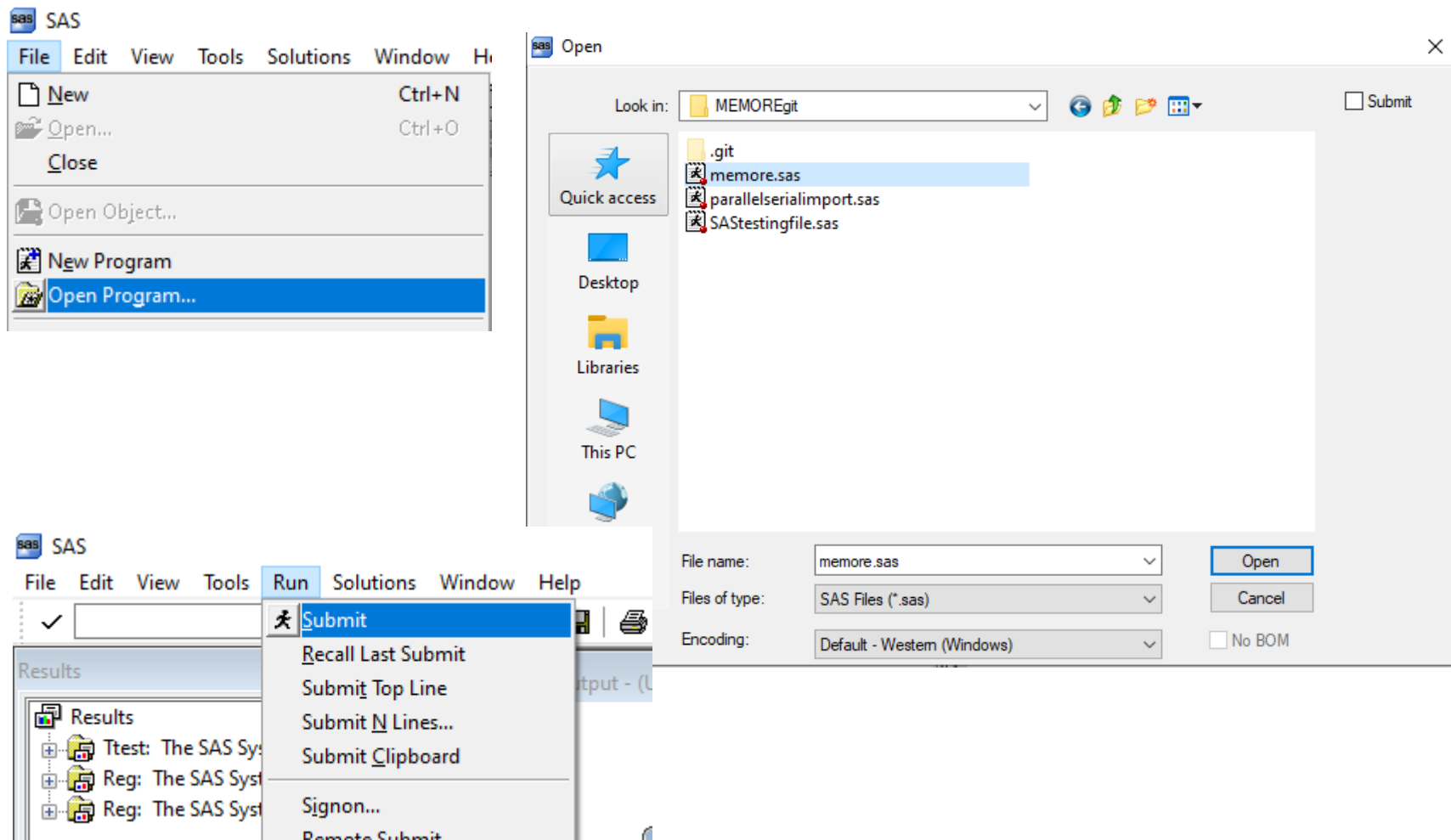
MEMORE is a command which must be taught and re-taught to your statistical package (SPSS) every time you open the package. To teach your program the MEMORE command, open the memore.sps file and run the script exactly as is.



SPSS now knows a new command called
MEMORE

Teaching your package MEMORE (SAS)

MEMORE is a command which must be taught and re-taught to your statistical package (SAS) every time you open the package. To teach your program the MEMORE command, open the memore.sas file and run the script exactly as is.



Writing MEMORE Syntax

MEMORE has 2 required arguments: **Y** and **M**, `data` is required for SAS

```
MEMORE m= comm_G comm_I /y = int_G int_I /normal=1/samples=10000  
/conf = 90 /model = 1.
```

```
%memore(m = comm_G comm_I, y = int_G int_I, normal = 1, samples =  
10000, conf = 90, model = 1, data = CompSci_WS);
```

M is your list of mediators (order matters)

Y is your list of outcomes (order should be matched to the order in the M list)

Arguments:

model specifies the model you are interested. The default is 1, mediation.
Moderation models are 2 and 3, 4+ are moderation.

normal = 1 asks for Sobel test

samples corresponds to the number of bootstrap/MC samples you would like

conf specifies level of confidence you want (default is 95)

mc = 1 asks for Monte Carlo confidence intervals

bc = 1 asks for bias corrected bootstrap confidence intervals

Using MEMORE for CompSci WS data

```
MEMORE m= comm_G comm_I /y = int_G int_I /model = 1.
```

```
%memore(m=comm_G comm_I, y=int_G int_I, model=1, data=CompSci_WS);
```

```
***** MEMORE Procedure for SPSS Version 2.Beta *****
```

```
Written by Amanda Montoya
```

```
Documentation available at akmontoya.com
```

```
*****
```

```
Model:
```

```
1
```

```
Variables:
```

```
Y = int_G int_I
```

```
M = comm_G comm_I
```

```
Computed Variables:
```

```
Ydiff = int_G - int_I  
Mdiff = comm_G - comm_I  
Mavg = ( comm_G + comm_I ) /2 Centered
```

```
Sample Size:
```

```
51
```

```
*****
```

First part of output repeats what you told MEMORE to do. Always double check that this is correct!

Using MEMORE for CompSci WS data

```
MEMORE m= comm_G comm_I /y = int_G int_I /model = 1.
```

```
%memore(m=comm_G comm_I, y=int_G int_I, model=1, data=CompSci_WS);
```

```
*****
Outcome: Ydiff = int_G - int_I Outcome variable
```

Model	Effect	SE	t	p	LLCI	ULCI
'X'	.3725	.2795	1.3330	.1886	-.1888	.9339

```
Degrees of freedom for all regression coefficient estimates: c = .37
50
```

```
*****
Outcome: Mdiff = comm_G - comm_I
```

Model	Effect	SE	t	p	LLCI	ULCI
'X'	2.2941	.2491	9.2108	.0000	1.7938	2.7944

a = 2.29

```
Degrees of freedom for all regression coefficient estimates:
50
```

```
*****
```

First few sections are regression models involved in the mediation analysis. This is the model of Y from X, therefore this is the model which produces the estimate of c

Using MEMORE for CompSci WS data

```
MEMORE m= comm_G comm_I /y = int_G int_I /model = 1.
```

```
%memore(m=comm_G comm_I, y=int_G int_I, model=1, data=CompSci_WS);
```

```
*****
```

```
Outcome: Ydiff = int_G - int_I
```

Model Summary

R	R-sq	MSE	F	df1	df2	p
.5639	.3180	2.8299	11.1909	2.0000	48.0000	.0001

Model

	coeff	SE	t	p	LLCI	ULCI
'x'	-.9814	.3884	-2.5269	.0149	-1.7623	-.2005
Mdiff	.5902	.1346	4.3845	.0001	.3195	.8608
Mavg	-.5505	.4328	-1.2718	.2096	-1.4208	.3198

This is the model predicting Y_D from a constant, M_D , and M_{avg}^c therefore this model gives us an estimate of b and c'

Degrees of freedom for all regression coefficient estimates:

48

$c' = -.98$

$b = .590$

Using MEMORE for CompSci WS data

```
MEMORE m= comm_G comm_I /y = int_G int_I /model = 1.
```

```
%memore(m=comm_G comm_I, y=int_G int_I, model=1, data=CompSci_WS);
```

***** TOTAL, DIRECT, AND INDIRECT EFFECTS *****

Total effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
.3725	.2795	1.3330	50.0000	.1886	-.1888	.9339

Direct effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
-.9814	.3884	-2.5269	48.0000	.0149	-1.7623	-.2005

Indirect Effect of X on Y through M

	Effect	BootSE	BootLLCI	BootULCI
Ind1	1.3540	.3260	.6827	1.9653

Indirect Key

Ind1 X -> Mldiff -> Ydiff

Important effects
for mediation and
inference about
these effects

Based on a 95% bootstrap
confidence interval we
have evidence of
mediation!

Turning off the XM interaction

$$Y_{2i} - Y_{1i} = c' + b(M_{2i} - M_{1i}) + d(M_{2i} + M_{1i} - (\overline{M_2} + \overline{M_1})) + \epsilon_{Y_i}$$

When we estimate this regression model, we allow the relationship between M and Y to differ by instance (X). This is like allowing for an interaction between X and M when estimating Y .

We do this by including the sum/average term in the regression model.

d estimates the difference in the relationship between $M_1 \rightarrow Y_1$ and $M_2 \rightarrow Y_2$. If we fix this coefficient to zero (do not include the sum term in the model) we fix the interaction to zero.

```
MEMORE m= comm_G comm_I /y = int_G int_I /xmint = 0/model = 1.
```

```
%memore(m=comm_G comm_I, y=int_G int_I, model=1, xmint = 0,  
data=CompSci_WS) ;
```

Turning off the XM interaction

```
MEMORE m= comm_G comm_I /y = int_G int_I /xmint = 0/model = 1.
```

```
%memore(m=comm_G comm_I, y=int_G int_I, model=1, xmint = 0,  
data=CompSci_WS) ;
```

No interaction

Interaction

```
Outcome: Ydiff = int_G - int_I
```

$c = .3725 (.2795)$

Model

	Effect	SE	t	p	LLCI	ULCI
'X'	.3725	.2795	1.3330	.1886	-.1888	.9339

```
Degrees of freedom for all regression coefficient estimates:  
50
```

```
*****
```

```
Outcome: Mdiff = comm_G - comm_I
```

$a = 2.2941 (.2491)$

Model

	Effect	SE	t	p	LLCI	ULCI
'X'	2.2941	.2491	9.2108	.0000	1.7938	2.7944

```
Degrees of freedom for all regression coefficient estimates:  
50
```

```
*****
```

Turning off the XM interaction

```
MEMORE m= comm_G comm_I /y = int_G int_I /xmint = 0/model = 1.
```

```
%memore(m=comm_G comm_I, y=int_G int_I, model=1, xmint = 0,  
data=CompSci_WS);
```

No interaction

Interaction

Outcome: Ydiff = int_G - int_I

Model Summary

R	R-sq	MSE	F	df1	df2	p
.5432	.2950	2.8655	20.5060	1.0000	49.0000	.0000

Model

	coeff	SE	t	p	LLCI	ULCI
'X'	-1.0257	.3893	-2.6349	.0112	-1.8079	-.2434
Mdiff	.6095	.1346	4.5284	.0000	.3390	.8799

$c = -.9814(.2795)$

$b = .5902 (.1346)$

$d = -.5505 (.4328)$

Indirect Effect of X on Y through M

	Effect	BootSE	BootLLCI	BootULCI
Ind1	1.3982	.3082	.8034	2.0156

$ab = 1.3540 [.6827,1.9653]$

Ultimately results are mostly unchanged, but that is not always the case.

Testing the *XM* interaction

There are three potential approaches to testing the *XM* interaction that you might take:

1. Include the *XM* interaction by default

- This is what is recommended in causal mediation analysis (Vo et al., 2020)
- Use the `xmint=1` option in MEMORE always (this is the default)

2. Test/Evaluation *XM* interaction as a pre-step

- Estimate coefficient and use a hypothesis test or effect size measure to evaluate whether it should be included
- Include your threshold in your preregistration
- Final model will depend on whether *XM* is included or not

3. Robustness Check

- Fit the model as hypothesized (*XM* in or out)
- Afterwards fit the other model and evaluate whether the results are sensitive to the choice (change in effect size or significance level)

Writing up a Repeated Measures Mediation Analysis

Tips:

- Walk the reader through the steps of the mediation in a way that is intuitive.
 - Include interpretations of the results: b.e.g. “The total effect was significant, $p < .05$ ”
- Use equations and numbers *where helpful*.
- Avoid using computational variable names (e.g. RESPAPPR)
- Avoid causal language if it is not supported by your research design.
- Pick one inferential method and report it
- Read the write ups of other’s mediation analyses

Is the effect of group work on class interest mediated by communal goal affordance of the class?

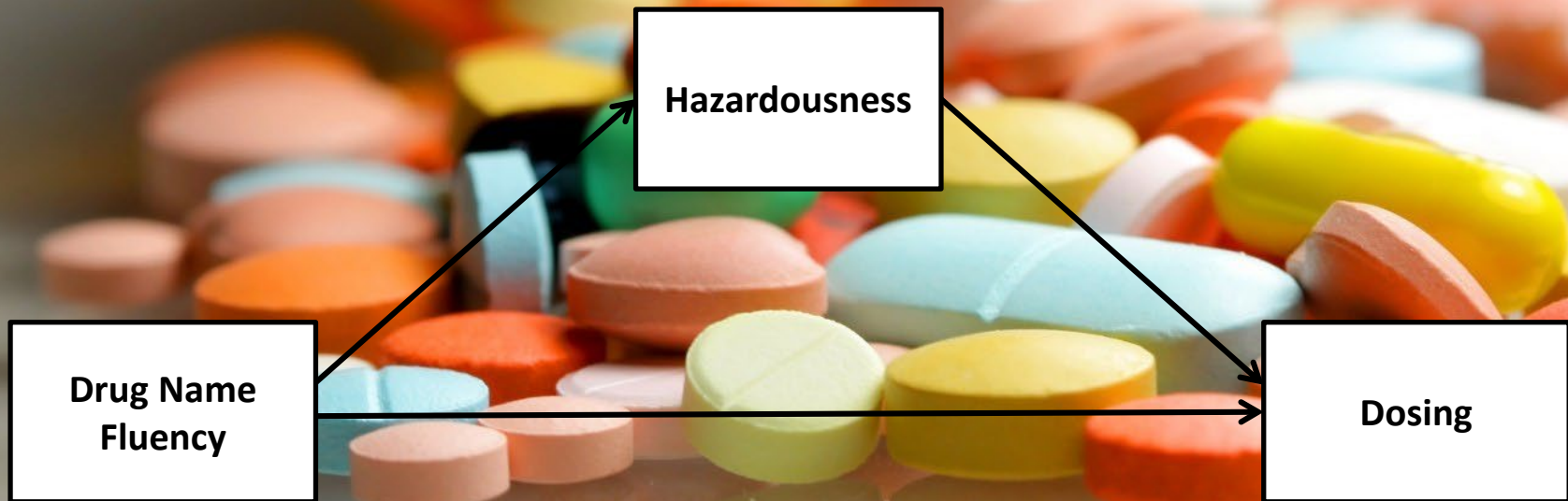
Overall there was no evidence of a total effect of group work on interest in computer science classes, we estimate that individuals were .37 units higher on interest in group work than individual work classes ($p = .19$). The class with group work was rated 2.29 units higher on communal goal affordance than the class with individual work ($p < .001$). A one unit increase in perception of communal goal affordance increased interest in the class by .59 units ($p = .0001$), and the relationship between communal goal affordance and interest in a class did not depend on condition ($p = .21$). The effect of group work on interest through communal goal fulfillment was different from zero ($ab = 1.35$, 95% Bootstrap CI [.68, 1.96]). This means that we expect women to be 1.35 units more interested in a computer science class with group work compared to one without group work, through the effect of group work on communal goal affordance, and the subsequent effect of communal goal affordance on interest. There was a significant direct effect between group work and interest ($c' = -.98$, $p = .01$). This indicates that there may be some other process, separate from communal goal affordance, which is actually deterring women from computer science classes with group work.

Example: Drug Name Fluency

Dohle, S., & Montoya, A. K. (2017). The dark side of fluency: Fluent names increase drug dosing. *Journal of Experimental Psychology: Applied*, 23(3), 231 – 239.

Research Question: Can the name of drugs impact how hazardous they seem and how much people are willing to dose the drugs?

Imagine you have a cold, and there are a variety of medications available including (a) Fastinorbine and (b) Cytrigmcmium. Which seems more hazardous? Which are you willing to dose more of?



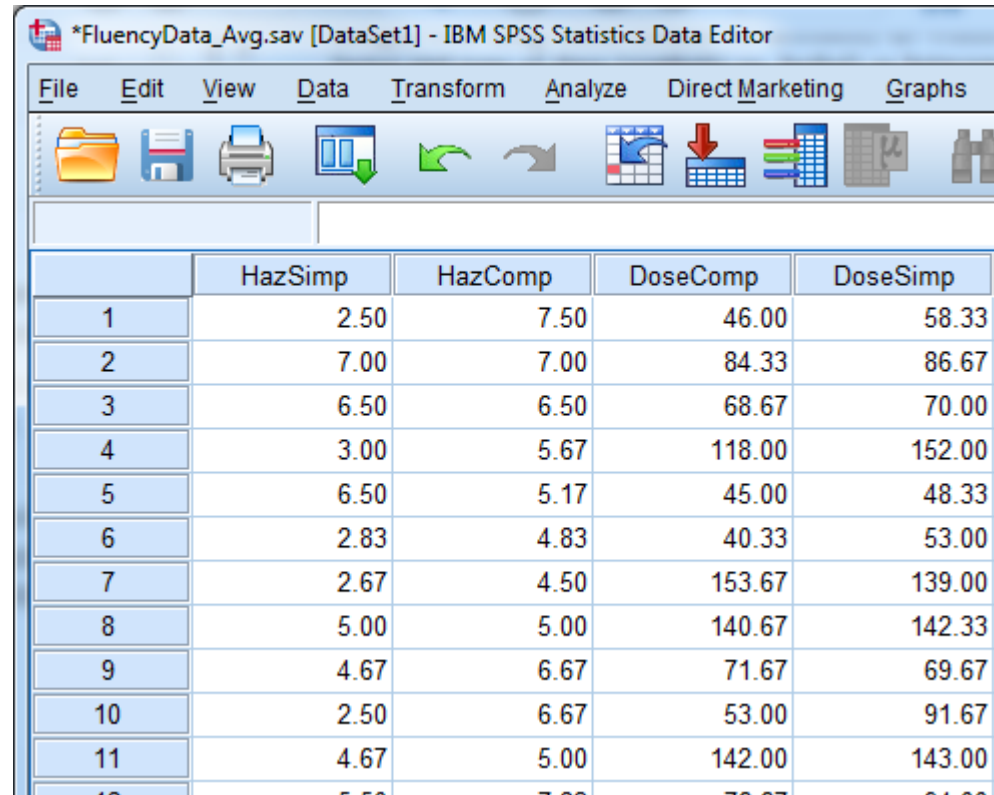
Example: Drug Name Fluency

Dohle, S., & Montoya, A. K. (2017). The dark side of fluency: Fluent names increase drug dosing. *Journal of Experimental Psychology: Applied*, 23(3), 231 – 239.

Participants (N = 70) were asked to imagine they had the flu, and 6 different drugs were provided to treat the drug. Participants poured the dose they would feel comfortable taking at maximum into a plastic cup. Each person judged drugs with simple or complex names (3 of each). Responses on the measured variables were averaged across the 3 drugs (but later we'll look at what happens when we treat these separately).

Measured Variables:

- Dosage in mL
 - Variable name: Dose
 - 0 mL – 200mL
- Hazardousness of drug
 - Variable name: Haz
 - Average of two questions:
 - Hazardousness (1-7)
 - Dangerousness (1-7)



	HazSimp	HazComp	DoseComp	DoseSimp
1	2.50	7.50	46.00	58.33
2	7.00	7.00	84.33	86.67
3	6.50	6.50	68.67	70.00
4	3.00	5.67	118.00	152.00
5	6.50	5.17	45.00	48.33
6	2.83	4.83	40.33	53.00
7	2.67	4.50	153.67	139.00
8	5.00	5.00	140.67	142.33
9	4.67	6.67	71.67	69.67
10	2.50	6.67	53.00	91.67
11	4.67	5.00	142.00	143.00

Example: Drug Name Fluency

Dohle, S., & Montoya, A. K. (2017). The dark side of fluency: Fluent names increase drug dosing. *Journal of Experimental Psychology: Applied*, 23(3), 231 – 239.

1. Estimate the proposed model (Fluency → Hazardousness → Dosage) using MEMORE
2. Turn off the XM interaction
3. Find estimates of the following paths: a , b , c , c'
4. Of the following inferential methods, which support the hypothesized mediation model (use $\alpha = 0.05$ or 95% confidence intervals):
Percentile bootstrap CIs, Monte Carlo CIs, Sobel Test / Normal Theory
5. Practice writing up some of the results explored above.

Activity before Tomorrow

Using MEMORE for Drug Fluency data

```
MEMORE m= HazSimp HazComp /y = DoseSimp DoseComp /xmint = 0 /model = 1.
```

```
%memore(m=HazSimp HazComp,y=DoseSimp DoseComp,xmint=0,  
        model=1,data=Fluency);
```

```
*****  
Outcome: Ydiff =   DoseSimp   -      DoseComp
```

```
Model
```

	Effect	SE	t	p	LLCI	ULCI
'X'	11.0476	1.5770	7.0055	.0000	7.9016	14.1937

c = 11.05

```
Degrees of freedom for all regression coefficient estimates:  
69
```

```
*****  
Outcome: Mdiff =   HazSimp   -      HazComp
```

```
Model
```

	Effect	SE	t	p	LLCI	ULCI
'X'	-2.1048	.1848	-11.3893	.0000	-2.4734	-1.7361

a = -2.10

```
Degrees of freedom for all regression coefficient estimates:  
69
```

On average, participants dosed 11.05 mL more of the simply named drug than the complex named drug ($t(69) = 7.06$, $p < .001$).

Participants thought the complex drug was 2.10 points more hazardous than the simply named drug, on average ($t(69) = 11.39$, $p < .001$).

Using MEMORE for Drug Fluency data

```
MEMORE m= HazSimp HazComp /y = DoseSimp DoseComp /xmint = 0 /model = 1.
```

```
%memore(m=HazSimp HazComp,y=DoseSimp DoseComp,xmint=0,  
        model=1,data=Fluency);
```

```
*****
```

```
Outcome: Ydiff = DoseSimp - DoseComp
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.4020	.1616	148.1029	13.1047	1.0000	68.0000	.0006

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
'X'	3.8280	2.4684	1.5508	.1256	-1.0978	8.7537
Mdiff	-3.4302	.9475	-3.6200	.0006	-5.3210	-1.5393

$c' = 3.83$

$b = -3.43$

```
Degrees of freedom for all regression coefficient estimates:
```

```
68
```

After controlling for hazardousness, participants were expected to dose 3.8 mL more of the simple drug. This effect was not significantly different than zero ($t(68) = 1.55$, $p = .13$).

A one unit increase in the difference in perceived hazardousness between conditions results in a 3.43 unit decrease in the difference in dosage ($t(68) = 3.62$, $p < .001$).

A one unit increase in perceived hazardousness results in a 3.43 unit decrease in dosage, averaged across fluency conditions ($t(68) = 3.62$, $p < .001$).

Using MEMORE for Drug Fluency data

```
MEMORE m= HazSimp HazComp /y = DoseSimp DoseComp /xmint = 0 /model = 1.
```

```
%memore(m=HazSimp HazComp,y=DoseSimp DoseComp,xmint=0,  
        model=1,data=Fluency);
```

```
***** TOTAL, DIRECT, AND INDIRECT EFFECTS *****
```

Total effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
11.0476	1.5770	7.0055	69.0000	.0000	7.9016	14.1937

Direct effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
3.8280	2.4684	1.5508	68.0000	.1256	-1.0978	8.7537

Indirect Effect of X on Y through M

	Effect	BootSE	BootLLCI	BootULCI
Ind1	7.2197	1.8940	3.8590	11.1609

ab = 7.22 Interpretation?

Indirect Key

Ind1	'X'	->	Mldiff	->	Ydiff

Participants were dosed simple drugs 7.22 mL more, through the effect of simple drugs on hazardousness and the subsequent effect of hazardousness on dosage (Percentile CI = [3.86, 11.16]).

Drug name fluency increased dosage indirectly effect through hazardousness by 7.22 mL (Percentile CI = [3.86, 11.16]).

Simple names were perceived as less hazardous, which then increased dosage, resulting in an indirect effect of 7.22 mL on dosage (Percentile CI = [3.86, 11.16]).

Using MEMORE for Drug Fluency data

Percentile Bootstrap CI

```
MEMORE m= HazSimp HazComp /y = DoseSimp DoseComp /xmint = 0.
```

```
%MEMORE(m= HazSimp HazComp,y = DoseSimp DoseComp, xmint = 0);
```

Indirect Effect of X on Y through M

	Effect	BootSE	BootLLCI	BootULCI
Ind1	7.2197	1.8940	3.8590	11.1609

Monte Carlo CI

```
MEMORE m= HazSimp HazComp /y = DoseSimp DoseComp /xmint = 0 /mc = 1.
```

```
%MEMORE(m= HazSimp HazComp,y=DoseSimp DoseComp,xmint = 0,mc = 1);
```

Indirect Effect of X on Y through M

	Effect	MCSE	MCLLCI	MCULCI
Ind1	7.2197	2.0916	3.2369	11.3834

Sobel Test / Normal Theory

```
MEMORE m= HazSimp HazComp /y = DoseSimp DoseComp /xmint = 0 /normal = 1.
```

```
%MEMORE(m= HazSimp HazComp,y=DoseSimp DoseComp,xmint = 0,normal=1);
```

Normal Theory Tests for Indirect Effect

	Effect	SE	Z	p
Ind1	7.2197	2.0927	3.4500	.0006

Causality

Mediation hypotheses and indirect effects are inherently causal.

Even when there is no an experiment or random assignment, if your interest is in mediation/indirect effects you're interest is in causal effects.

There are important assumptions to consider which would allow our estimate of an indirect effect to be an **unbiased estimate of a causal effect** even without random assignment.

Temporal Precedence

The assumption of temporal precedence assumes that the **cause precedes the effect**.

This issue is often of concern with mediators and outcomes measured at the same time points.

Study design elements can be used to improve temporal precedence:

- Measure mediators before outcomes in single sitting surveys (but there are some reasons not to)
- Measure mediators at a time point earlier than outcomes (e.g., Mediator at baseline and 3 months, Outcome at baseline and 6 months)

Omitted Confounder Assumptions

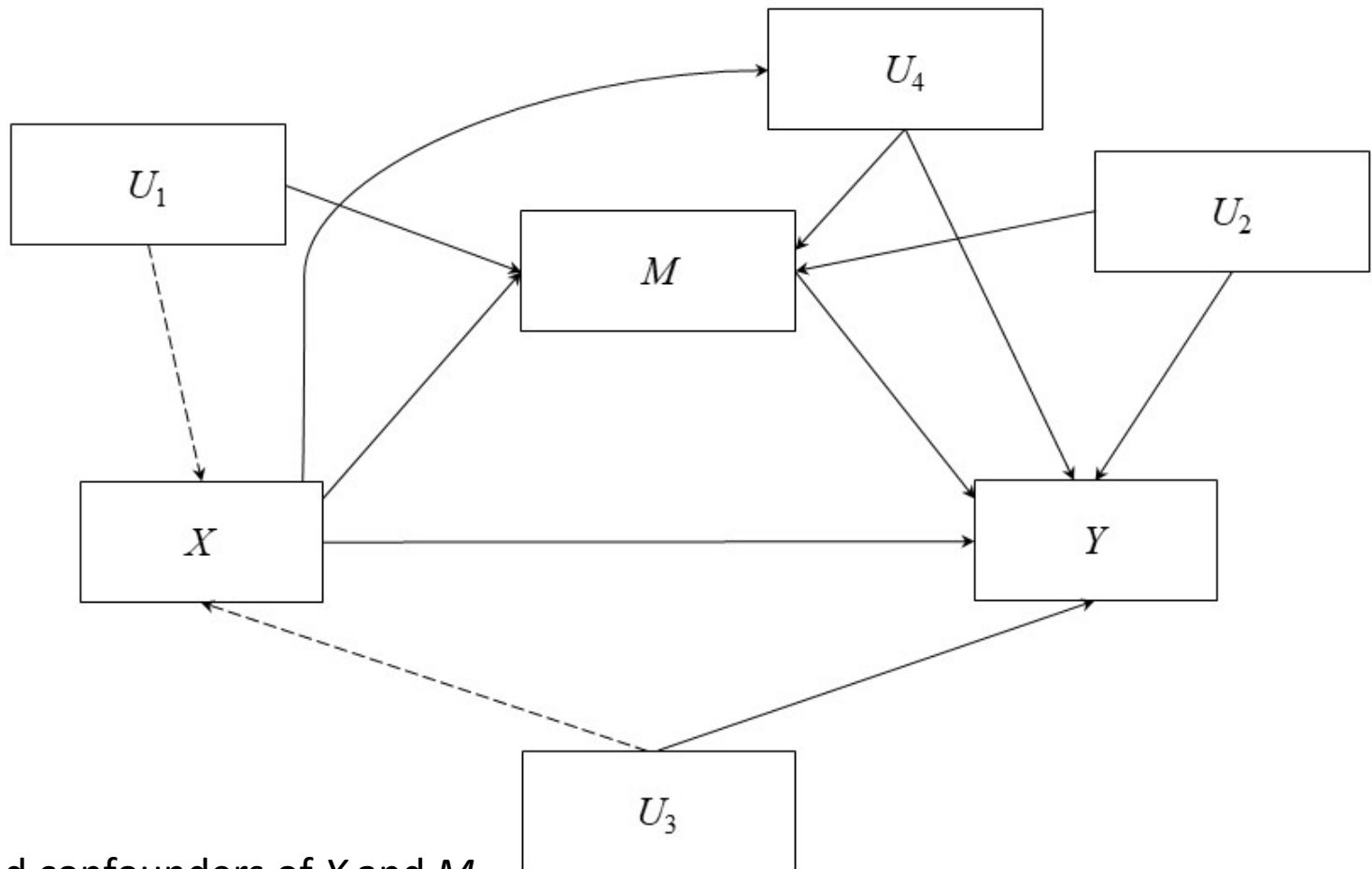
What's an “omitted confounder”?

It is any variable that is not included in the model (omitted) that causally affects two variables in the model

These are the classic assumptions for mediation:

1. No omitted confounders of X and M
2. No omitted confounders of M and Y
3. No omitted confounders of X and Y
4. No omitted or measured confounders of M and Y that are affected by X

Omitted Confounders DAG



1. No omitted confounders of X and M
2. No omitted confounders of M and Y
3. No omitted confounders of X and Y
4. No omitted or measured confounders of M and Y that are affected by X

Confounder Levels

In within-subject designs confounders can exist at two different levels:

1. Person level: Constant across instances, but varies across subjects

Academic year, Gender, Personality

2. Instance level: Varies across instances within subjects

Expected teacher gender, perceived difficulty of the course

Good news: If upper-level confounders of M - Y relationship are additive (no interaction) then they do not affect causal identification (Joseph et al., 2015)

Carry-Over Effects

Carry-over effects are when measures from one instance affect measures from another instance.

Assumed models:

$$M_{1i} = a_1 + \epsilon_{M_{1i}} \quad M_2 \text{ or } Y_2 \text{ are **not** in this model}$$

$$M_{2i} = a_2 + \epsilon_{M_{2i}} \quad M_1 \text{ or } Y_1 \text{ are **not** in this model}$$

$$Y_{1i} = g_{10} + g_{11}M_{1i} + \epsilon_{Y_{1i}} \quad M_2 \text{ or } Y_2 \text{ are **not** in this model}$$

$$Y_{2i} = g_{20} + g_{21}M_{2i} + \epsilon_{Y_{2i}} \quad M_1 \text{ or } Y_1 \text{ are **not** in this model}$$

If the TRUE data generating model includes these predictors, then our estimates of the indirect effect will be biased.

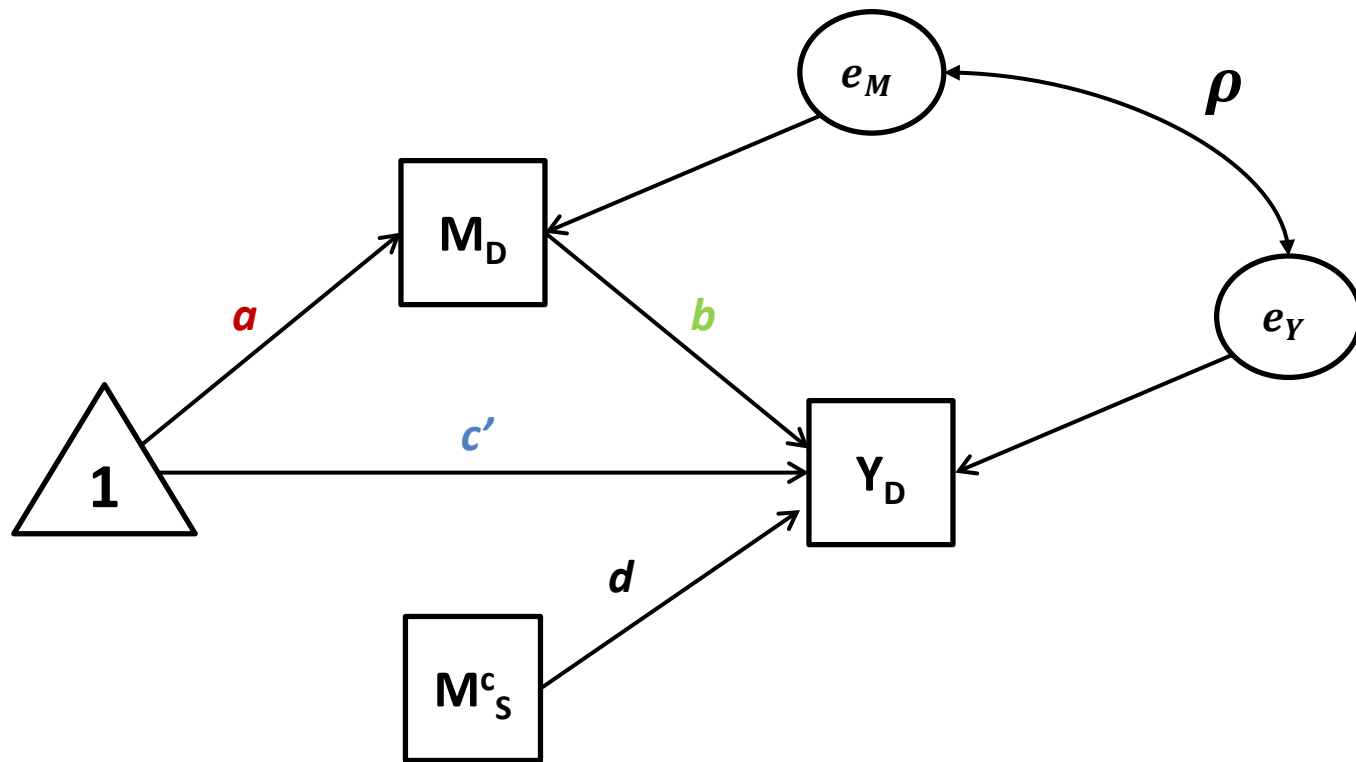
Sensitivity Analysis

An analytical approach for evaluating *how much confounding* would need to be present to reduce indirect effect.

Focus is typically on M - Y confounders since these can be so problematic for mediation analysis.

When M - Y is confounded, they have correlated residuals, focus is on the size of that correlation.

Sensitivity Analysis

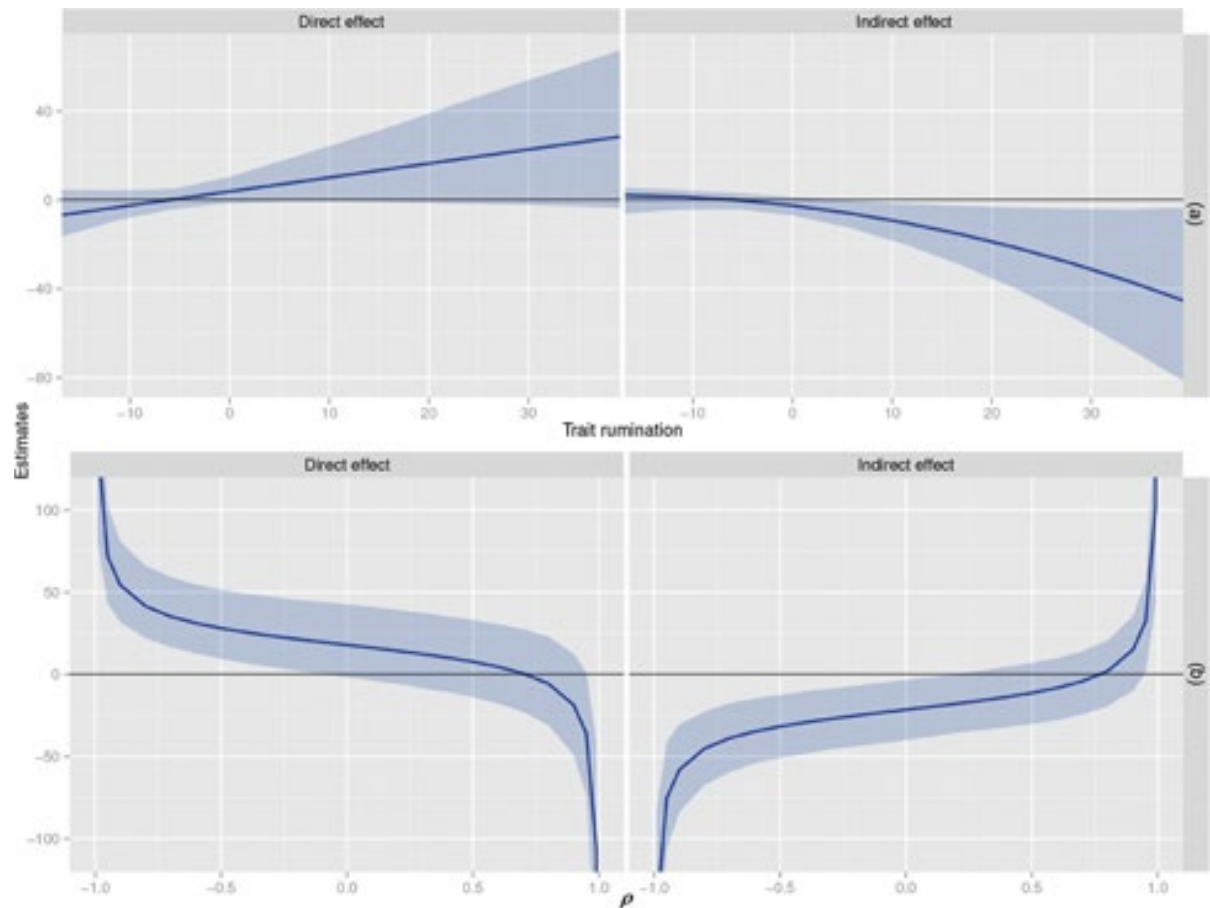


Estimate the size of ρ that would...

- Make the indirect effect non-significant
- Make the indirect effect zero

Sensitivity Analysis

No existing tools for conducting sensitivity analyses in these designs (but can do with *mediation* package in R for multilevel mediation).



STUDY PLANNING

Power Analysis & Preregistration

Power Analysis

R script is available for power analysis for within-subject mediation at this [GitHub Page](#).

Run script in R, then revise the following code for analysis:

```
WSmedpower(nsims = 5000, boots = 0, MCsamples = 0,  
alpha = 0.05, N = 100, corm = .22, cory = .32, aest  
= -.44, best = -.34, cest = .39, dest = 0)
```

Can set boots and MCsamples to zero to get close with Joint Significance method, run time is long when non-zero

JointSig	Bootstrap	MonteCarlo
0.895	NA	NA

Power Analysis

Arguments

- nsims: number of simulations to run (default is 1000)
- boots: number of bootstrap samples (default is 0, which doesn't run bootstrap analysis)
- MCsamples: number of monte carlo samples (default is 0, which does run Monte Carlo analysis)
- alpha: level of test, acceptable type I error rate (default is 0.05)
- N: sample size to test
- corm: correlation among repeated measurements of mediator
- cory: correlation among repeated measurements of outcome
- aest: a-path estimates standardized by $SD(M)$
- best: b-path estimate standardized by $SD(M)$ and $SD(Y)$
- cest: c-path estimate standardized by $SD(Y)$
- dest: d-path estimate standardized by $SD(M)$ and $SD(Y)$

Preregistration

Preregistration: A process where you create a time-stamped, publicly accessible record of your plan for a specific study.

Current preregistrations of mediation analysis are light on the details:
"We plan to conduct mediation tests (including multiple mediation) using the MEMORE macro (Montoya & Hayes, 2017) and the PROCESS macro (Hayes, 2017)."

- ❑ Planned sample size
- ❑ α -level/CI-level for each test
- ❑ Role of different variables in the analysis (e.g., independent variable, mediator, outcome, covariate), and how they are computed
- ❑ Estimation Method
- ❑ Inferential method and any important specifications for that method (e.g., how many bootstraps, seed number)
- ❑ Plans for sensitivity analysis: Benchmarks for acceptable correlations for evaluating confounding
- ❑ *Plans for XM interaction*
- ❑ **Tools and specifications**

Example Preregistration

We will estimate a simple mediation analysis where protest condition (X) affects liking (Y) through response appropriateness (M) using a sample of 500 participants. Protest condition is assigned in a random order where 50% will be assigned to the protest condition first and 50% to the no protest condition first. Liking is measured using 4 Likert type items evaluating the lawyers positive characteristics. We will check the reliability of our scale prior to analysis and if the reliability (alpha) is greater than 0.7, we will proceed. If it is not we may consider dropping items. Similarly, response appropriateness is measured using 5 Likert type items. We will use the same procedure for measurement evaluation as with liking. In our analysis we will not control for any covariates.

We will use the **MEMORE for SPSS (version 3.0)** to estimate our model, which uses ordinary least squares regression and generates percentile bootstrap confidence intervals for the indirect effect. We will use 5,000 bootstraps. *We plan to test whether the XM interaction is statistically significant at $\alpha = 0.10$, if it is, we will use the $xmint=1$ option in MEMORE to estimate the indirect effect (using 232714). If the XM interaction is not statistically significant, we will use $xmint = 0$, to estimate the indirect effect. We will test hypothesis 1 by examining the 95% bootstrap confidence interval for the indirect effect includes zero.*

After our analysis we will use the mediation package in R to conduct sensitivity analysis, specifically for the relationship between response appropriateness and liking. Correlations greater than .5 would suggest robustness to reasonable levels of confounding.

Finding Examples

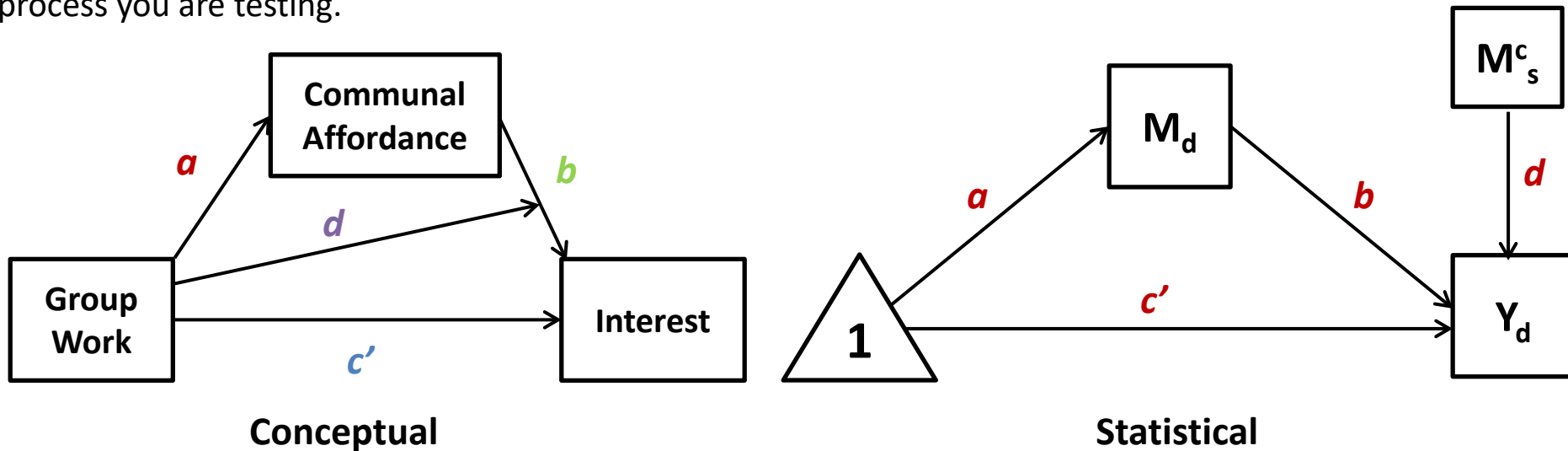
DataStudio for finding examples

<https://lookerstudio.google.com/s/gFgefAkOjKA>

Publication Year Research Areas ModelNumb... (1) Xtype levelsX numYvars											
Journal Title		Covariates		Sample Size		Article Count 50					
Article Title (link)	Authors	Journal Title	Publication...	Research Areas	Study Num...	SampleSize	ModelNumb...	numXvars	Xtype	levelsX	numYvars
When should retail...	Jeong, H; Ye,...	JOURNAL O...	2021	Business & Economics	1	271	1 (Mediation)	1	Pre-post	2	1
What hinders resi...	Dong, XJ	TOURISM A...	2022	Social Sciences - Other...	2	130	1 (Mediation)	1	Experimental	2	1
The role of metad...	Hartley, S; P...	GROUP PRO...	2022	Psychology	3	239	1 (Mediation)	1	Experimental	2	2
The connotative ...	Motoki, K; P...	JOURNAL O...	2022	Business & Economics	1	154	1 (Mediation)	1	Experimental	2	3
The Self-Other Div...	Ring, C; Kav...	JOURNAL O...	2020	Social Sciences - Other...	1	100	1 (Mediation)	1	Experimental	2	1
The Impact of Mix...	Huang, XZ; Z...	INTERNATIO...	2022	Environmental Science...	2	434	1 (Mediation)	1	Experimental	2	1
The Hypoalgesic E...	Song, JS; Ka...	RESEARCH ...	2022	Social Sciences - Other...	1	40	1 (Mediation)	1	Experimental	2	1
Testing the effecti...	Brochu, PM	JOURNAL O...	2023	Psychology	1	45	1 (Mediation)	1	Pre-post	2	2
Suicidality and so...	Breitborde, ...	EARLY INTE...	2021	Psychiatry	1	38	1 (Mediation)	1	Pre-post	2	1
Putting the Me in ...	Hamilton, K...	NEW MEDIA...	2021	Communication	1	119	1 (Mediation)	1	Experimental	2	1
Positive reputatio...	Inoue, Y; Mif...	FRONTIERS ...	2023	Psychology	2	293	1 (Mediation)	1	Experimental	2	1
Perceptions of a P...	Pals, AM; Go...	JOURNAL O...	2022	Psychology; Family Stu...	1	52	1 (Mediation)	3	Experimental	2	1
Pandemic Pedago...	Armstrong, ...	SOUTHERN ...	2022	Communication	1	163	1 (Mediation)	1	Pre-post	2	4
Mind the ad: How ...	Kocak, A; Ro...	JOURNAL O...	2022	Psychology; Business ...	1	123	1 (Mediation)	1	Experimental	2	1
Mind the ad: How ...	Kocak, A; Ro...	JOURNAL O...	2022	Psychology; Business ...	2	151	1 (Mediation)	1	Experimental	2	1
Mediation and Mo...	Wong, CL; C...	JOURNAL O...	2020	Public, Environmental ...	1	1001	1 (Mediation)	1	Pre-post	2	1

Visualizations

I suggest using both a conceptual and statistical visualization in order to help the reader understand the process you are testing.



Tips:

- Providing a conceptual diagram helps the readers understand the process you are interested in.
- Providing a statistical diagram helps readers understand how you estimated the model, and that you did it correctly.
- Provide path estimates on statistical diagram or in a table.
- Don't forget to report the path estimates and statistics for the d path. It's important!

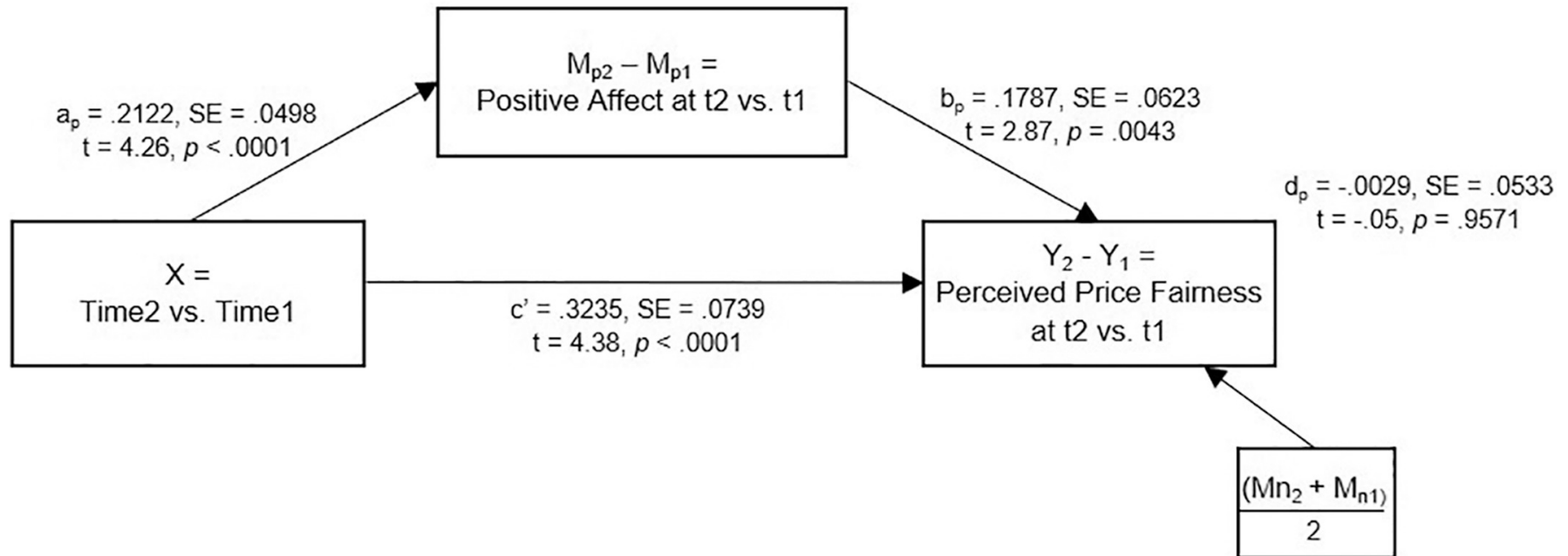
Visualization Examples



Figure 1. Self-brand association as a mediator of the relationship between condition (self-endorsed vs other-endorsed) and brand attitude ($n = 119$). *Indirect Effect* = 0.35 CI = [0.23, 0.51]. Used 5000 sample bootstraps to calculate indirect effect. *** $p < .001$.

Visualization Examples

Indirect effect of X (t1 vs. t2) on Y through Positive Affect = .0379, boot SE = .0174, 95% CI: [.0086, .0767])



Indirect effect of X (t1 vs. t2) on Y through Negative Affect = -.0284, boot SE = .0167, 95% CI: [-.0629, .0026])

Visualization Examples

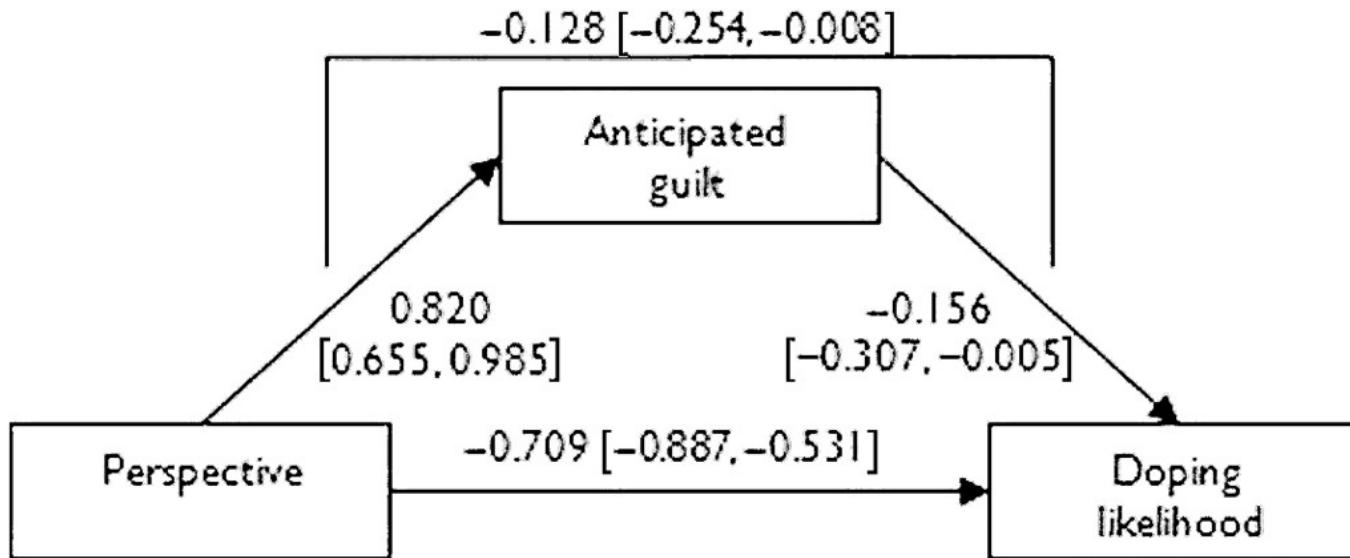
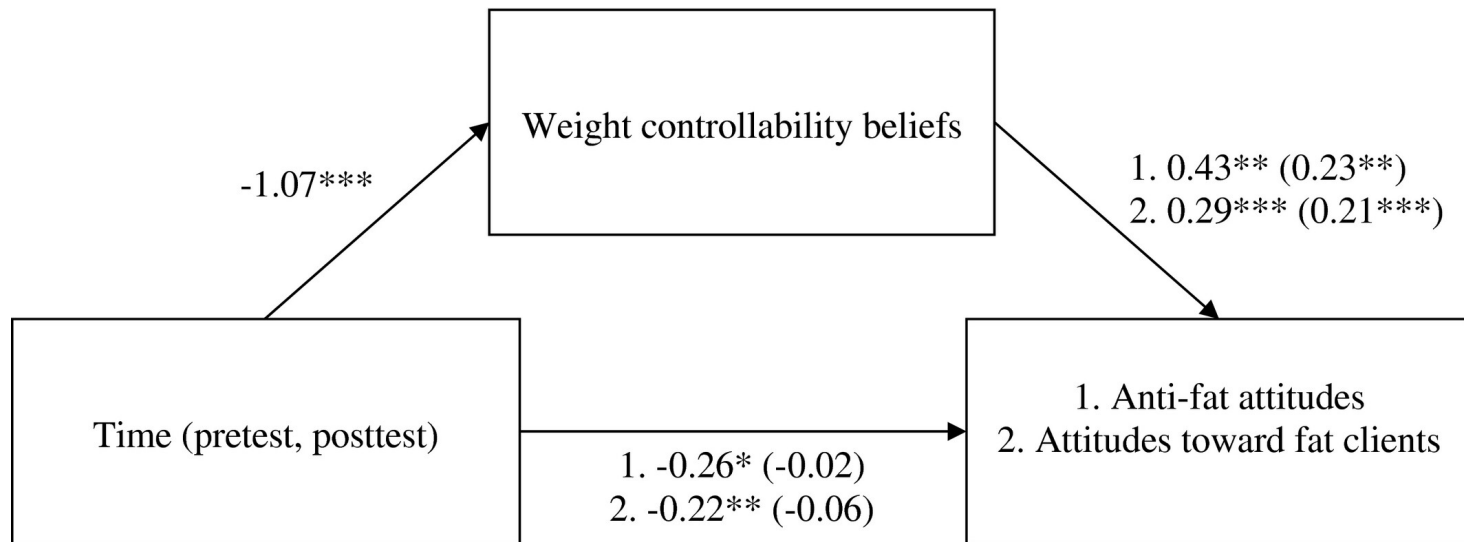


Figure 1—The direct effects of perspective (self minus other) on doping likelihood and guilt, and the indirect effect of perspective on doping likelihood via guilt. Unstandardized coefficients are reported, with 95% confidence intervals in brackets. Solid lines indicate significant paths.

Visualization Examples



Weight controllability beliefs mediate the effect of the weight bias seminar (time: pretest, posttest) on (1) anti-fat attitudes and (2) attitudes toward fat clients. *Note.* Unstandardized betas are presented on figure paths. Numbers in parentheses represent betas when the mediator and all model variables were included. * $p < .05$. ** $p < .01$. *** $p < .001$

Visualization Examples

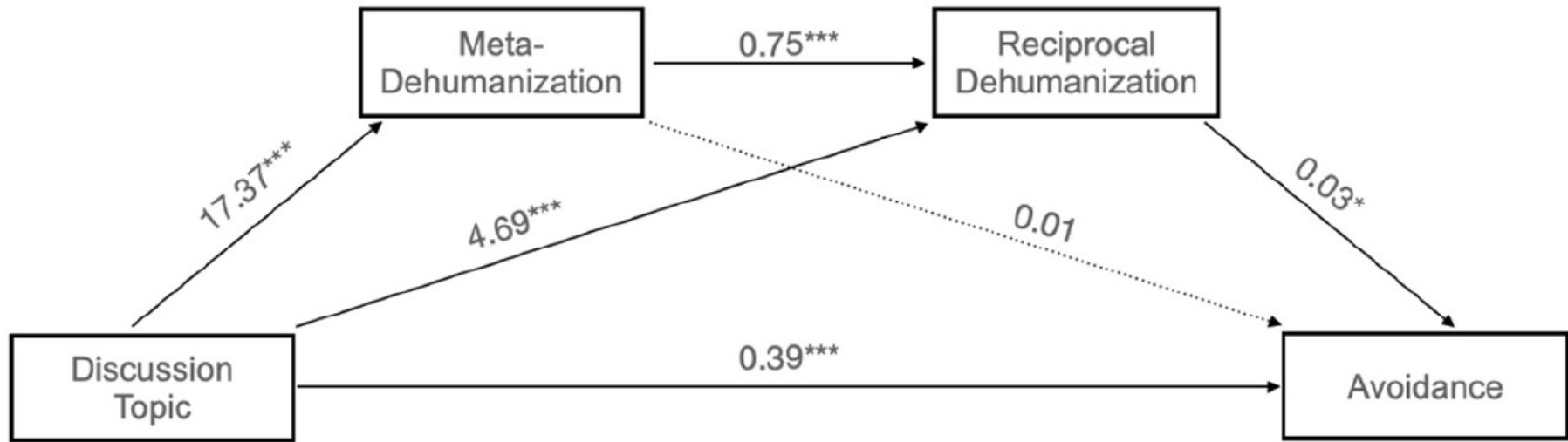


Figure 6. Within-person serial mediation: Study 3.

Note. All reported coefficients are unstandardized effect sizes because standardized betas could not be computed using this analysis.

Common Questions

Can this method be used for more than two conditions?

YES! Judd, Kenny, and McClelland (2001) describe a system for setting up contrasts among conditions, and testing the indirect effects of those contrasts.

I recommend reading Hayes & Preacher (2014) on mediation analysis with a multicategorical IV if you want to try this out. I am happy to give instructions on how to trick MEMORE into doing this. There will be functionality (soonish) for MEMORE to do this.

ALTERNATIVES: Some of the other repeated-measures mediation options are more appropriate if you have more than two conditions (especially longitudinal), so take a look at those when thinking about these options.

Can I use multiple mediators?

YES! MEMORE is already set up to do parallel mediation with up to 10 sets of mediators and serial mediation with up to **five** sets of mediators (See Montoya & Hayes 2017 for instructions).

Can we do conditional process models?

YES! We'll review this briefly on the second day, after covering moderation

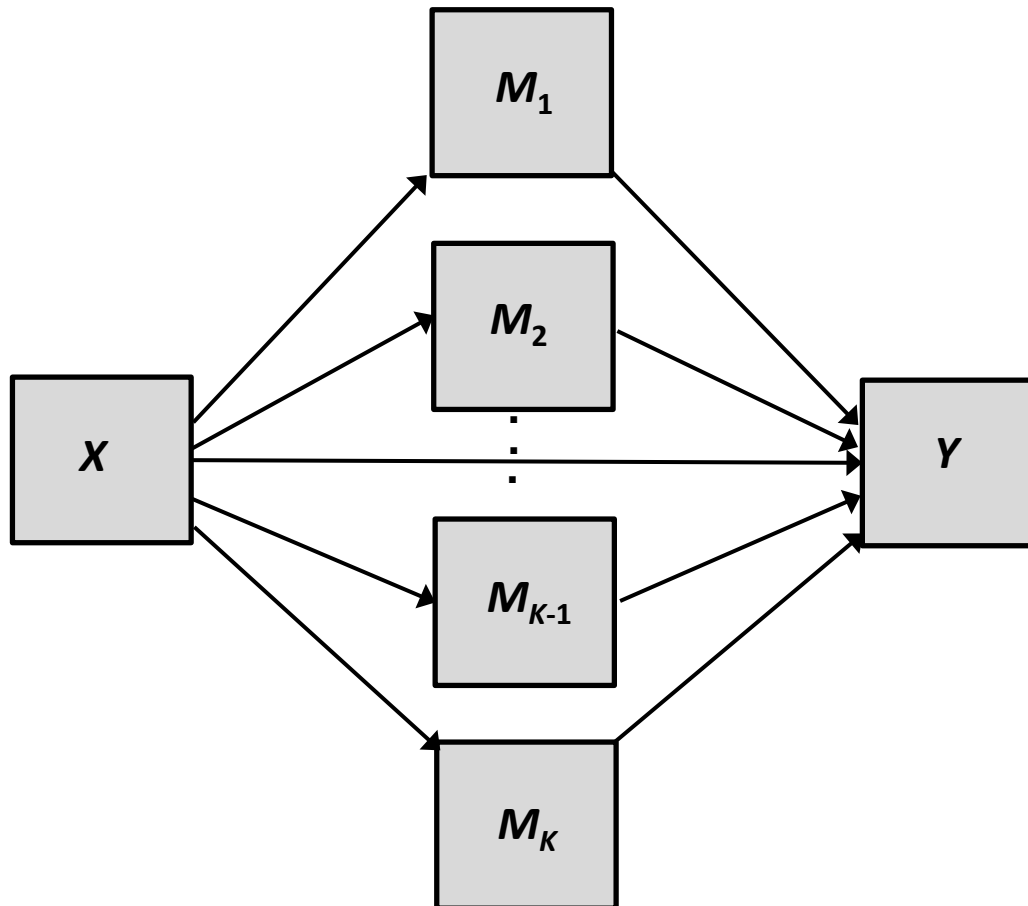
How do I control for covariates?

All of MEMORE's mediation analyses are within-person models, so you do not need to control for any between subjects variables such as age, gender, big-5.

Sometimes there are covariates which change within a person across conditions that you want to account for, this can be done by treating this additional variable as another set of mediators.

Models with More Than One Mediator

A parallel multiple mediator model



Why estimate such a model?

- ❑ Many causal effects probably operate through multiple mechanisms simultaneously. Better to estimate a model **consistent with such real-world** complexities.
- ❑ If your proposed mediator is correlated with the real mediator but not caused by the independent variable, a model with only your proposed mediator in it will be a **misspecification** and will potentially misattribute the process to your proposed mediator rather than the real mediator—“epiphenomenality.”
- ❑ Different theories may postulate different mediators as mechanisms. Including them all in a model simultaneously allows for a formal statistical comparison of indirect effects **representing different theoretical mechanisms**.

A Note on Notation

Thus far all measures have had 2 subscripts:

M_{ji} is the mediator M measured at time j for person i

Now we will have 3 subscripts:

M_{kji} is the k^{th} mediator measured at time j for person i

If you ever get confused by these, feel free to pipe up!

Two Mediator Case

The Total Effect model is **unchanged** because it doesn't involve the mediators

$$Y_{2i} - Y_{1i} = c + e_{Y_i^*}$$

Each mediator (pair) gets it's own model

$$\begin{aligned} M_{12i} - M_{11i} &= a_1 + e_{M_{1i}} \\ M_{22i} - M_{21i} &= a_2 + e_{M_{2i}} \end{aligned}$$

These residuals are NOT assumed to be independent, and may be correlated

Model for outcome incorporates both pairs of mediators

$$\begin{aligned} Y_{2i} - Y_{1i} &= c' + b_1(M_{12i} - M_{11i}) + b_2(M_{22i} - M_{21i}) \\ &\quad + d_1(M_{12i} + M_{11i} - (\overline{M_{12}} + \overline{M_{11}})) + d_2(M_{22i} + M_{21i} - (\overline{M_{22}} + \overline{M_{21}})) \\ &\quad + e_{Y_i} \end{aligned}$$

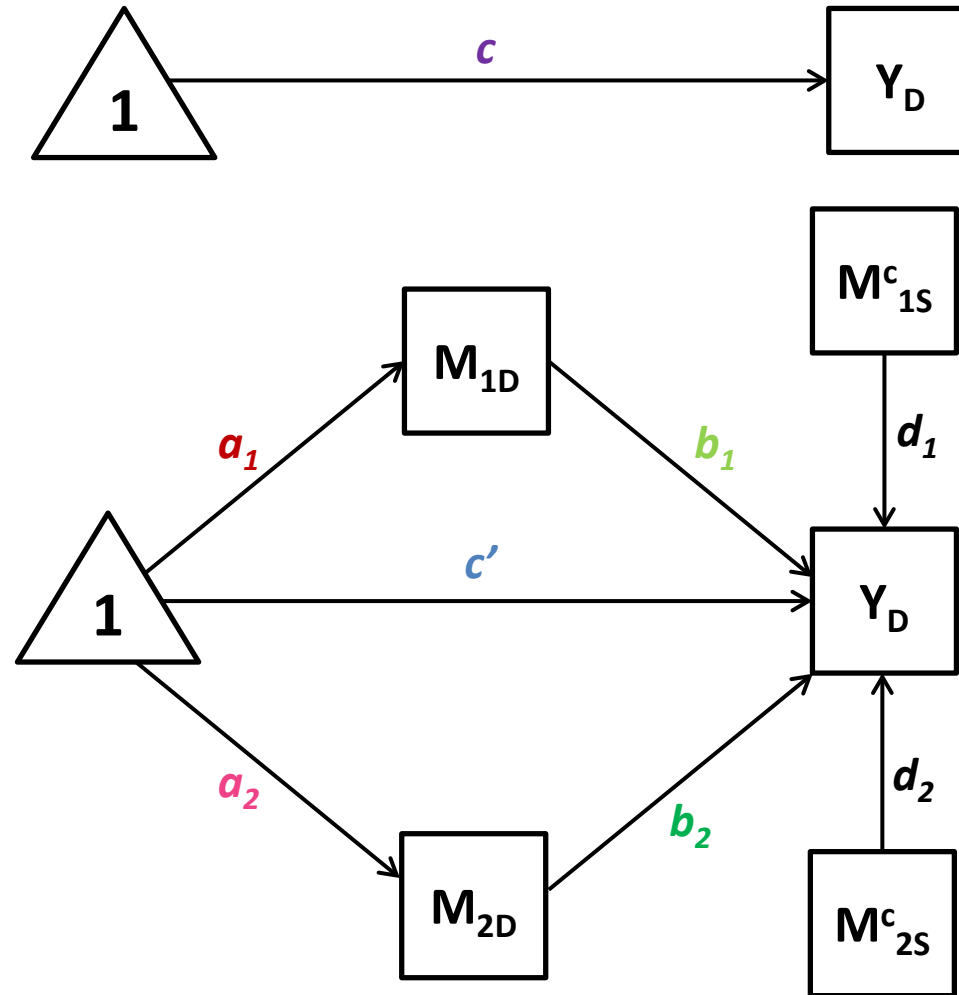
c = "total effect" of X on Y

$a_k \times b_k$ = "specific indirect effect" of X on Y through M_k

$\Sigma(a_k \times b_k)$ = "total indirect effect" of X on Y

c' = "direct effect" of X on Y

Path Diagram: Total, Direct, and Indirect Effects



^c Indicates mean centered

K Mediator Case

The Total Effect model is **unchanged** because it doesn't involve the mediators

$$Y_{2i} - Y_{1i} = c + e_{Y_i^*}$$

Each mediator (pair) gets its own model

$$M_{k2i} - M_{k1i} = a_k + e_{M_{ki}}$$

Model for outcome incorporates all pairs of mediators

$$Y_{2i} - Y_{1i} = c' + \sum_{k=1}^K b_k (M_{k2i} - M_{k1i}) + \sum_{k=1}^K d_k (M_{k2i} + M_{k1i} - (\overline{M_{k2}} + \overline{M_{k1}})) + e_{Y_i}$$

c = "total effect" of X on Y

$a_k \times b_k$ = "specific indirect effect" of X on Y through M_k

$\Sigma (a_k \times b_k)$ = "total indirect effect" of X on Y

c' = "direct effect" of X on Y

total effect = direct effect + total indirect effect

$$c = c' + \Sigma (a_k \times b_k)$$

Example: Group work in Computer Science

Participants read two syllabi for computer science classes. The syllabi had of two policies: **procollaboration** or **no collaboration**.

Participants were randomly assigned read each syllabus in a random order

Participants completed questionnaire (Higher = greater):

- (1) **interest in the class** (this is the primary DV).
- (2) how much they felt the class would help them in achieving **communal goals** (helping others, working with others)
- (3) how **difficult** they expected the class to be.

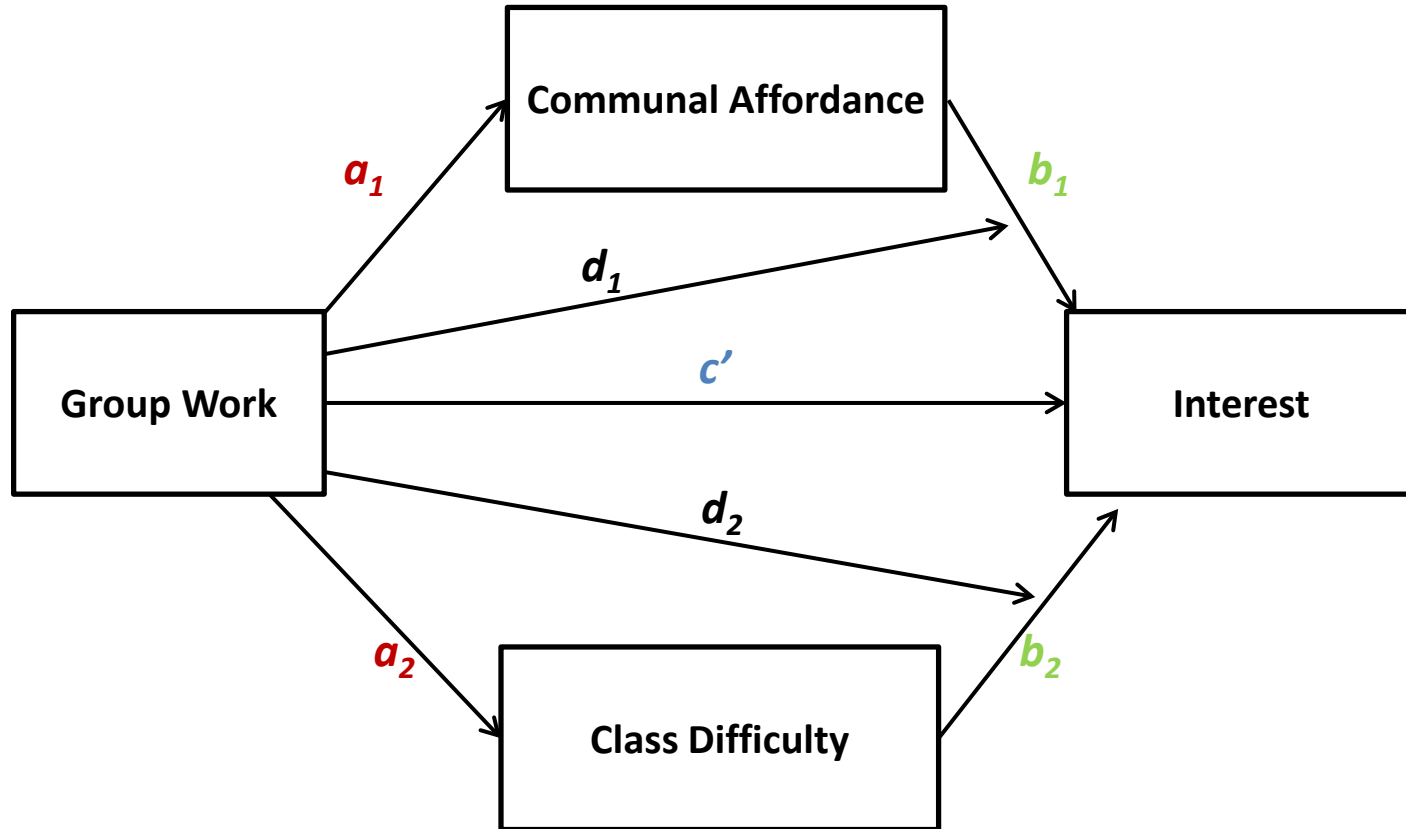


Question: Does group work in computer science classes increase interest in the class indirectly through perceived communal goal fulfillment, through class difficulty, or both?

Would people who read about the procollaboration policy think the class is more communal and would that communality then predict greater interest? Would the procollaboration policy make students think the course is easier, and this would increase interest?

Using MEMORE for CompSci WS data

Do people just like group work classes because they are easier?



Multiple Parallel Mediators

To add mediator pairs list them in the m= list, in the same order as the other pair and outcome variables

```
MEMORE m = comm_G comm_I diff_G diff_I /y = int_G int_I /model = 1 /seed = 74572.
```

```
%memore(m = comm_G comm_I diff_G diff_I, y = int_G int_I, model = 1, seed = 74572, data = CompSci_WS);
```

```
***** MEMORE Procedure for SPSS Version 3.0 *****
```

```
Written by Amanda Montoya
```

```
Documentation available at akmontoya.com
```

```
*****
```

```
Model:
```

```
1
```

```
Variables:
```

```
Y = int_G int_I
```

```
M1 = comm_G comm_I
```

```
M2 = diff_G diff_I
```

Check that variables are in correct pairs and in correct order

```
Computed Variables:
```

```
Ydiff = int_G - int_I
```

```
M1diff = comm_G - comm_I
```

```
M2diff = diff_G - diff_I
```

```
M1avg = ( comm_G + comm_I ) /2 Centered
```

```
M2avg = ( diff_G + diff_I ) /2 Centered
```

```
Sample Size:
```

```
51
```


Multiple Parallel Mediators

```
MEMORE m = comm_G comm_I diff_G diff_I /y = int_G int_I /model = 1 /seed = 74572.
```

```
%memore(m = comm_G comm_I diff_G diff_I, y = int_G int_I, model = 1, seed = 74572, data = CompSci_WS);
```

Results for Total Effect and Model of M1 (Communal Goals) unchanged

```
*****
```

```
Outcome: Ydiff = int_G - int_I
```

```
Model
```

	Effect	SE	t	p	LLCI	ULCI
constant	.3725	.2795	1.3330	.1886	-.1888	.9339

```
Degrees of freedom for all regression coefficient estimates:
```

```
50
```

```
*****
```

```
Outcome: Mldiff = comm_G - comm_I
```

```
Model
```

	Effect	SE	t	p	LLCI	ULCI
constant	2.2941	.2491	9.2108	.0000	1.7938	2.7944

```
Degrees of freedom for all regression coefficient estimates:
```

```
50
```

Multiple Parallel Mediators

```
MEMORE m = comm_G comm_I diff_G diff_I /y = int_G int_I /model = 1 /seed = 74572.
```

```
%memore(m = comm_G comm_I diff_G diff_I, y = int_G int_I, model = 1, seed = 74572, data = CompSci_WS);
```

Model of Difficulty

Outcome: M2diff = diff_G - diff_I

Model	Effect	SE	t	p	LLCI	ULCI
constant	-.4314	.1905	-2.2648	.0279	-.8139	-.0488

Degrees of freedom for all regression coefficient estimates:
50

Negative coefficient means $\text{diff}_I > \text{diff}_G$

The individual work class is perceived as 0.43 units more difficult than the group work class, and this effect is statistically significant ($p < .05$).

Multiple Parallel Mediators

The model of interest now includes both communal goals and difficulty variables

```
MEMORE m = comm_G comm_I diff_G diff_I /y = int_G int_I /model = 1 /seed = 74572.
```

```
%memore(m = comm_G comm_I diff_G diff_I, y = int_G int_I, model = 1, seed = 74572, data = CompSci_WS);
```

```
*****
```

```
Outcome: Ydiff = int_G - int_I
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.6307	.3978	2.6073	7.5978	4.0000	46.0000	.0001

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	-.9172	.3815	-2.4042	.0203	-1.6851	-.1493
M1diff	.4847	.1448	3.3460	.0016	.1931	.7762
M2diff	-.4123	.1878	-2.1952	.0332	-.7904	-.0342
M1avg	-.5160	.4157	-1.2411	.2209	-1.3528	.3209
M2avg	.3781	.2879	1.3133	.1956	-.2014	.9577

```
Degrees of freedom for all regression coefficient estimates:
```

```
46
```

Notice that we are now **controlling** for difficulty of the class when estimating the effect of communal goal affordance on interest!

The effect is still significant and positive

Multiple Parallel Mediators

The model of interest now includes both communal goals and difficulty variables

```
MEMORE m = comm_G comm_I diff_G diff_I /y = int_G int_I /model = 1 /seed = 74572.
```

```
%memore(m = comm_G comm_I diff_G diff_I, y = int_G int_I, model = 1, seed = 74572, data = CompSci_WS);
```

```
*****
```

```
Outcome: Ydiff = int_G - int_I
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.6307	.3978	2.6073	7.5978	4.0000	46.0000	.0001

Higher levels of difficulty correspond lower levels of interest in the course.

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	-.9172	.3815	-2.4042	.0203	-1.6851	-.1493
M1diff	.4847	.1448	3.3460	.0016	.1931	.7762
M2diff	-.4123	.1878	-2.1952	.0332	-.7904	-.0342
M1avg	-.5160	.4157	-1.2411	.2209	-1.3528	.3209
M2avg	.3781	.2879	1.3133	.1956	-.2014	.9577

The effect is statistically significant

```
Degrees of freedom for all regression coefficient estimates:
```

```
46
```

Multiple Parallel Mediators

The model of interest now includes both communal goals and difficulty variables

```
MEMORE m = comm_G comm_I diff_G diff_I /y = int_G int_I /model = 1 /seed = 74572.
```

```
%memore(m = comm_G comm_I diff_G diff_I, y = int_G int_I, model = 1, seed = 74572, data = CompSci_WS);
```

Outcome: Ydiff = int_G - int_I

Model Summary

R	R-sq	MSE	F	df1	df2	p
.6307	.3978	2.6073	7.5978	4.0000	46.0000	.0001

Direct effect is significant and negative

Model

	coeff	SE	t	p	LLCI	ULCI
constant	-.9172	.3815	-2.4042	.0203	-1.6851	-.1493
M1diff	.4847	.1448	3.3460	.0016	.1931	.7762
M2diff	-.4123	.1878	-2.1952	.0332	-.7904	-.0342
M1avg	-.5160	.4157	-1.2411	.2209	-1.3528	.3209
M2avg	.3781	.2879	1.3133	.1956	-.2014	.9577

After controlling for communal goals and difficulty, students are significantly more interested in the individual work class compared to group work.

Degrees of freedom for all regression coefficient estimates:

46

Multiple Parallel Mediators

```
MEMORE m = comm_G comm_I diff_G diff_I /y = int_G int_I /model = 1 /seed = 74572.
```

```
%memore(m = comm_G comm_I diff_G diff_I, y = int_G int_I, model = 1, seed = 74572, data = CompSci_WS);
```

***** TOTAL, DIRECT, AND INDIRECT EFFECTS *****

Total effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
.3725	.2795	1.3330	50.0000	.1886	-.1888	.9339

Direct effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
-.9172	.3815	-2.4042	46.0000	.0203	-1.6851	-.1493

Indirect Effect of X on Y through M

	Effect	BootSE	BootLLCI	BootULCI
Ind1	1.1119	.3871	.3597	1.8710
Ind2	.1779	.1148	.0000	.4347
Total	1.2897	.3567	.5690	1.9766

Indirect Key

Ind1	'X'	->	M1diff	->	Ydiff
Ind2	'X'	->	M2diff	->	Ydiff

Controlling for difficulty, there is still a significant indirect effect through communal affordance!

Controlling for communal goals, there is still a significant indirect effect through difficulty. (Just barely)

Total Indirect Effect

```
MEMORE m = comm_G comm_I diff_G diff_I /y = int_G int_I /model = 1 /seed = 74572.
```

```
%memore(m = comm_G comm_I diff_G diff_I, y = int_G int_I, model = 1, seed = 74572, data = CompSci_WS);
```

***** TOTAL, DIRECT, AND INDIRECT EFFECTS *****

Total effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
.3725	.2795	1.3330	50.0000	.1886	-.1888	.9339

Direct effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
-.9172	.3815	-2.4042	46.0000	.0203	-1.6851	-.1493

Indirect Effect of X on Y through M

	Effect	BootSE	BootLLCI	BootULCI
Ind1	1.1119	.3871	.3597	1.8710
Ind2	.1779	.1148	.0000	.4347
Total	1.2897	.3567	.5690	1.9766

Indirect Key

Ind1	'X'	->	M1diff	->	Ydiff
Ind2	'X'	->	M2diff	->	Ydiff

Total indirect effect can be useful when mediators are conceptually similar, and you want to estimate their combined effect.

For example, communal goals and agentic goals could both be mediators, and total indirect effect would reflect the indirect effect of “goals”

Summary of Results

Is the effect of group work on class interest mediated by communal goal affordance of the class, difficulty or the class, both or neither?

Overall there was no evidence of a total effect of group work on interest in computer science classes, we estimate that individuals were .37 units higher on interest in group work than individual work classes ($p = .19$). The class with group work was rated 2.29 units higher on communal goal affordance than the class with individual work ($p < .001$). The class with group work was rated 0.4314 units lower on difficulty ($p = 0.027$). Controlling for difficulty, a one unit increase in perception of communal goal affordance increased interest in the class by 0.48 units ($p = .002$), and the relationship between communal goal affordance and interest in a class did not depend on condition ($d_1 = -0.5160$, $p = .22$). Controlling for communal goal affordance, a one unit difference in difficulty corresponds to lower interest in the class by 0.41 units ($p = .03$), and the relationship between difficulty and interest in a class did not significantly depend on condition ($d_2 = 0.3781$, $p = .19$). The indirect effect of group work on interest through communal goal fulfillment, controlling for difficulty, was significantly different from zero ($a_1b_1 = 1.11$, 95% Bootstrap CI [0.36, 1.87]). This means that we expect women to be 1.11 units more interested in a computer science class with group work compared to one without group work that is equally difficult, through the effect of group work on communal goal affordance, and the subsequent effect of communal goal affordance on interest. The indirect effect of group work on interest through difficulty, controlling for communal goal affordance, was significantly different from zero ($a_2b_2 = 0.18$, 95% Bootstrap CI [0.00, 0.43]). This means that we expect women to be 0.18 units more interested in a computer science class with group work compared to one without group work that is equally affording of communal goals, through the effect of group work on difficulty, and the subsequent effect of difficulty on interest. There was a significant direct effect between group work and interest ($c' = -.92$, $p = .02$). This indicates that there may be some other process, separate from communal goal affordance and difficulty, which is actually deterring women from computer science classes with group work.

A note on the d-paths

```
*****
Outcome: Ydiff =  int_G      -      int_I

Model Summary
      R      R-sq      MSE      F      df1      df2      p
      .6307      .3978      2.6073      7.5978      4.0000      46.0000      .0001

Model
      coeff      SE      t      p      LLCI      ULCI
constant      -.9172      .3815      -2.4042      .0203      -1.6851      -.1493
M1diff      .4847      .1448      3.3460      .0016      .1931      .7762
M2diff      -.4123      .1878      -2.1952      .0332      -.7904      -.0342
M1avg      -.5160      .4157      -1.2411      .2209      -1.3528      .3209
M2avg      .3781      .2879      1.3133      .1956      -.2014      .9577
```

This model provides a good opportunity to discuss the interpretation of the d -paths, because we have one in each direction.

Interpretation depends on the order of subtraction: $M_2 - M_1$

- Positive: Effect of M on Y is more positive in condition 2 than condition 1
- Negative: Effect of M on Y is more positive in condition 1 than condition 2

$d_1 = -0.52$: The effect of communal goal affordance on interest is 0.52 higher in the individual work class than the group work class.

$$\frac{g_{21} + g_{11}}{2} = b_1 = 0.4847$$

$$g_{21} - g_{11} = d_1 = -0.5160$$

A note on the d-paths

```
*****
Outcome: Ydiff =  int_G      -      int_I

Model Summary
      R      R-sq      MSE      F      df1      df2      p
      .6307      .3978      2.6073      7.5978      4.0000      46.0000      .0001

Model
      coeff      SE      t      p      LLCI      ULCI
constant      -.9172      .3815      -2.4042      .0203      -1.6851      -.1493
M1diff         .4847      .1448      3.3460      .0016      .1931      .7762
M2diff        -.4123      .1878      -2.1952      .0332      -.7904      -.0342
M1avg         -.5160      .4157      -1.2411      .2209      -1.3528      .3209
M2avg         .3781      .2879      1.3133      .1956      -.2014      .9577
*****
```

g_{21} : Effect of communal goals on interest in group work class

g_{11} : Effect of communal goals on interest in individual work class

$$\frac{g_{21} + g_{11}}{2} = b_1 = 0.4847$$

$$g_{21} - g_{11} = d_1 = -0.5160$$

$$g_{21} = 0.2267$$

$$g_{11} = 0.7427$$

A note on the d-paths

```
*****
Outcome: Ydiff =  int_G      -      int_I

Model Summary
      R      R-sq      MSE      F      df1      df2      p
      .6307      .3978      2.6073      7.5978      4.0000      46.0000      .0001

Model
      coeff      SE      t      p      LLCI      ULCI
constant      -.9172      .3815      -2.4042      .0203      -1.6851      -.1493
M1diff      .4847      .1448      3.3460      .0016      .1931      .7762
M2diff      -.4123      .1878      -2.1952      .0332      -.7904      -.0342
M1avg      -.5160      .4157      -1.2411      .2209      -1.3528      .3209
M2avg      .3781      .2879      1.3133      .1956      -.2014      .9577
*****
```

$d_2 = 0.38$: The effect of difficulty on interest is 0.38 *more positive* in the group work class than the individual work class. (NOTE: this does not necessarily mean stronger)

$$\frac{g_{21} + g_{11}}{2} = b_1 = -0.4123 \quad g_{21} - g_{11} = d_1 = 0.3781$$

$$g_{21} = -0.223 \quad g_{11} = -0.601$$

CI that seem too close to zero

The CI for the indirect effect of difficulty seems very very close to zero. Could be hard to tell if it's significant.

Indirect Effect of X on Y through M				
	Effect	BootSE	BootLLCI	BootULCI
Ind1	1.1119	.3871	.3597	1.8710
Ind2	.1779	.1148	.0000	.4347
Total	1.2897	.3567	.5690	1.9766

Indirect Effect of X on Y through M				
	Effect	BootSE	BootLLCI	BootULCI
Ind1	1.1119	.3795	.3692	1.8557
Ind2	.1779	.1159	-.0014	.4459
Total	1.2897	.3488	.5942	1.9772

Indirect Effect of X on Y through M				
	Effect	BootSE	BootLLCI	BootULCI
Ind1	1.1119	.3801	.3483	1.8516
Ind2	.1779	.1134	.0001	.4368
Total	1.2897	.3481	.5789	1.9500

CI's that seem too close to zero

Confidence intervals will vary from run to run if you do not set the seed in your analysis.

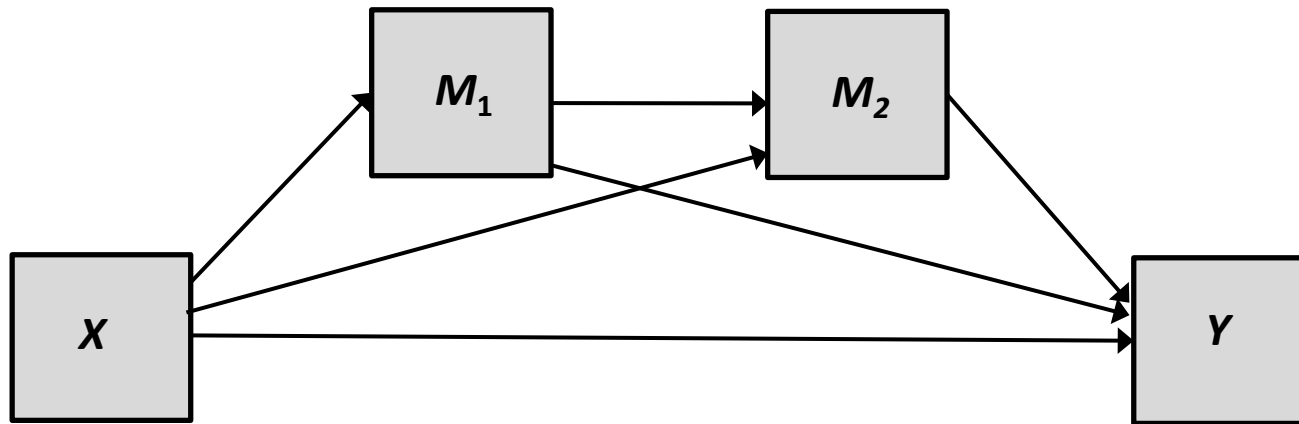
Recommendations:

1. Preregister your seed and rely on the result that you get
`seed = 74572`
2. Increase the number of bootstraps until the CI is reliably on one side of zero
`samples = 10000`
3. Increase the number of decimal points to see more exact results
`results decimals = F10.8`

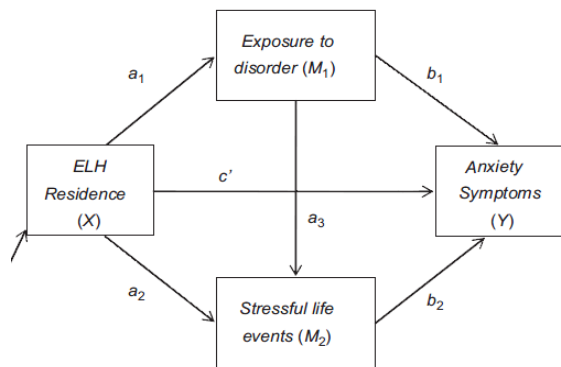
It may seem like overkill to preregister a seed but it can make a big difference especially when results are so borderline

Serial Mediation Models

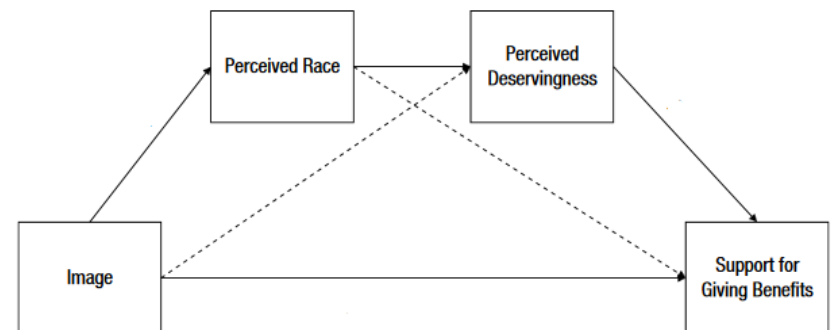
A serial multiple mediator model with two mediators and all possible direct and indirect effects freely estimated.



Some examples in the literature:



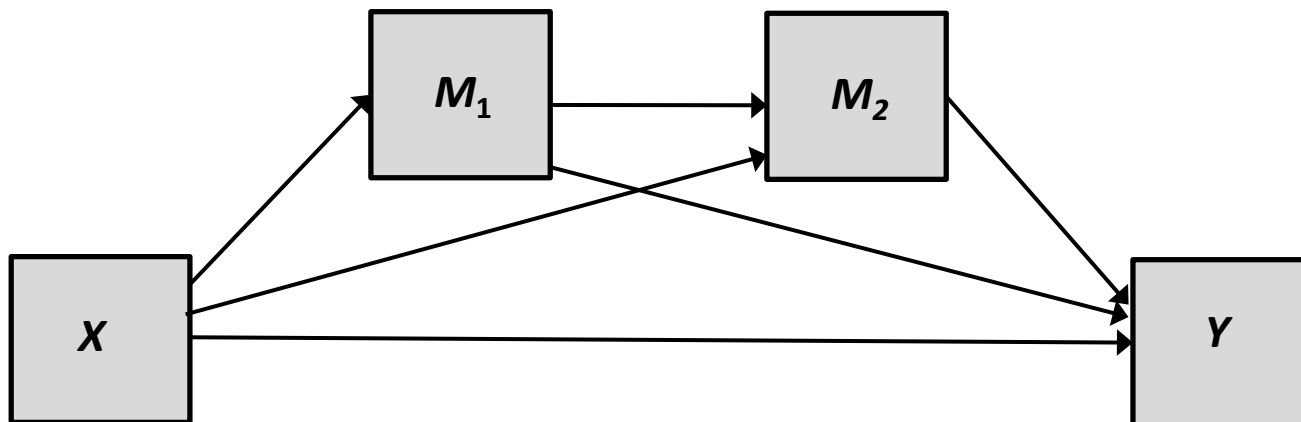
Casciano, R., & Massey, D. S. (2012). Neighborhood disorder and anxiety symptoms: New evidence from a quasiexperimental study. *Health and Place*, 18, 180-190.



Brown-Iannuzzi, J. L., Dotsch, R., Cooley, E., & Payne, B. K. (2017). The relationship between mental representations of welfare recipients and attitudes toward welfare. *Psychological Science*, 28, 92-103.

Why estimate such a model?

- ❑ Many causal effects probably operate through multiple steps. Estimation of a model that is **aligned with the real world** is ideal.
- ❑ Existence of serial mediators is a **violation of the causal assumptions** of a simple mediation, **biasing** the estimate of the indirect effect.
 - ❑ M_1 is also a confounder of the effect of M_2 on Y
 - ❑ Controlling for M_1 would bias the effect of X on M_2
- ❑ Many theories postulate a set of mediators acting in series. Fitting a model consistent with these theories is one way to test them.



Two Mediator Case

The Total Effect model is **unchanged** because it doesn't involve the mediators

$$Y_{2i} - Y_{1i} = c + e_{Y_i^*}$$

Each mediator (pair) gets it's own model, all mediators predict downstream mediators

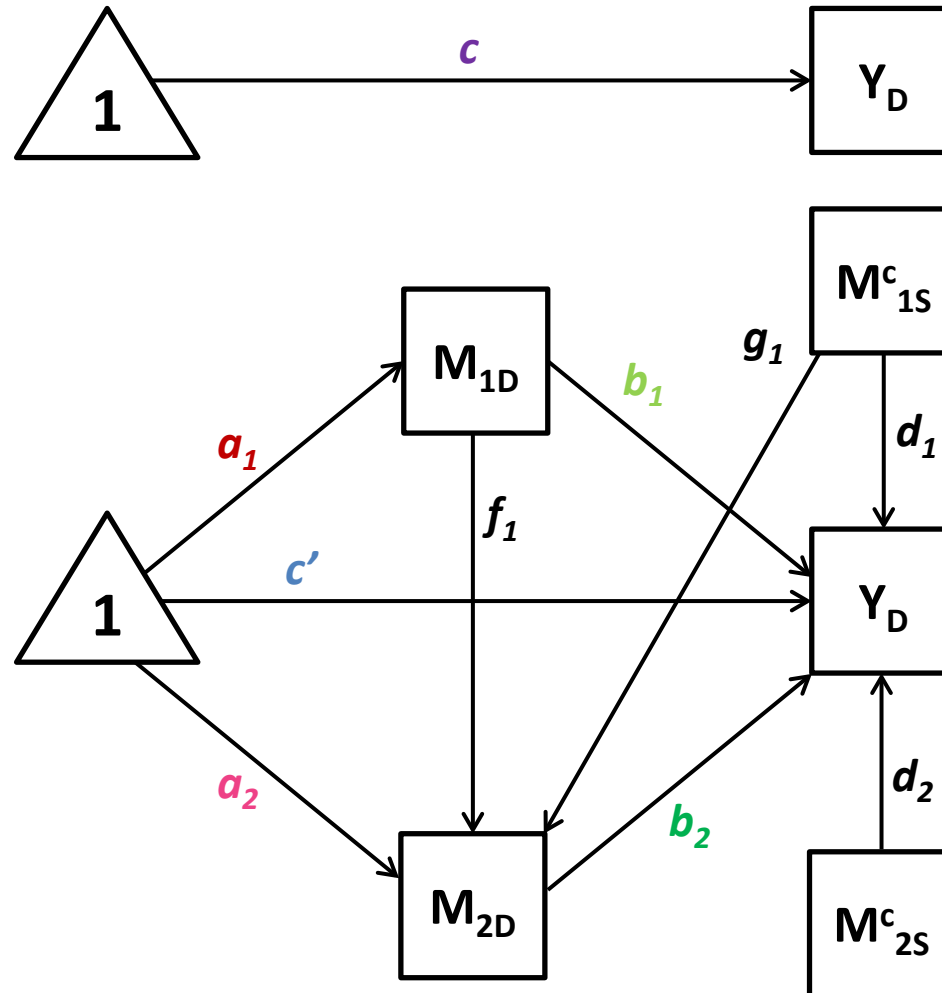
$$M_{12i} - M_{11i} = a_1 + e_{M_{1i}}$$

$$M_{22i} - M_{21i} = a_2 + f_1(M_{12i} - M_{11i}) \\ + g_1(M_{12i} + M_{11i} - (\overline{M_{12} + M_{11}})) + e_{M_{2i}}$$

Model for outcome incorporates both pairs of mediators, is identical to parallel model

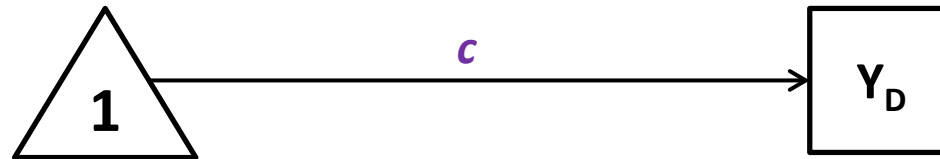
$$Y_{2i} - Y_{1i} = c' + b_1(M_{12i} - M_{11i}) + b_2(M_{22i} - M_{21i}) \\ + d_1(M_{12i} + M_{11i} - (\overline{M_{12} + M_{11}})) \\ + d_2(M_{22i} + M_{21i} - (\overline{M_{22} + M_{21}})) + e_{Y_i}$$

Path Diagram: Total, Direct, and Indirect Effects

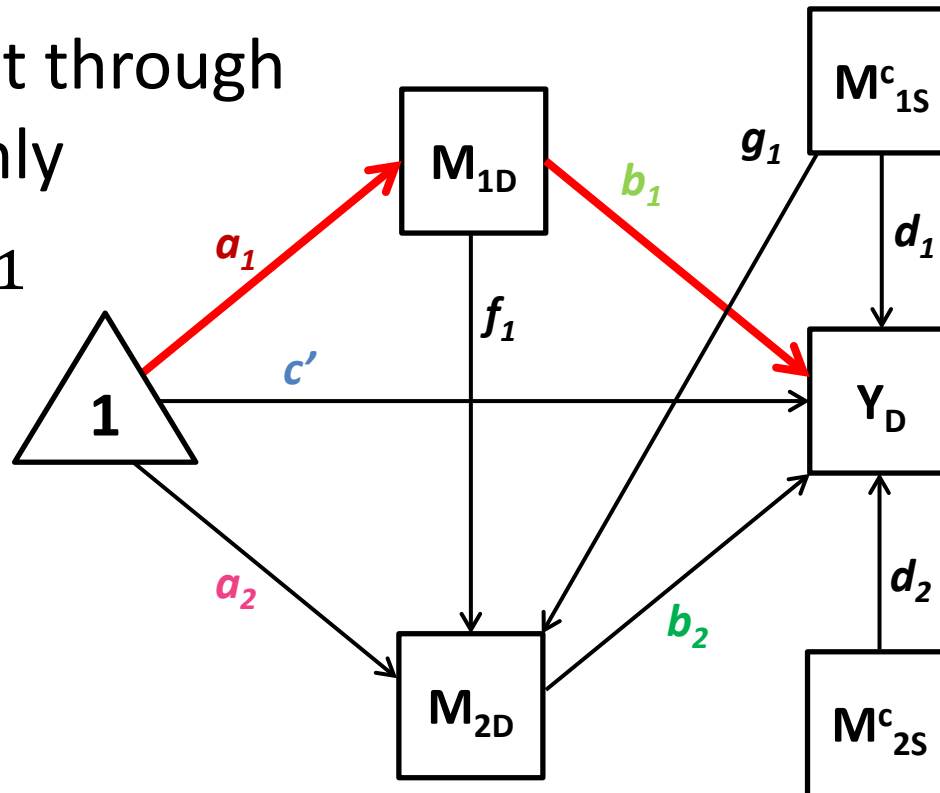


^c Indicates mean centered

Indirect Effects

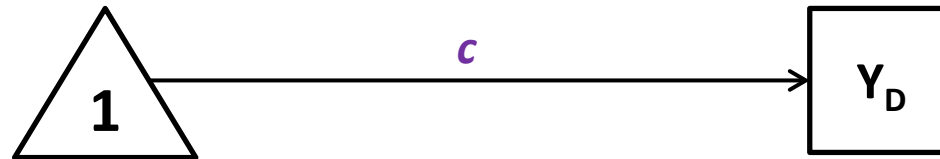


Indirect effect through
 M_1 only
 $a_1 b_1$

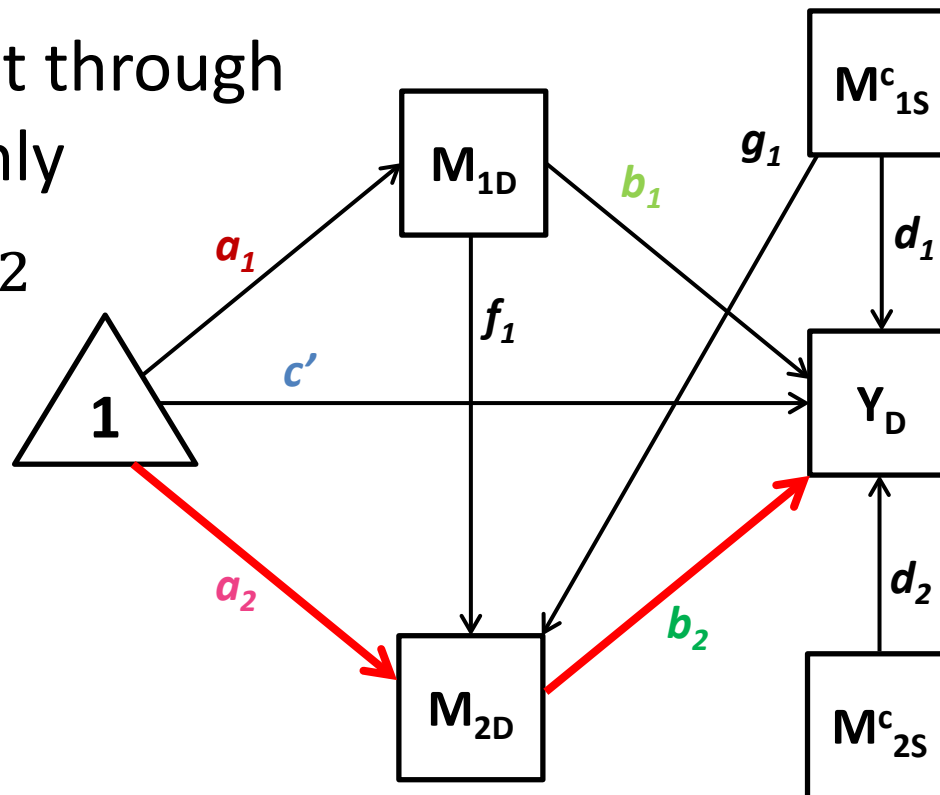


^c Indicates mean centered

Indirect Effects



Indirect effect through
 M_2 only
 $a_2 b_2$



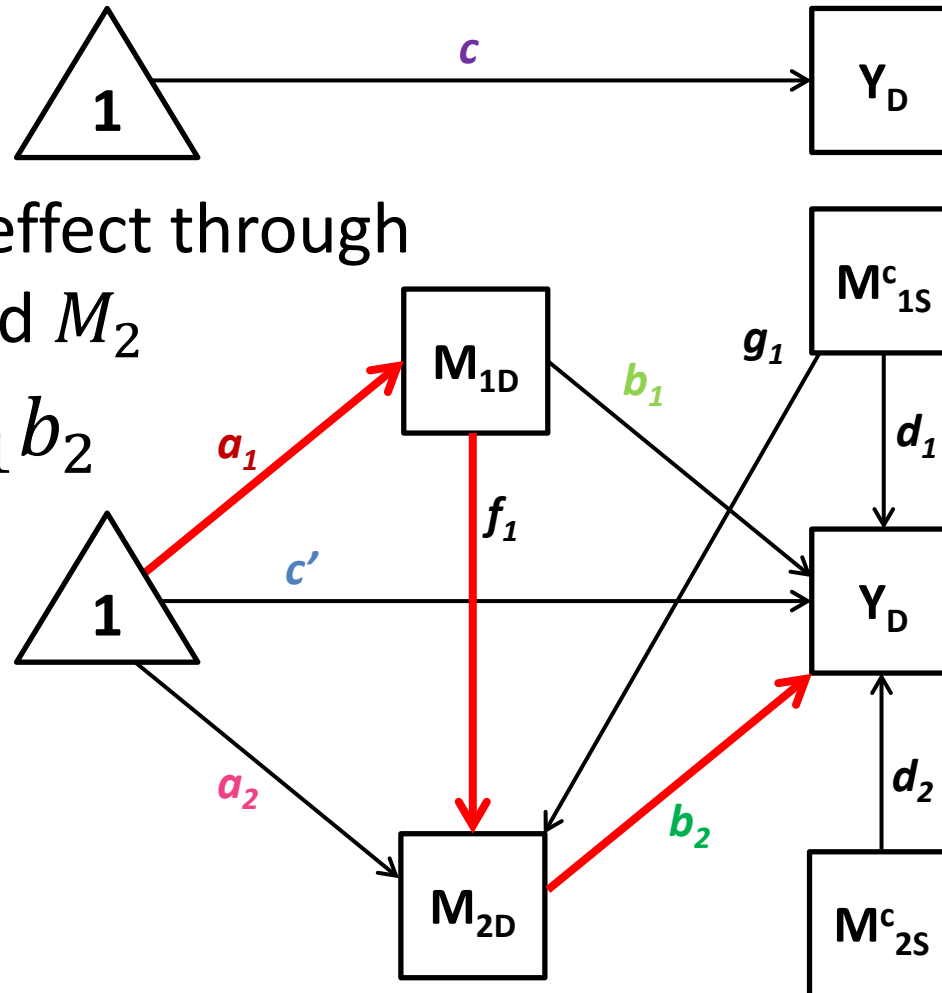
^c Indicates mean centered

Indirect Effects

Serial indirect effect through

M_1 and M_2

$a_1 f_1 b_2$



^c Indicates mean centered

Path Analysis Algebra

c = “total effect” of X on Y

a_1b_1 = “specific indirect effect” of X on Y through M_1

a_2b_2 = “specific indirect effect” of X on Y through M_2

$a_1f_1b_2$ = “serial indirect effect” of X on Y through M_1 and M_2

$a_1b_1 + a_2b_2 + a_1f_1b_2$ = “total indirect effect” of X on Y

c' = “direct effect” of X on Y

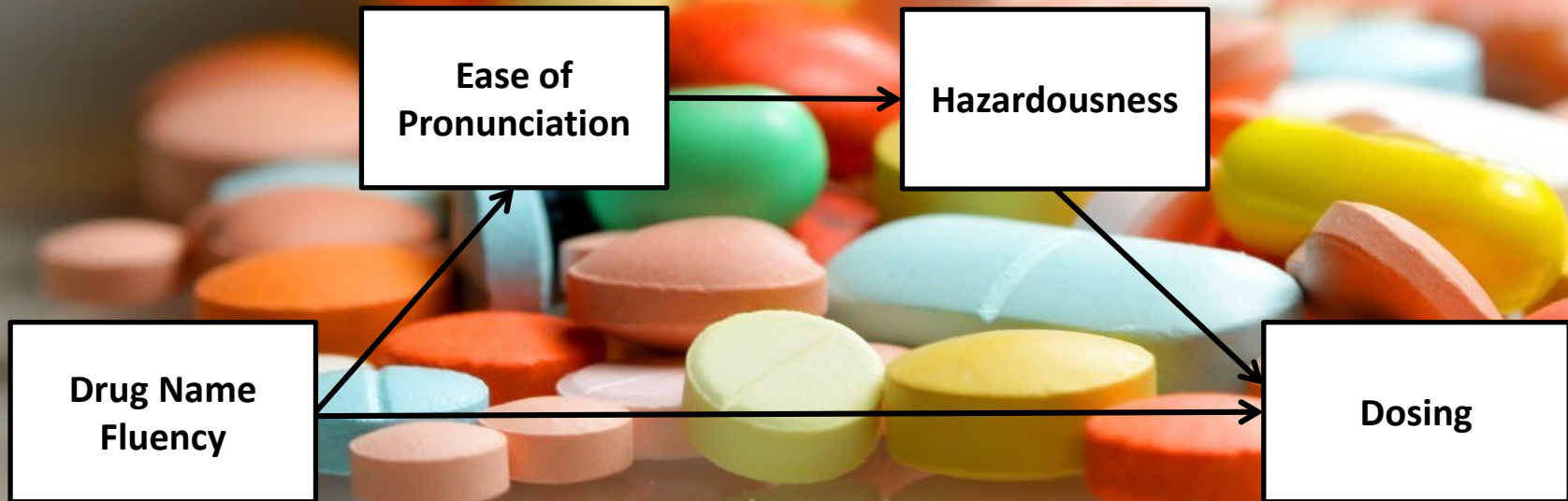
total effect = direct effect + total indirect effect

Example: Drug Name Fluency

Dohle, S., & Montoya, A. K. (2017). The dark side of fluency: Fluent names increase drug dosing. *Journal of Experimental Psychology: Applied*, 23(3), 231 – 239.

Research Question: Can the name of drugs impact ease of pronunciation which effects how hazardous they seem and how much people are willing to dose the drugs?

Imagine you have a cold, and there are a variety of medications available including (a) Fastinorbine and (b) Cytrigmcmium. Which is easier to pronounce? Which seems more hazardous? Which are you willing to dose more of?



Example: Drug Name Fluency

Dohle, S., & Montoya, A. K. (2017). The dark side of fluency: Fluent names increase drug dosing. *Journal of Experimental Psychology: Applied*, 23(3), 231 – 239.

N = 70

Measured Variables:

- Dosage in mL
 - Variable name: Dose
 - 0 mL – 200mL
- Hazardousness of drug
 - Variable name: Haz
 - Average of two questions:
 - Hazardousness (1-7)
 - Dangerousness (1-7)
- Ease of Pronunciation
 - Variable name: Ease
 - Single item per drug

FluencyData_Avg.sav [DataSet2] - IBM SPSS Statistics Data Editor

	HazSimp	HazComp	DoseComp	DoseSimp	EaseComp	EaseSimp
1	2.50	7.50	46.00	58.33	1.33	7.67
2	7.00	7.00	84.33	86.67	1.33	8.00
3	6.50	6.50	68.67	70.00	2.67	9.00
4	3.00	5.67	118.00	152.00	2.00	6.33
5	6.50	5.17	45.00	48.33	3.00	8.00
6	2.83	4.83	40.33	53.00	2.33	7.00
7	2.67	4.50	153.67	139.00	1.33	7.67
8	5.00	5.00	140.67	142.33	4.33	5.33
9	4.67	6.67	71.67	69.67	1.00	9.00
10	2.50	6.67	53.00	91.67	2.67	8.00
11	4.67	5.00	142.00	143.00	2.33	7.33
12	5.50	7.83	76.67	94.00	4.33	9.00
13	2.33	4.67	76.67	107.67	1.67	5.67
14	3.33	6.17	145.00	140.00	1.67	8.33
15	3.00	6.50	73.00	95.67	2.67	7.33
16	3.33	6.00	91.00	85.67	2.33	7.00
17	2.67	8.00	61.00	83.67	2.00	7.00
18	4.67	5.00	79.00	92.00	1.33	8.00
19	5.50	6.83	38.33	59.00	1.67	7.00

Serial Mediators

To add mediator pairs list them in the m= list, in the same order as the other pair and outcome variables

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1  
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =  
1, serial = 1, data = Fluency).
```

```
***** MEMORE Procedure for SPSS Version 3.0 *****
```

Written by Amanda Montoya

Documentation available at akmontoya.com

```
*****
```

Model:

1

Variables:

Y = DoseSimp DoseComp

M1 = EaseSimp EaseComp

M2 = HazSimp HazComp

Check that variables are in correct pairs and in correct order

Computed Variables:

Vdiff = DoseSimp - DoseComp

M1diff = EaseSimp - EaseComp

M2diff = HazSimp - HazComp

M1avg = (EaseSimp + EaseComp) /2 Centered

M2avg = (HazSimp + HazComp) /2 Centered

Sample Size:

70

Serial Mediators

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1  
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =  
1, serial = 1, data = Fluency).
```

```
*****
```

```
Outcome: Ydiff = DoseSimp - DoseComp
```

Model	Effect	SE	t	p	LLCI	ULCI
constant	11.0476	1.5770	7.0055	.0000	7.9016	14.1937

```
Degrees of freedom for all regression coefficient estimates:
```

```
69
```

Positive coefficient means DoseSimp > DoseComp

Participants dosed the simple drugs 11.05 mL more on average than the drugs with complex names, and this is a statistically significant difference ($t(69) = 7.01$, $p < .001$).

Serial Mediators

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1  
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =  
1, serial = 1, data = Fluency).
```

```
*****
```

```
Outcome: Mldiff = EaseSimp - EaseComp
```

Model	Effect	SE	t	p	LLCI	ULCI
constant	5.1143	.1679	30.4562	.0000	4.7793	5.4493

```
Degrees of freedom for all regression coefficient estimates:
```

```
69
```

Positive coefficient means EaseSimp > EaseComp

Participants perceived simple drugs 5.11 units more on easy to pronounce than the drugs with complex names, and this is a statistically significant difference ($t(69) = 30.46$, $p < .001$).

Serial Mediators

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1  
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =  
1, serial = 1, data = Fluency).
```

```
*****
```

```
Outcome: M2diff = HazSimp - HazComp
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.2380	.0566	2.3226	2.0109	2.0000	67.0000	.1419

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	-.7642	.6934	-1.1021	.2744	-2.1482	.6199
Mldiff	-.2621	.1308	-2.0037	.0491	-.5232	-.0010
Mlavg	.0084	.2296	.0367	.9708	-.4499	.4668

```
Degrees of freedom for all regression coefficient estimates:
```

```
67
```

Negative intercept means HazSimp < HazComp after controlling for Ease

No significant impact of drug name on hazardousness after controlling for ease of pronunciation. (Could be a red flag if this was significant)

Higher ease of pronunciation corresponds to lower hazardousness ($f_1 = -.26$, $p = .05$)

No significant difference in effect of ease on hazardousness across conditions ($g_1 = 0.01$, $p = .97$)

Serial Mediators

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =
1, serial = 1, data = Fluency).
```

```
*****
```

```
Outcome: Ydiff = DoseSimp - DoseComp
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.4170	.1739	152.6672	3.4200	4.0000	65.0000	.0134

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	4.7395	5.6984	.8317	.4086	-6.6410	16.1200
M1diff	-.2158	1.1011	-.1960	.8452	-2.4149	1.9832
M2diff	-3.5215	.9943	-3.5418	.0007	-5.5072	-1.5358
M1avg	-1.9940	2.1296	-.9363	.3526	-6.2472	2.2592
M2avg	1.1574	1.6968	.6821	.4976	-2.2313	4.5462

```
Degrees of freedom for all regression coefficient estimates:
```

```
65
```

Intercept is positive meaning DoseSimp > DoseComp after controlling for ease of pronunciation and hazardousness, but the effect is not statistically significant.

Serial Mediators

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1  
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =  
1, serial = 1, data = Fluency).
```

```
*****
```

```
Outcome: Ydiff = DoseSimp - DoseComp
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.4170	.1739	152.6672	3.4200	4.0000	65.0000	.0134

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	4.7395	5.6984	.8317	.4086	-6.6410	16.1200
M1diff	-.2158	1.1011	-.1960	.8452	-2.4149	1.9832
M2diff	-3.5215	.9943	-3.5418	.0007	-5.5072	-1.5358
M1avg	-1.9940	2.1296	-.9363	.3526	-6.2472	2.2592
M2avg	1.1574	1.6968	.6821	.4976	-2.2313	4.5462

```
Degrees of freedom for all regression coefficient estimates:
```

```
65
```

Higher ease corresponds with lower dosage after controlling for hazardousness, but this is not a statistically significant effect ($b_1 = -.22$, $p = .85$)

Serial Mediators

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =
1, serial = 1, data = Fluency).
```

```
*****
```

```
Outcome: Ydiff = DoseSimp - DoseComp
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.4170	.1739	152.6672	3.4200	4.0000	65.0000	.0134

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	4.7395	5.6984	.8317	.4086	-6.6410	16.1200
M1diff	-.2158	1.1011	-.1960	.8452	-2.4149	1.9832
M2diff	-3.5215	.9943	-3.5418	.0007	-5.5072	-1.5358
M1avg	-1.9940	2.1296	-.9363	.3526	-6.2472	2.2592
M2avg	1.1574	1.6968	.6821	.4976	-2.2313	4.5462

```
Degrees of freedom for all regression coefficient estimates:
```

```
65
```

Higher hazardousness corresponds with lower dosage after controlling for ease of pronunciation, but this is not a statistically significant effect ($b_2 = -3.52$, $p < .001$)

Serial Mediators

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1  
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =  
1, serial = 1, data = Fluency).
```

```
*****
```

```
Outcome: Ydiff = DoseSimp - DoseComp
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.4170	.1739	152.6672	3.4200	4.0000	65.0000	.0134

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	4.7395	5.6984	.8317	.4086	-6.6410	16.1200
M1diff	-.2158	1.1011	-.1960	.8452	-2.4149	1.9832
M2diff	-3.5215	.9943	-3.5418	.0007	-5.5072	-1.5358
M1avg	-1.9940	2.1296	-.9363	.3526	-6.2472	2.2592
M2avg	1.1574	1.6968	.6821	.4976	-2.2313	4.5462

```
Degrees of freedom for all regression coefficient estimates:
```

```
65
```

Neither the relationship between ease of pronunciation and dosage or hazardousness and dosage seems to differ between the two conditions ($d_1 = -1.99$, $p = .35$; $d_2 = 1.16$, $p = .50$)

Serial Mediators

```
MEMORE m= EaseSimp EaseComp HazSimp HazComp /y = DoseSimp DoseComp /model = 1
/serial = 1.
```

```
%MEMORE(m= EaseSimp EaseComp HazSimp HazComp, y = DoseSimp DoseComp, model =
1, serial = 1, data = Fluency).
```

***** TOTAL, DIRECT, AND INDIRECT EFFECTS *****

Total effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
11.0476	1.5770	7.0055	69.0000	.0000	7.9016	14.1937

Direct effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
4.7395	5.6984	.8317	65.0000	.4086	-6.6410	16.1200

Indirect Effect of X on Y through M

	Effect	BootSE	BootLLCI	BootULCI
Ind1	-1.1039	5.4275	-12.0719	9.1937
Ind2	2.6911	2.1809	-1.2069	7.4292
Ind3	4.7210	2.5278	.3695	10.2421
Total	6.3081	5.5363	-5.0427	17.2221

Indirect Key

Ind1	'X'	->	M1diff	->	Ydiff
Ind2	'X'	->	M2diff	->	Ydiff
Ind3	'X'	->	M1diff	->	M2diff -> YDiff

Neither specific indirect effects are statistically significant

Serial indirect effect is positive meaning dosage is higher for simple (vs. complex) drugs through effect on ease of pronunciation then hazardousness

Other Types of Within-Subject Mediation

Mediation Analysis with Multilevel Models

- MEMORE method is equivalent for two repeated-measurements
- Can accommodate multiple trials per condition, longitudinal data, continuous X , random slopes
- MLMED is an SPSS macro for multilevel mediation
- Kris Preacher's MLM courses cover mediation for multilevel models.



Resources:

[Bauer, Preacher, & Gil \(2006\)](#)

[Hayes & Rockwood \(2020\)](#)

[Rockwood & Hayes \(2022\)](#)

Other Types of Within-Subject Mediation

Latent Growth Curve models

- MEMORE is equivalent for two time points
- Model trajectories over time
- Accommodate measurement models, longitudinal data, between-subject treatment variables
- Mplus or lavaan in R tend to be the best tools
- Statistical Horizons courses on Longitudinal Data Analysis may cover

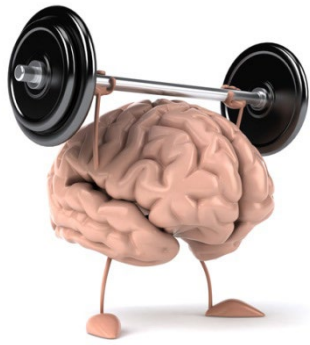
Resources:

[Cheong, MacKinnon, and Khoo, 2003](#)

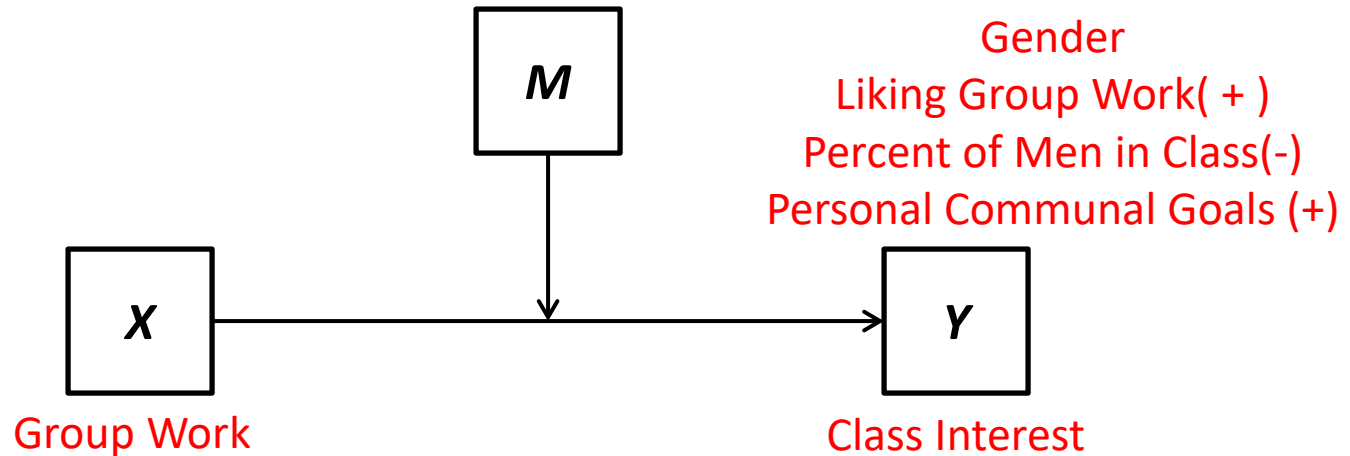
[Cole & Maxwell \(2003\)](#)

Mediation Summary

- Estimation with Regression Equations
- Path analysis approach
- Inference for indirect effects
- Visualizations
- Study Planning
- Parallel & Serial Mediator Models
- Next Steps: MLM and SEM



Moderation



The relationship between the focal predictor (X) and an outcome (Y) is said to be moderated when the size or direction depends on M . Moderation helps us understand boundary conditions of effect: for whom on when is the effect large or small, present or absent, positive or negative.

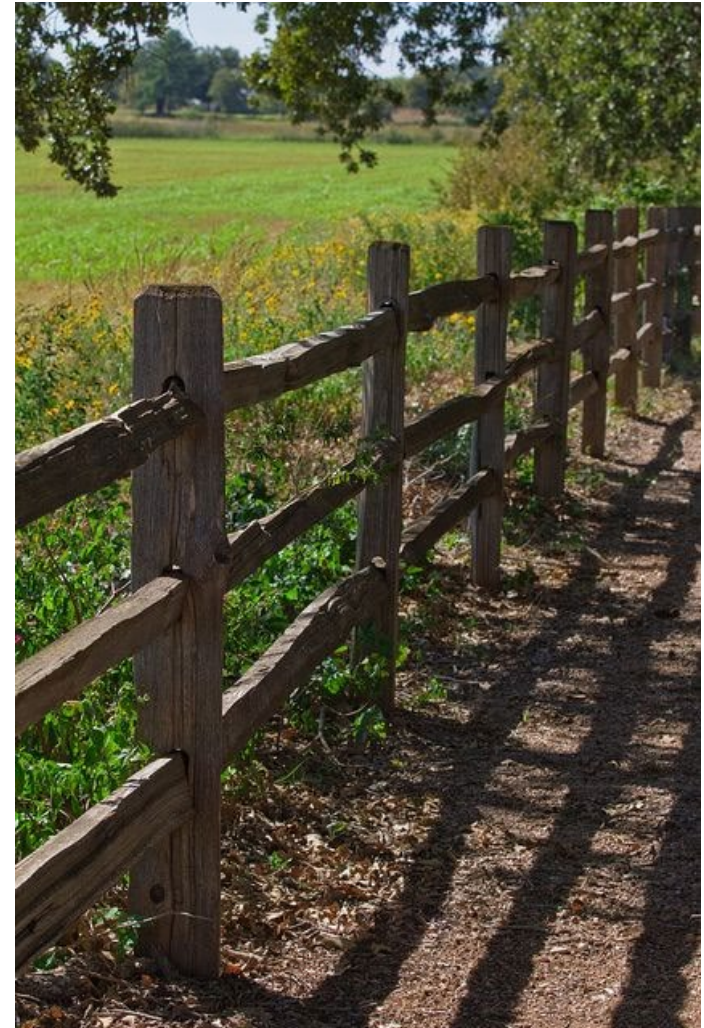
X and M are frequently described as “interacting” in their prediction of Y .

Many different kind of variables may act as moderators. Emotional variables, situational, individual level variables, cognitive variables, environmental variables, etc.

A quick example: Name some possible moderators!

Moderation

- Between-Subject Moderation
- Two-Condition Within Subjects Moderation
 - Judd Kenny and McClelland (2001, 1996)
 - Interpretations
 - Probing
 - MEMORE
 - Reporting (Writing and Figures)
 - Common Questions
 - Multiple Moderator Models



Moderation in Between-Subject Designs

Moderation analysis is **very common** with between-subject designs.

We focus on regression based moderation analysis, but it can also be examined using ANOVA (and these methods are largely equivalent)

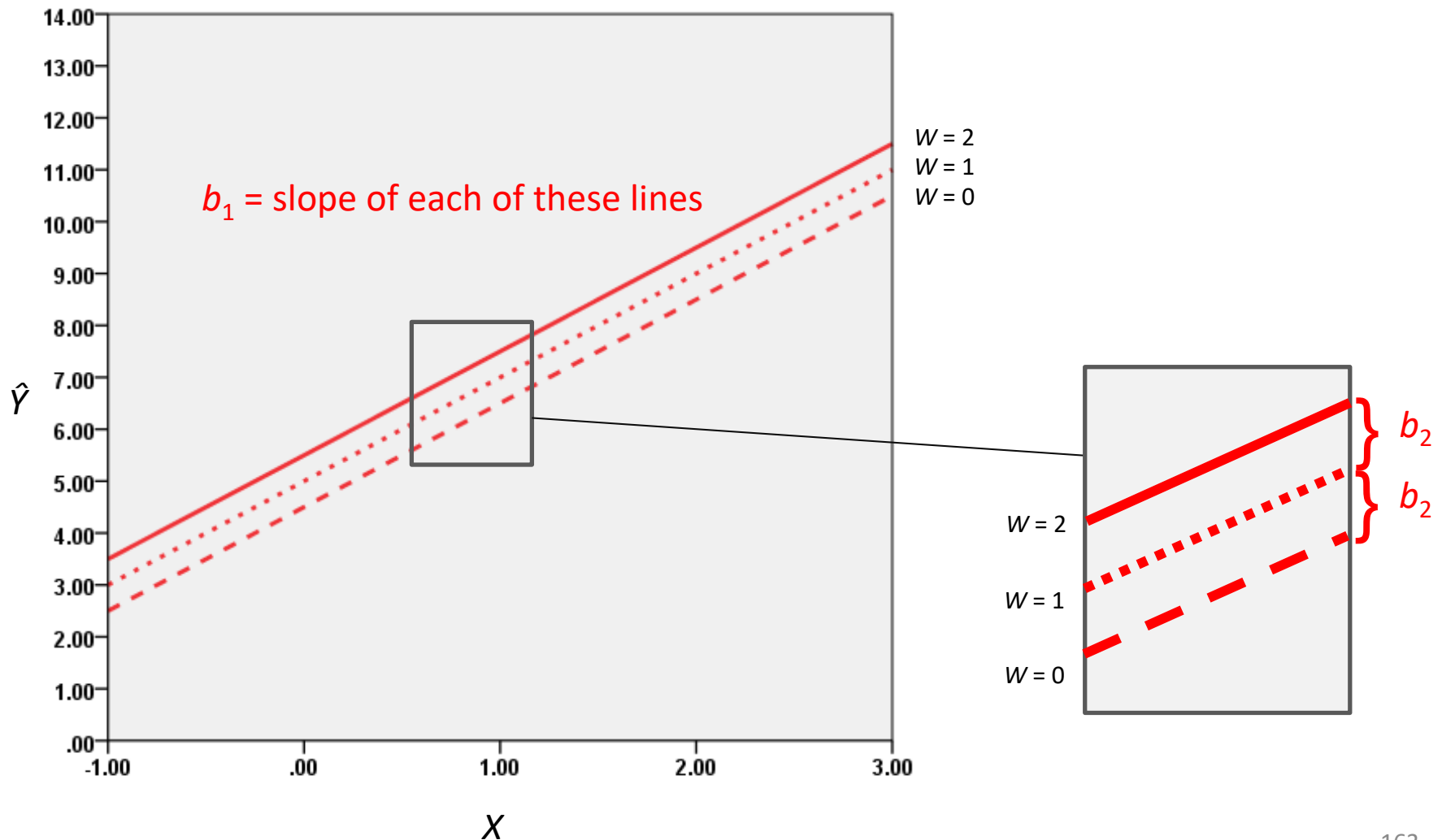
X : observed or assigned once per subject

W : observed or assigned once per subject

Y : observed once per subject

Partial regression coefficients as *unconditional* effects

$$\hat{Y}_i = 4.50 + 2.00X_i + 0.50W_i$$



Releasing this constraint

Suppose we let X 's effect be a function of W , $f(W)$, as in

$$\hat{Y}_i = b_0 + f(W_i)X_i + b_2W_i$$

For instance, let $f(W)$ be a linear function of W , $b_1 + b_3W$. Thus,

$$\hat{Y}_i = b_0 + (b_1 + b_3W_i)X_i + b_2W_i$$

This can be rewritten in an equivalent form as

$$\hat{Y}_i = b_0 + b_1X_i + b_2W_i + b_3X_iW_i$$

This model, the “simple moderation model,” allows X 's effect on Y to depend linearly on W . Other forms of moderation are possible, but this form is the one most frequently estimated.

X 's effect on Y as a function of W

$$b_0 = 6.00$$

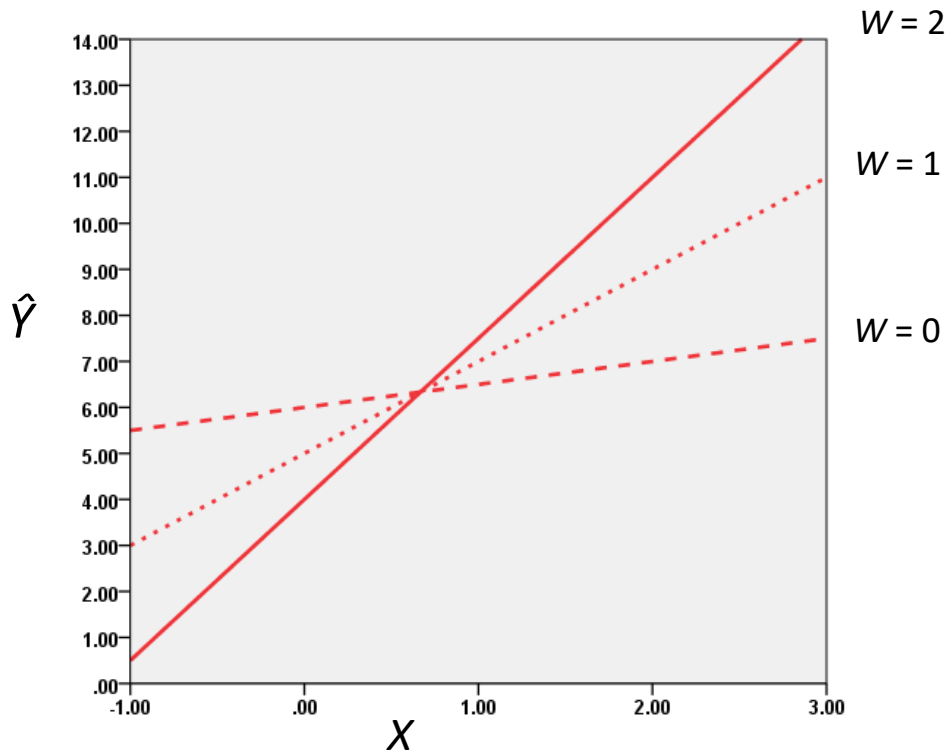
$$b_1 = 0.50$$

$$b_2 = -1.00$$

$$b_3 = 1.50$$

$$\hat{Y}_i = 6.00 + 0.50X_i - 1.00W_i + 1.50X_iW_i$$

Observe that the amount by which two cases that differ by one unit on X are estimated to differ on Y **depends on W** .



X	W	$\hat{Y}$
-1	0	5.50
-1	1	3.00
-1	2	0.50
0	0	6.00
0	1	5.00
0	2	4.00
1	0	6.50
1	1	7.00
1	2	7.50
2	0	7.00
2	1	9.00
2	2	11.00

Differences in interpretation

$$\hat{Y}_i = b_0 + b_1X_i + b_2W_i$$

b_0

The estimated value of Y when X and $W = 0$.

b_1

The effect of X on Y holding W constant. This is a *partial* effect.

b_2

The effect of W on Y holding X constant. This is a *partial* effect.

b_3

$$\hat{Y}_i = b_0 + b_1X_i + b_2W_i + b_3X_iW_i$$

The estimated value of Y when X and $W = 0$.

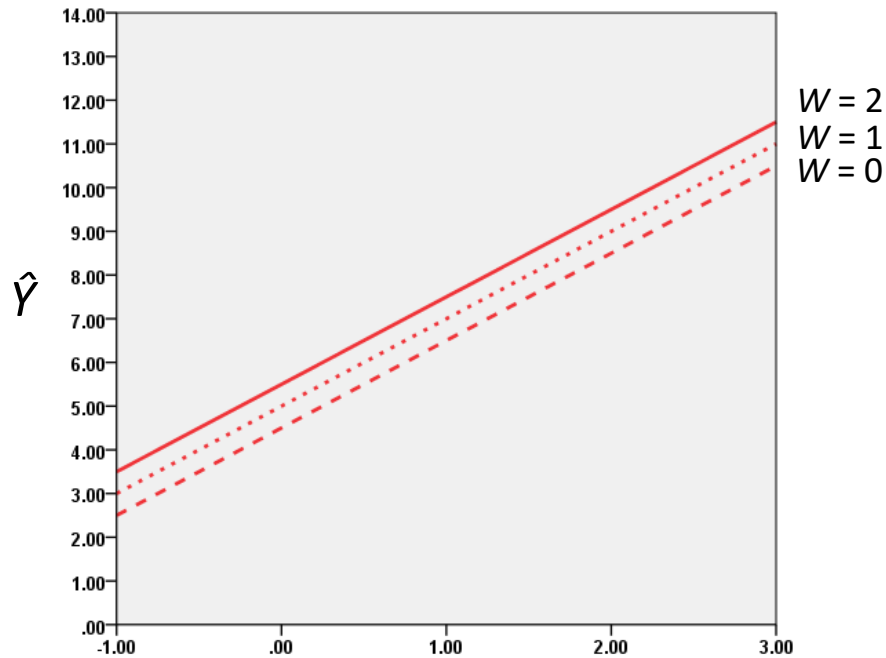
The effect of X on Y when $W = 0$. This is a *conditional* effect. It is Not a “main effect” or “average effect” of X .

The effect of W on Y when $X = 0$. This is a *conditional* effect. It is not a “main effect” or “average effect” of W .

How much the effect of X on Y changes as W changes by 1 unit.

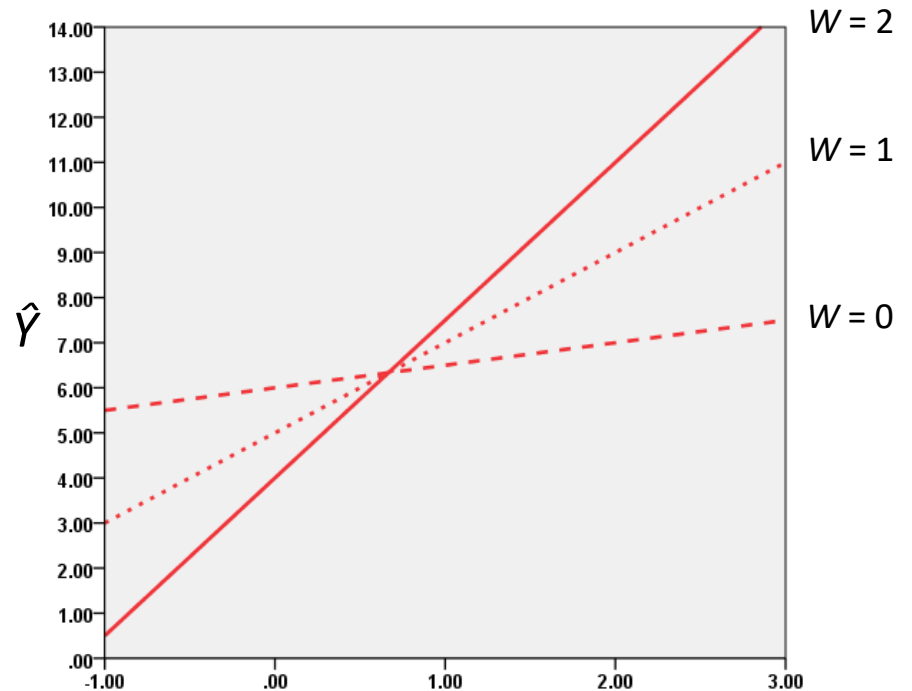
Importance of b_3

$$\hat{Y}_i = 4.50 + 2.00X_i + 0.50W_i + 0X_iW_i$$



$$\begin{aligned} \theta_{X \rightarrow Y} &= b_1 + b_3 W & \theta_{W \rightarrow Y} &= b_2 + b_3 X \\ &= 2.00 + 0W & &= 0.50 + 0X \end{aligned}$$

$$\hat{Y}_i = 6.00 + 0.50X_i - 1.00W_i + 1.50X_iW_i$$



$$\begin{aligned} \theta_{X \rightarrow Y} &= b_1 + b_3 W & \theta_{W \rightarrow Y} &= b_2 + b_3 X \\ &= 0.50 + 1.50W & &= -1.00 + 1.50X \end{aligned}$$

When $b_3 = 0$, a one unit change in X has the same effect on Y regardless of W , and a one unit change in W has the same effect on Y regardless of X . When $b_3 \neq 0$, the effect of a change in X on Y depends on W , and the effect of a change in W on Y depends on X . So we test a moderation hypothesis by testing whether b_3 is different from zero.

Probing an Interaction

The coefficient for the product term carries information about how changes in one variable are related to changes in the effect of the other. A picture helps to understand how the focal variable's effect changes as a function of the moderator variable.

It is typically desirable to conduct statistical tests of the focal predictor variable's effect at values of the moderator. This allows you to make more definitive claims about where the focal predictor variables effect is zero versus where it is not.

“Pick-a-Point” Approach

Select values of the moderator and estimate the conditional effect of the focal predictor at those values of the moderator, along with a hypothesis test or confidence interval.

Johnson-Neyman Technique

Derive mathematically where on the moderator variable continuum the focal variable's effect transitions between statistically significant and nonsignificant.

Pick-a-point approach

$$\hat{Y}_i = b_0 + b_1X_i + b_2W_i + b_3X_iW_i$$

Select a value of the moderator (W) at which you'd like to have an estimate of $\theta_{X \rightarrow Y}$, the focal predictor variable's (X) effect. Then derive its standard error. The ratio of the effect to its standard error is distributed as $t(df_{\text{residual}})$ under the null hypothesis that the effect of the focal predictor is zero at that moderator value, where df_{residual} is the residual degrees of freedom from the regression model.

We already know that

$$\theta_{X \rightarrow Y} = b_1 + b_3W$$

The estimated standard error of $\theta_{X \rightarrow Y}$ is

$$s_{\theta_{X \rightarrow Y}} = \sqrt{s_{b_1}^2 + 2W s_{b_1 b_3} + W^2 s_{b_3}^2}$$

Squared standard error of b_1

Covariance of b_1 and b_3

Squared standard error of b_3

You could do this by hand, and instructions are available in various books on regression analysis (e.g., Aiken and West, 1991; Cohen et al., 2003). But there is no reason to, and the potential for mistakes is high. It is made easier using “**regression centering**.”

The Johnson-Neyman Technique

The Johnson-Neyman technique seeks to find the value or values of the moderator (W) within the data, if they exist, such that the p -value for the ratio of the conditional effect of the focal predictor at that value or values of W is exactly equal to some chosen level of significance α

To do so, we ask what value of W produces a ratio exactly equal to the critical t value (t_{crit}) required to reject the null hypothesis that the conditional effect of X is equal to zero?

$$t_{crit} = \frac{b_1 + b_3 W}{\sqrt{s_{b_1}^2 + 2W s_{b_1 b_3}^2 + W^2 s_{b_3}^2}}$$

Isolate W and solve the polynomial that results. The quadratic formula finds the solutions:

$$W = \frac{-2(t_{crit}^2 s_{b_1 b_3}^2 - b_1 b_3) \pm \sqrt{(2t_{crit}^2 s_{b_1 b_3}^2 - 2b_1 b_3)^2 - 4(t_{crit}^2 s_{b_3}^2 - b_3^2)(t_{crit}^2 s_{b_1}^2 - b_1^2)}}{2(t_{crit}^2 s_{b_3}^2 - b_3^2)}$$

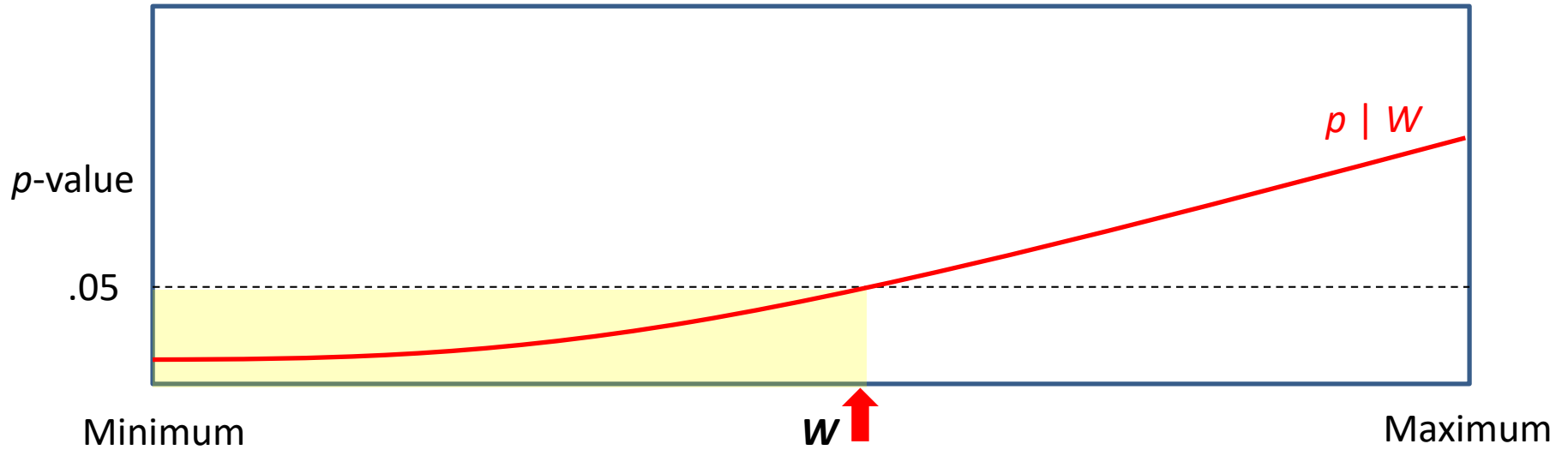
The Johnson-Neyman Technique

This will produce no values, one value, or two values of W that are within the range of the moderator variable data.

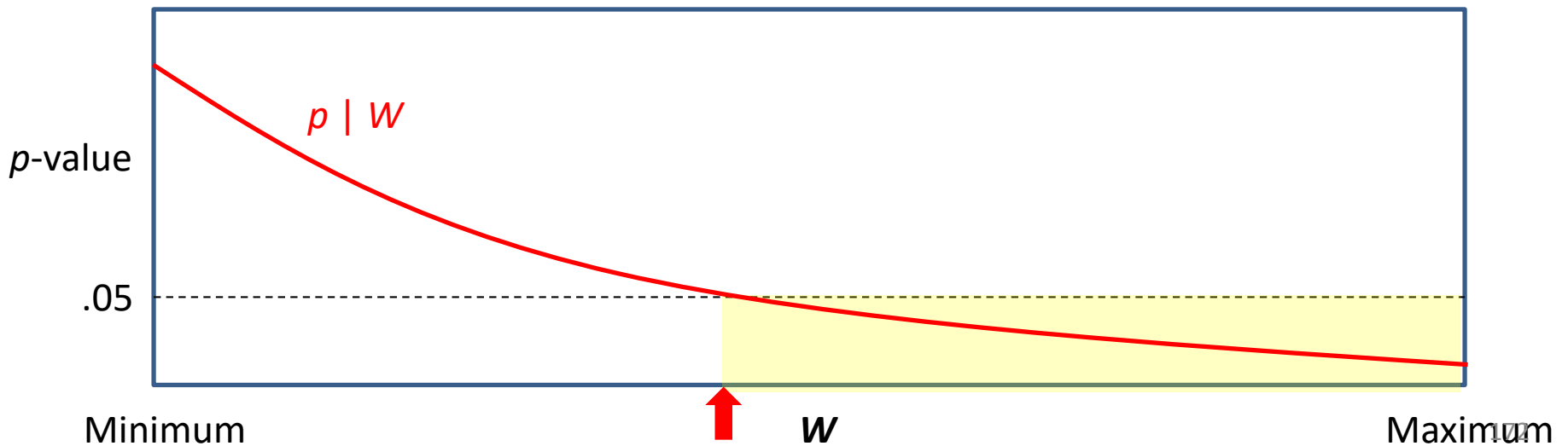
- If one value, this defines a single point of transition between a statistically significant and a statistically nonsignificant conditional effect of the focal predictor, such that $p \leq .05$ for either values of the moderator (1) equal to above W or (2) equal to and below W .
- If two values, this defines the two points of transition between a statistically significant and a statistically nonsignificant conditional effect of the focal predictor, such that the conditional effect is statistically significant for either (1) values of the moderator between the two values of W , or (2) values of the moderator at least as large as the larger W and at least as small as the smaller W .
- If no values, that means the conditional effect is statistically significant for ALL values of the moderator within the range of the data, or it NEVER is.

We would not attempt to do this by hand

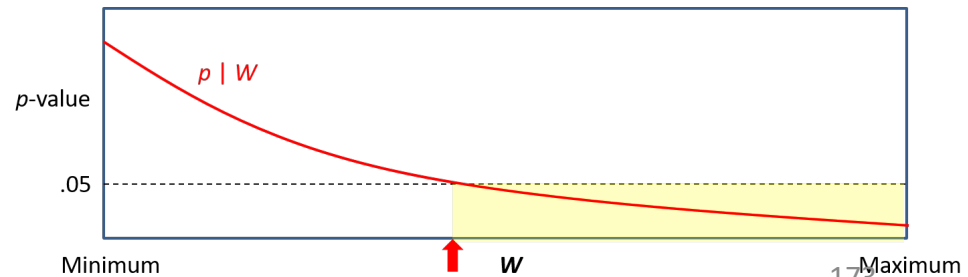
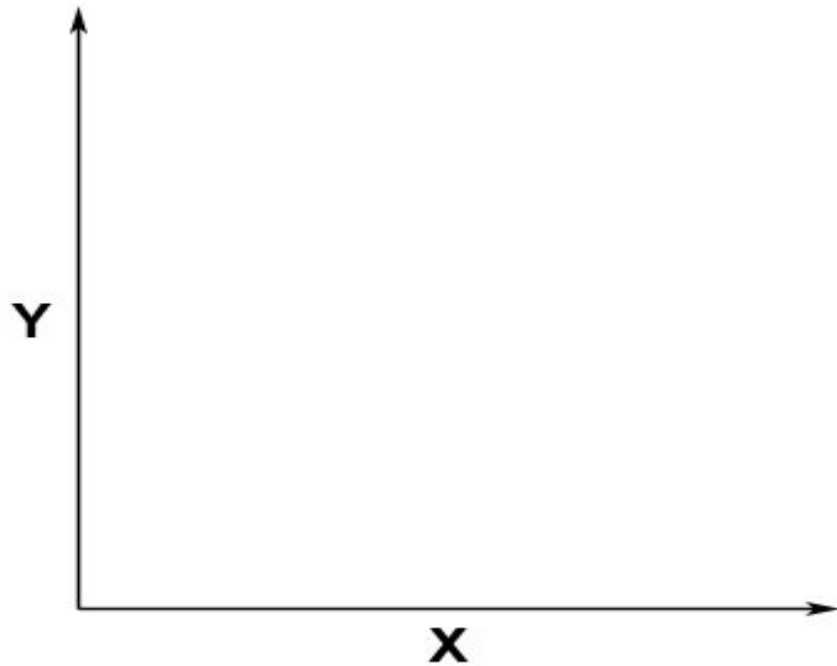
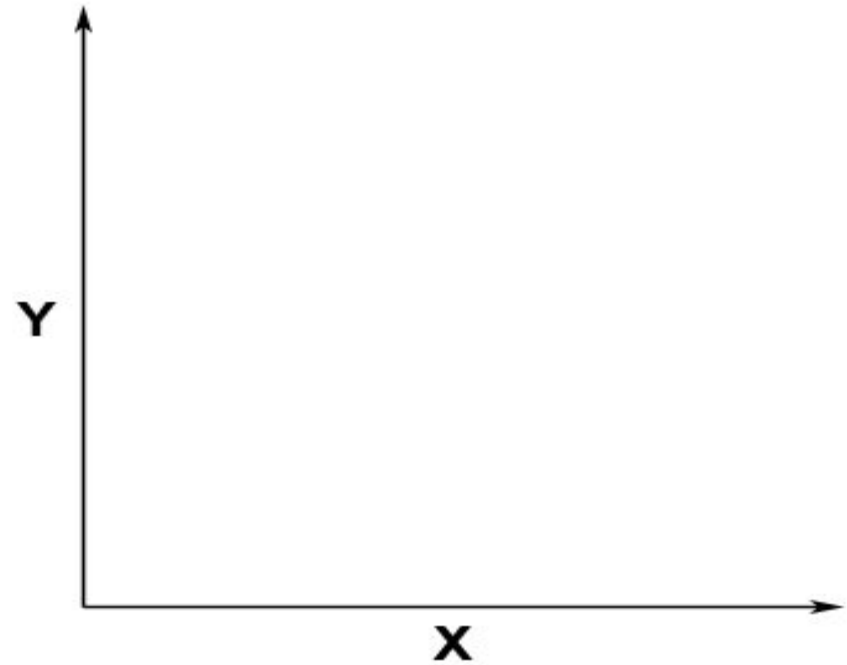
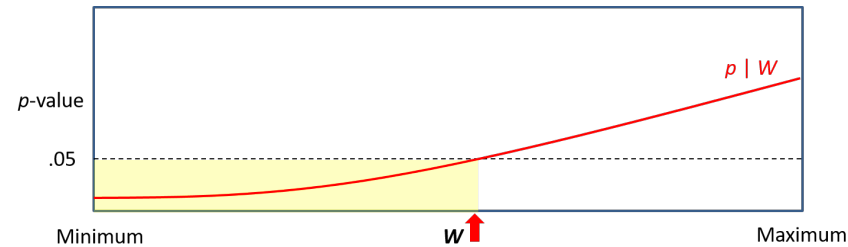
Examples of One Solution



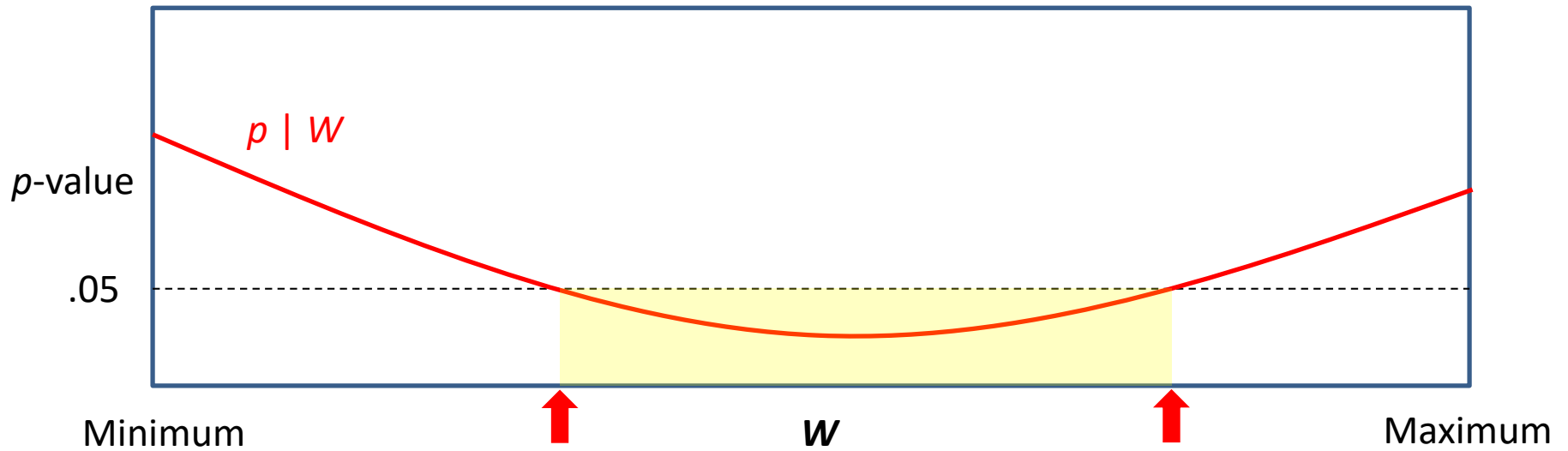
OR



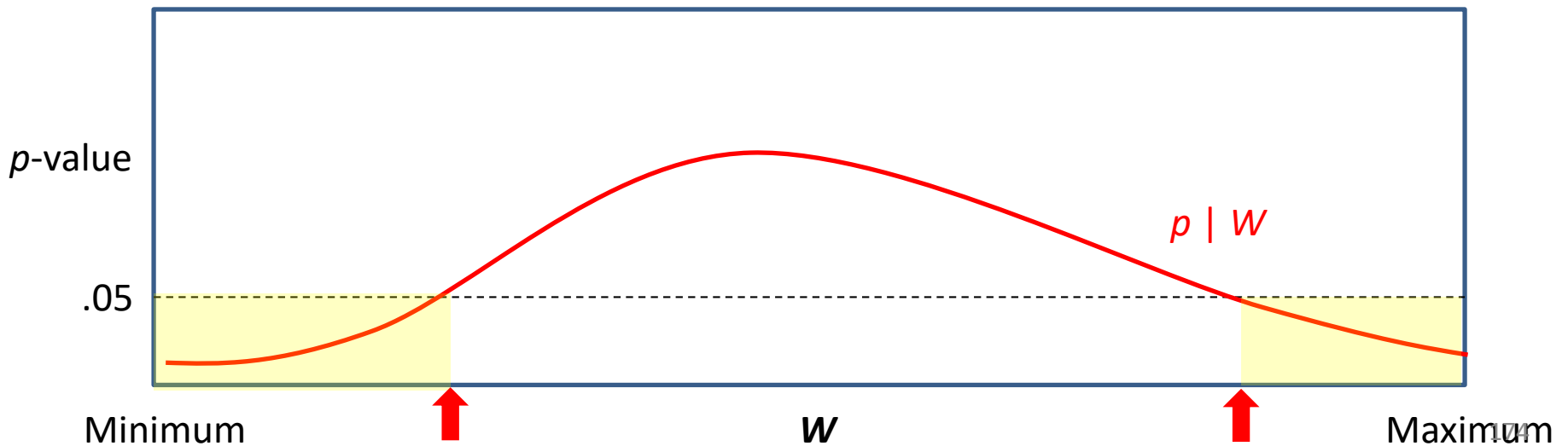
One Solution: What would the graph look like?



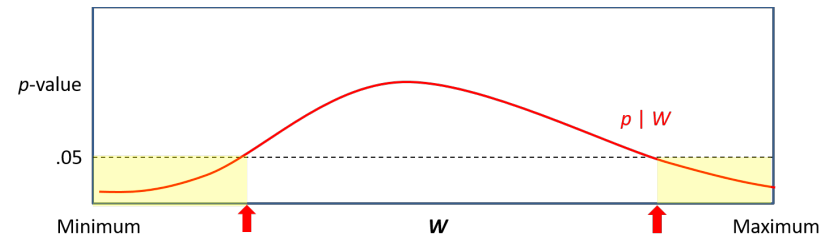
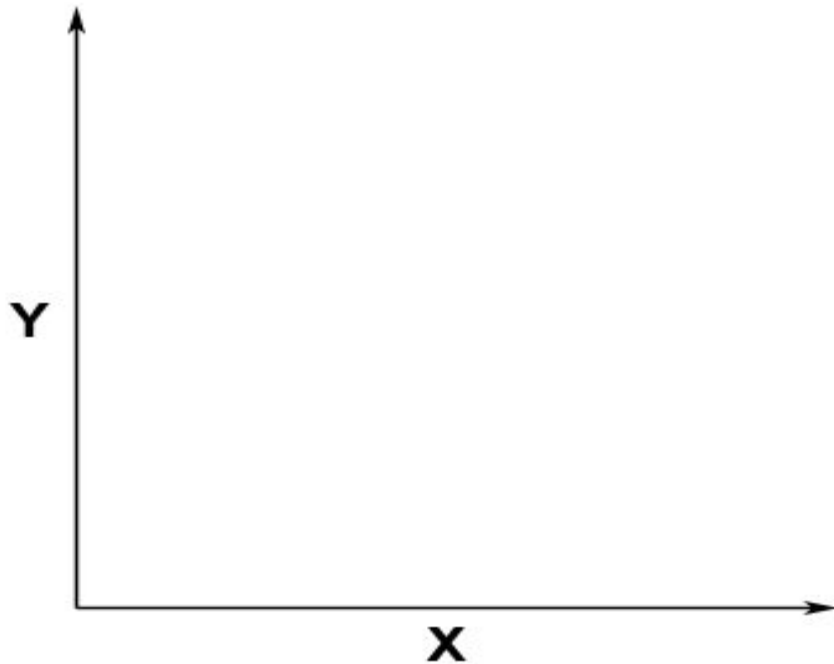
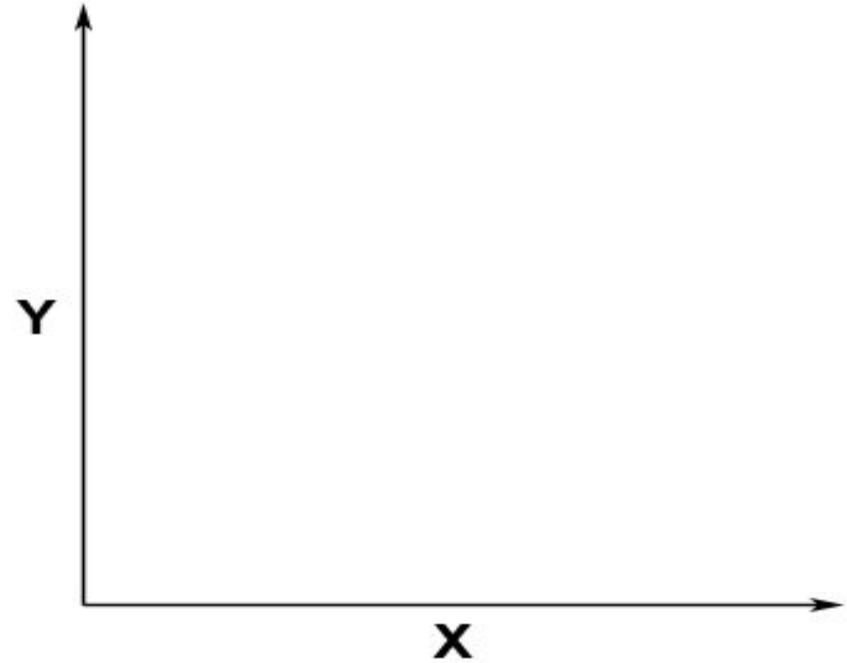
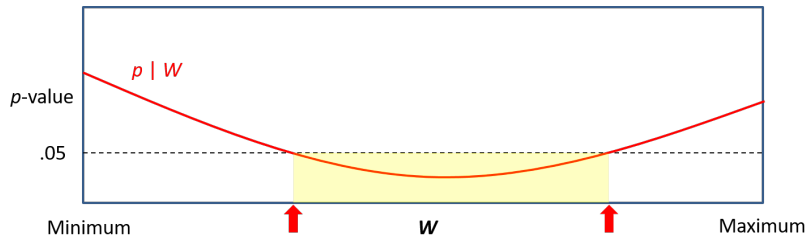
Examples of two solutions



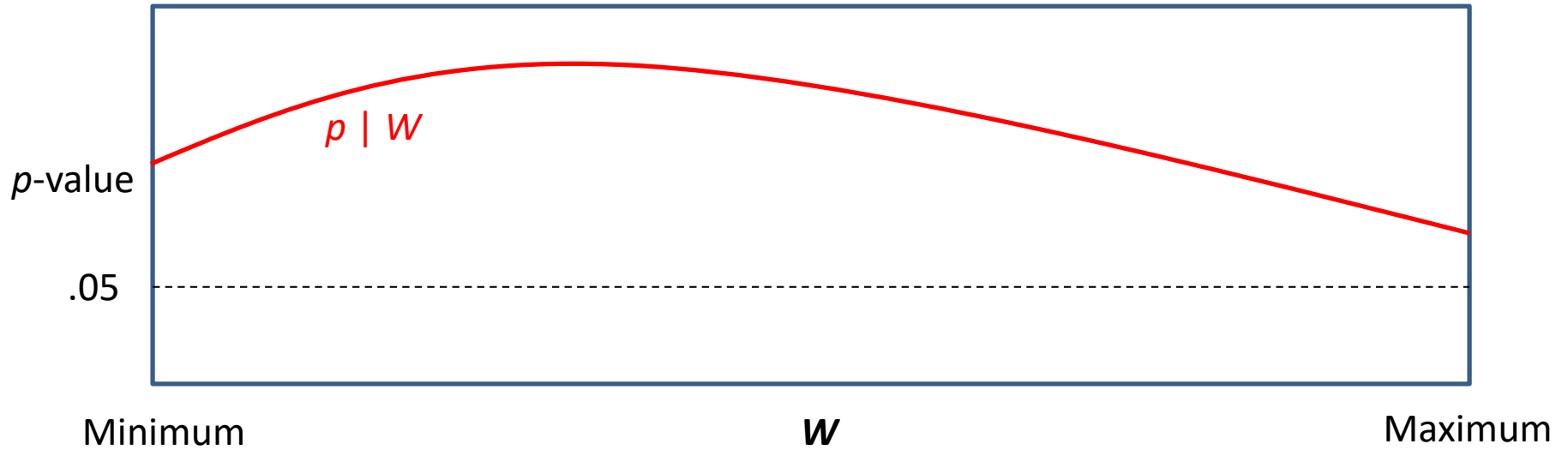
OR



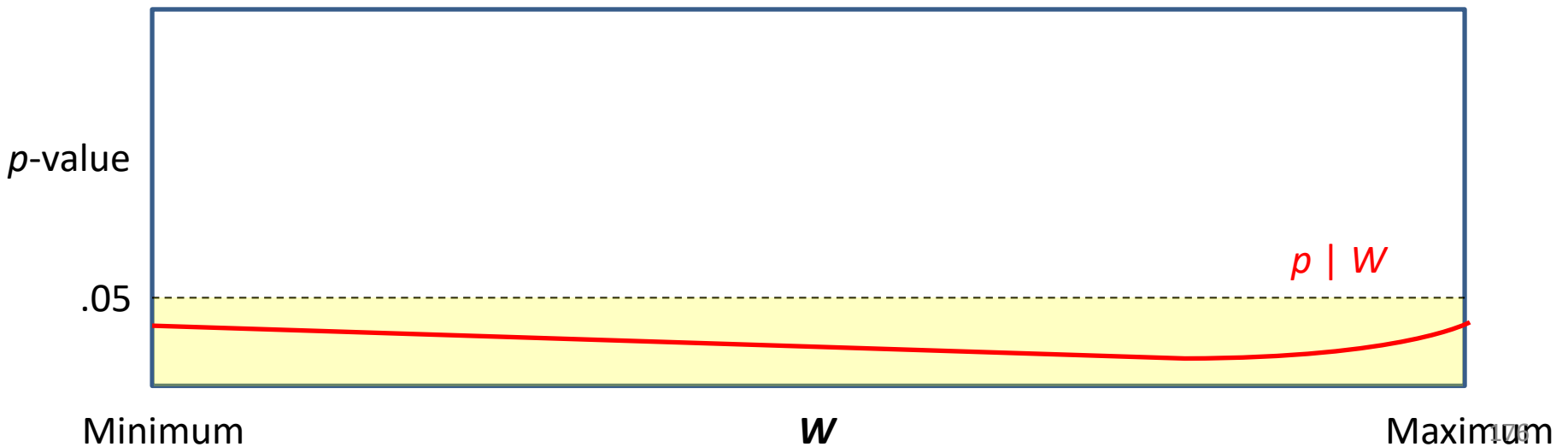
Two Solutions: What would the graph look like?



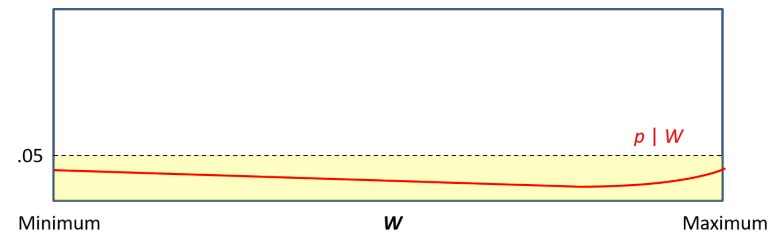
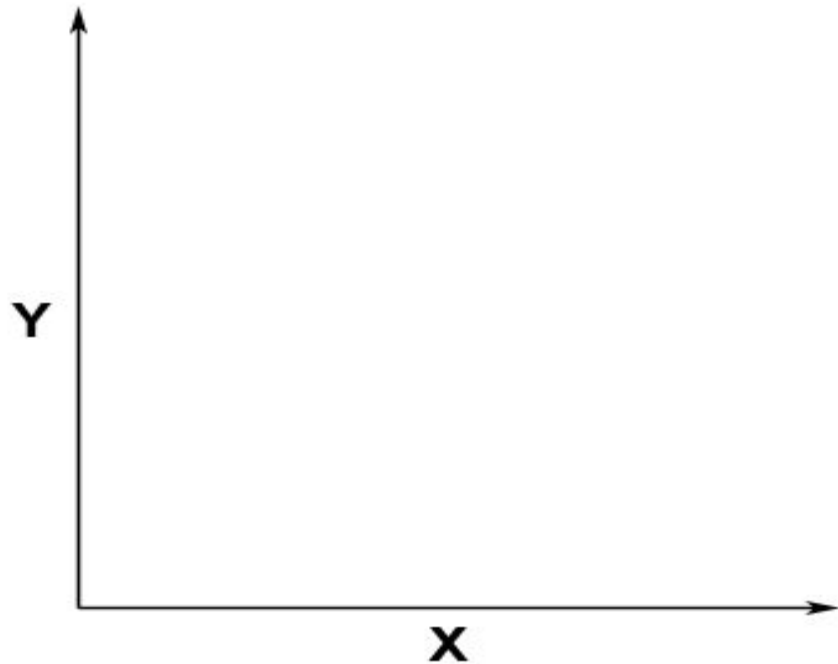
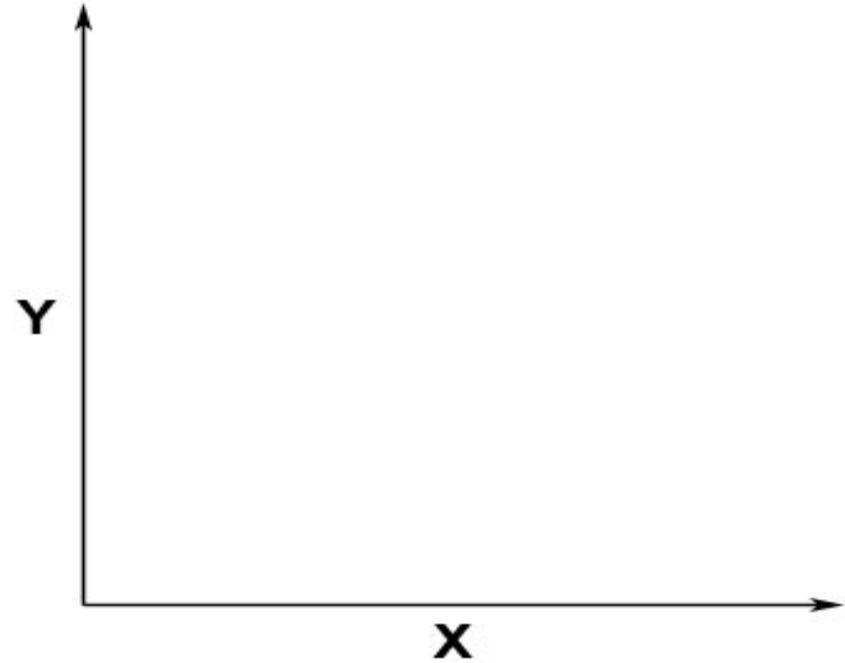
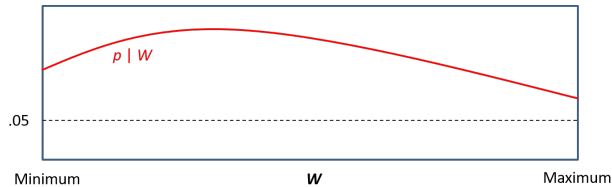
Examples of No Solutions



OR



No Solution: What would the graph look like?



MODERATION IN WITHIN-SUBJECT DESIGNS

Running Example: Group Work in Computer Science (WS)

Montoya, A. K. (2013) Increasing Interest in Computer Science through Group Work: A Goal Congruity Approach (Undergraduate Honors Thesis).

Within-Subjects Version (CompSci_WS.sav, CompSci_WS.sas) :

Female participants (N = 51) read two syllabi for a different computer science classes. One of the syllabi reported the class would have group projects throughout, and the other syllabi stated that individual project would be scheduled throughout.

- Syllabi also differed in professor's name (but not gender), and the primary programming language used in the class.

Measured Variables:

- Interest in each the class `int_i` `int_g`
- `Perscom` Personal Communal Goals ($\alpha = .87$)
- `Order`
 - 1 = Group First; 2 = Individual First

Modeling Non-Contingent Relationships

When we consider non-contingent relationships in a repeated-measures design, this means the relationship between a variable (W) and the outcome (Y) is the same across conditions.

$$Y_{1i} = b_{10} + b_1 W_i + \epsilon_{1i}$$

Example:

$$Y_{2i} = b_{20} + b_1 W_i + \epsilon_{2i}$$

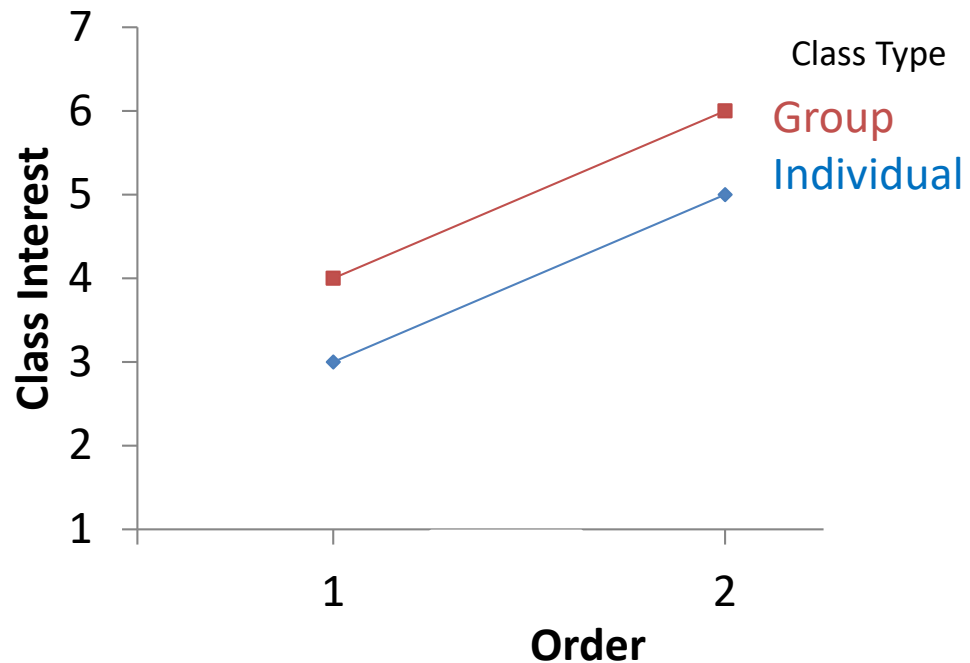
Y_1 : Interest in Individual Work Class (1-7)

Y_2 : Interest in Group Work Class

W : Order (1 = Group First, 2 = Individual First)

$\hat{Y}_1$	$\hat{Y}_2$	W
3	4	1
5	6	2

A one unit increase in order results in a 2 unit increase in interest, regardless of condition.



Modeling Contingent Relationships

What if instead we felt that the relationship between Order and Interest depends on condition? Thus the relationship between Order and Interest *differs* across the two conditions

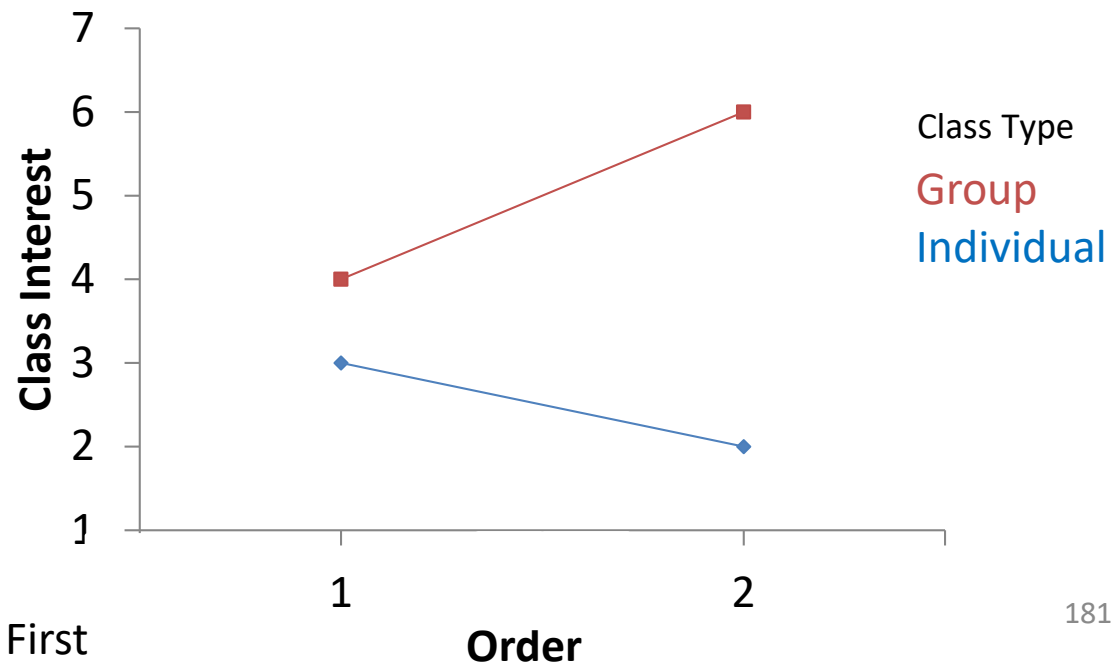
$$Y_{1i} = b_{10} + b_{11}W_i + e_{1i}$$

$$Y_{2i} = b_{20} + b_{21}W_i + e_{2i}$$

$$Y_{2i} - Y_{1i} = (b_{10} - b_{20}) + (b_{11} - b_{21})W_i + (e_{1i} - e_{2i}) = b_0 + b_1W_i + e_i$$

The difference between b_{11} and b_{21} tells us how much the relationship between W and Y differs across conditions. So b_1 tells us if there is moderation.

$\hat{Y}_1$	$\hat{Y}_2$	W
3	4	1
2	6	2



1 = Group First, 2 = Individual First

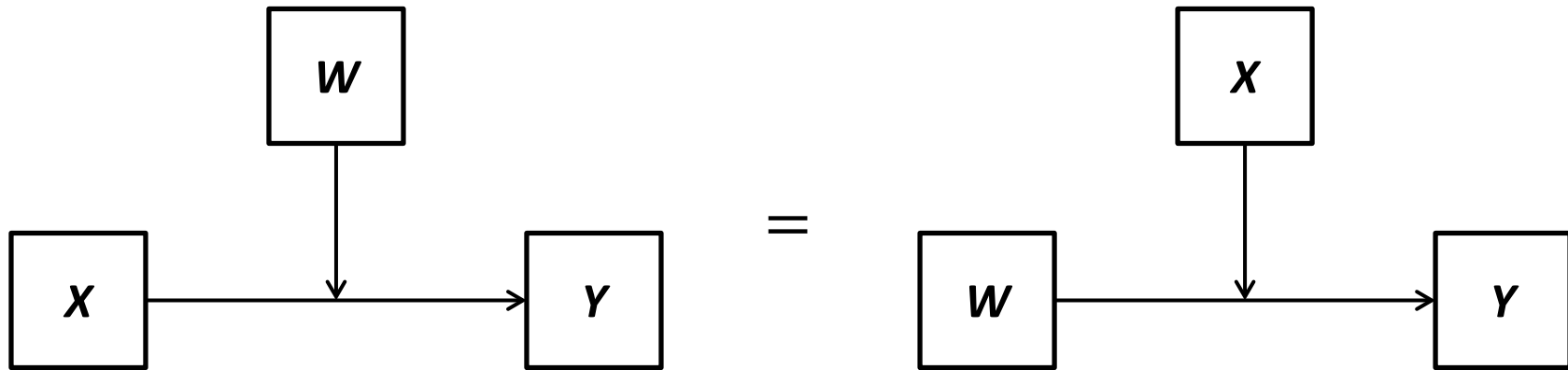
Symmetry in Within-Subjects Moderation

Does the effect of condition depend on W ?

$$Y_{2i} - Y_{1i} = (b_{10} - b_{20}) + (b_{11} - b_{21})W_i + (e_{1i} - e_{2i}) = b_0 + b_1M_i + \epsilon_i$$

$Y_{2i} - Y_{1i}$ is a quantification of the effect of condition, which means that if W predicts $Y_{2i} - Y_{1i}$ then the effect of condition depends on W .

b_1 is a test of exactly that!



Judd, McClelland, and Smith (1996)

Judd, C. M., McClelland, G. H., and Smith, E. R. (1996). Testing Treatment by Covariate Interactions When Treatment Varies Within Subjects. *Psychological Methods*, 1(4), 366-378.

A regression approach to considering a “cross level” interactions.

Approach is very simple:

1. Data should be a two-condition within-subjects design with a person level covariate.
2. Setup two regression equations, one for each condition
3. Take the difference between those two regression equations
4. Regression weight for person level covariate in Step 3 tests moderation.



Computer Science Within-Subjects Data Example

CompSci_WS.sav

Montoya, A. K. (2013) Increasing Interest in Computer Science through Group Work: A Goal Congruity Approach (Undergraduate Thesis).

1. Data should be a two-condition within-subjects design with a person level covariate.

Research Question: Does the degree to which class order predicts interest in computer science depend on whether the class has group work or not?

Or

Does effect of group work on interest in computer science classes depend on an the order they read the syllabi?

	Subject	int_I	int_G	Order	grp_PREF
1	300	1.50	4.00	1	6.67
2	301	2.75	3.25	1	6.33
3	325	5.75	2.50	1	2.67
4	342	3.50	5.75	1	6.00
5	349	2.25	2.00	1	4.00
6	350	1.50	1.75	1	3.67
7	305	2.50	4.25	1	4.00
8	348	6.00	1.75	1	2.33
9	318	3.00	2.00	1	4.67
10	320	4.00	5.25	1	4.00
11	332	5.00	5.00	1	3.67
12	338	2.00	1.75	1	3.00
13	310	1.00	1.75	1	3.00
14	304	1.25	4.50	2	5.67
15	306	5.75	4.50	2	4.00
16	308	3.25	4.75	2	4.00
17	315	2.75	2.25	2	4.33
18	322	5.50	2.00	2	2.33
19	343	1.75	5.25	2	6.00
20	314	4.00	5.50	2	3.00
21	319	2.25	4.00	2	5.00
22	330	4.00	6.50	2	5.67
23	334	5.00	4.50	2	3.33
24	309	5.00	3.75	2	1.00
25	329	4.75	5.25	2	4.00
26	333	1.75	5.25	2	6.33
27	336	4.50	2.25	2	3.67
28	341	1.00	3.75	2	4.33
29	302	1.75	1.75	2	4.00

Analysis using Judd et al. (1996)

2. Setup two regression equations, one for each condition

Setup a model of the outcome in each condition:

$$Y_{1i} = b_{10} + b_{11}W_i + e_{1i}$$

$$Y_{2i} = b_{20} + b_{21}W_i + e_{2i}$$

Is b_{11} different from b_{21} ?

3. Based on these models, setup a new model where you can directly estimate and conduct inference on what you are interested in (in this case $b_{11} - b_{21}$):

$$Y_{2i} - Y_{1i} = (b_{20} - b_{10}) + (b_{21} - b_{11})W_i + (e_{2i} - e_{1i}) = b_0 + b_1W_i + e_i$$

Use simple regression to conduct inference on $b_1 = b_{11} - b_{21}$

With the data: Does the relationship between order and interest depend on group work condition?

```
regression /dep = int_diff /method = enter order.
```

```
proc reg data=CompSci_WS;model int_diff=order;run;
```

```
summary(lm(int_diff~Order, data = CompSci_WS))
```



What sign do you expect b_1 to be? **Remember:** $\text{int_diff} = \text{int_G} - \text{int_i}$.

Analysis using Judd et al. (1996)

4. Regression weight for person level covariate in Step 3 tests moderation.

$$Y_{2i} = b_{20} + b_{21}W_i + e_{2i}$$

$$Y_{1i} = b_{10} + b_{11}W_i + e_{1i}$$

$$Y_{2i} - Y_{1i} = (b_{20} - b_{10}) + (b_{21} - b_{11})W_i + (e_{2i} - e_{1i}) = b_0 + b_1W_i + e_i$$

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-1.476	.880		-1.676	.100
	Order	1.193	.541	.300	2.205	.032

a. Dependent Variable: int_diff



What does it mean that b_1 is positive?

$$b_1 = b_{21} - b_{11} = 1.193$$

$$b_{21} > b_{11}$$

Practically, this means that the relationship between order and interest is significantly stronger (more positive) in the group work condition.

```
regression /dep = int_diff /method = enter order.
```

```
proc reg data=CompSci_WS;model int_diff=order;run;
```

```
summary(lm(int_diff~Order, data = CompSci_WS))
```

Interpreting the Coefficients

$$Y_{2i} - Y_{1i} = (b_{20} - b_{10}) + (b_{21} - b_{11})W_i + (e_{2i} - e_{1i}) = b_0 + b_1W_i + e_i$$

b_0 is the expected difference in Y when $W = 0$

We can think of this as the effect of “condition” on Y when W is zero.

In the Computer Science example, W can only be 1 or 2, so we do not interpret this parameter in this case.

b_1 is the degree to which the relationship between W and Y differs by condition.

Alternatively: the degree to which the effect of condition on Y depends on W . *i.e.*, if W increases by one unit the effect of condition on Y will increase by b_1 units

Conditional Effects in Within-Subjects Moderation

$$Y_{2i} - Y_{1i} = (b_{20} - b_{10}) + (b_{21} - b_{11})W_i + (e_{2i} - e_{1i}) = b_0 + b_1W_i + e_i$$

Given a value of W what is the effect of condition on the outcome?

$Y_{2i} - Y_{1i}$ is a quantification of the effect of condition, which means that the conditional effect of condition $\theta_{X \rightarrow Y}(W) = b_0 + b_1W$

Given a specific condition what is the effect of W on the outcome?

$$Y_{1i} = b_{10} + b_{11}W_i + e_{1i}$$

$$Y_{2i} = b_{20} + b_{21}W_i + e_{2i}$$

$$\theta_{W \rightarrow Y}(X) = b_{x1}$$

Conditional effects will become important when it comes to probing

Probing an Effect of Condition on Outcome: The “Pick-a-Point” Approach

$$\theta_{X \rightarrow Y}(W) = b_0 + b_1 W$$

Select a value of the moderator (W) at which you'd like to have an estimate of the condition's effect on Y . Then derive its standard error. The ratio of the effect to its standard error is distributed as $t(df_{residual})$ under the null hypothesis that the effect of condition is zero at that moderator value.

The estimated standard error of $\theta_{X \rightarrow Y}(W)$ is

$$s_{\theta_{X \rightarrow Y}(W)} = \sqrt{(s_{b_0}^2 + 2W s_{b_0 b_1} + W^2 s_{b_1}^2)}$$

Squared standard error of b_0

Covariance of b_0 and b_1

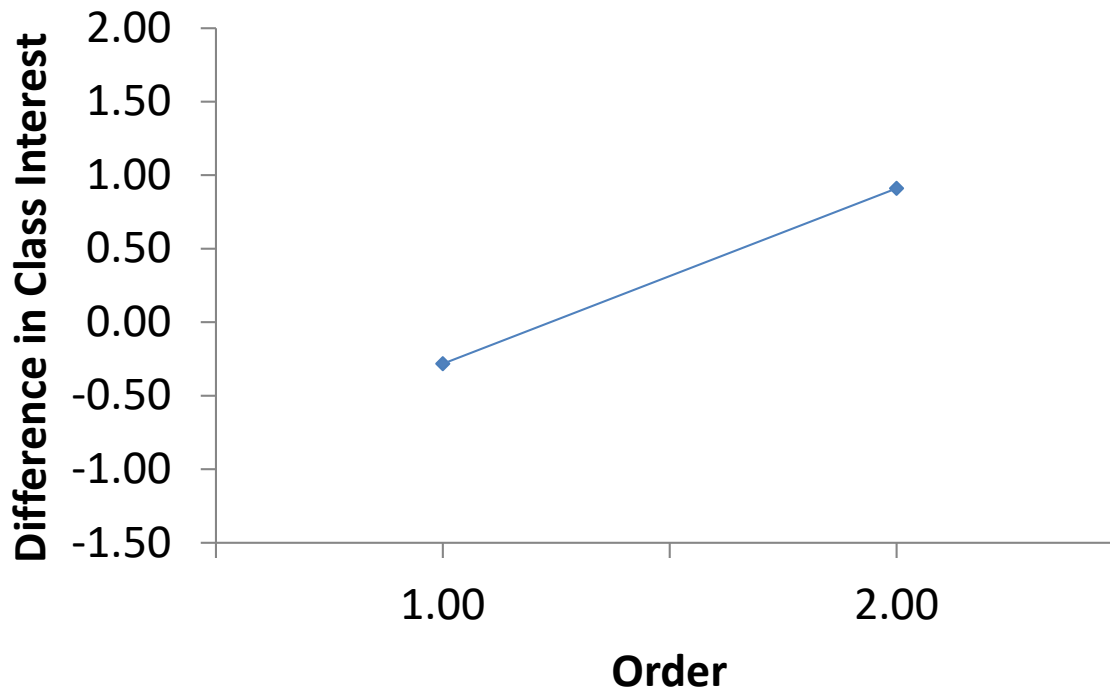
Squared standard error of b_1

Probing an Effect of Condition on Outcome: The “Pick-a-Point” Approach

You must choose the points along the moderator to “probe” the effect of condition on Y .

Let’s look at an example with our computer science data:

$$Y_{Di} = -1.476 + 1.193W_i$$



W	$\theta_{X \rightarrow Y W}$	$s\theta_{X \rightarrow Y W}$	p
1	-.2826	0.4010	.4843
2	.9107	.3634	.0156

Participants who saw the group work class first did not show a difference in interest between the two classes. However, those who saw the individual work class first showed a larger effect of condition such students were significantly more interested in the group class.

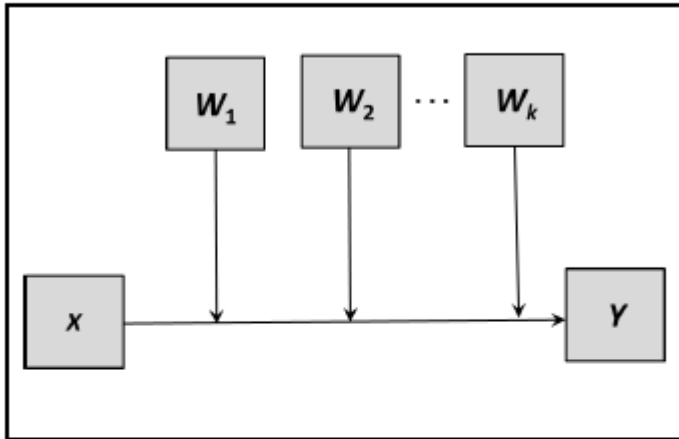
MEMORE

We can use MEMORE to estimate and probe this model.

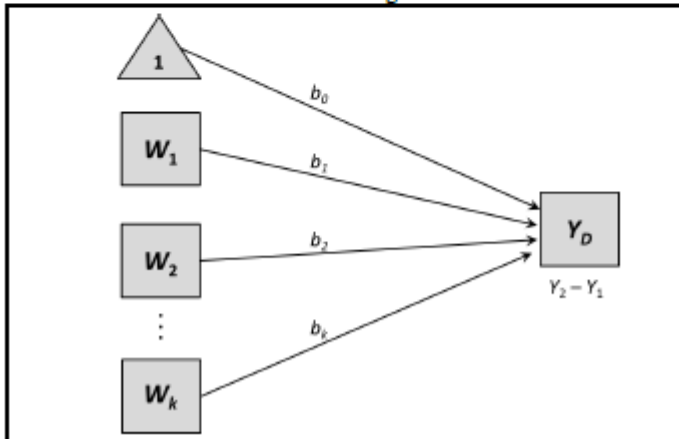
Model Templates for MEMORE V3.Beta

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Model 2 Additive Moderation
Conceptual Diagram



Statistical Diagram



Subscript k indicates moderators. Model can have up to 5 moderators

```
MEMORE w = order /y = int_G int_I  
/model = 2 /plot = 1.
```

```
%memore(w=order,y = int_G int_I,  
model = 2, plot = 1, data =  
CompSci_WS);
```

- List moderator(s) in the w list
- List outcomes in the y list
- Can use `model 2` or `model 3` when you have 1 moderator there is no difference.
- `PLOT` option calls a table of values for making a nice plot.

Using MEMORE for CASC WS data

```
MEMORE w = order /y = int_G int_I /model = 2 /plot = 1.
```

```
%memore(w=order,y = int_G int_I, model = 2, plot = 1, data =  
CompSci_WS) ;
```

```
***** MEMORE Procedure for SPSS Version 2.1 *****
```

Written by Amanda Montoya

Documentation available at akmontoya.com

```
*****
```

Model:

2

Variables:

Y = int_G int_I

W = Order

Computed Variables:

Ydiff = int_G - int_I

Sample Size:

51

First part of output repeats what you told MEMORE to do. Always double check that this is correct!

I double checked to make sure the order of subtraction was the same as when we did this by hand.

Using MEMORE for CASC WS data

```
MEMORE w = order /y = int_G int_I /model = 2 /plot = 1.
```

```
%memore(w=order,y = int_G int_I, model = 2, plot = 1, data =  
CompSci_WS);
```

```
Outcome: Ydiff = int_G - int_I
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.3005	.0903	3.6978	4.8629	1.0000	49.0000	.0322

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	-1.4759	.8804	-1.6764	.1000	-3.2452	.2934
Order	1.1933	.5411	2.2052	.0322	.1058	2.2808

```
Degrees of freedom for all regression coefficient estimates:
```

```
49
```

Regression results are the same as when we did this
using regression command

Using MEMORE for CASC WS data

```
MEMORE w = order /y = int_G int_I /model = 2 /plot = 1.
```

```
%memore(w=order,y = int_G int_I, model = 2, plot = 1, data =  
CompSci_WS) ;
```

Probing effect of condition on outcome at different values of the moderator

Conditional Effect of 'X' on Y at values of moderator(s)

Order	Effect	SE	t	p	LLCI	ULCI
1.0000	-.2826	.4010	-.7048	.4843	-1.0884	.5232
2.0000	.9107	.3634	2.5061	.0156	.1804	1.6410

Degrees of freedom for all conditional effects:

49

Values for quantitative moderators are the mean and plus/minus one SD from the mean.

Values for dichotomous moderators are the two values of the moderator.

Using MEMORE for CASC WS data

```
MEMORE w = order /y = int_G int_I /model = 2 /plot = 1.
```

```
%memore(w=order,y = int_G int_I, model = 2, plot = 1, data = CompSci_WS);
```

Conditional Effect of Moderator(s) on Y in each Condition

Condition 1 Outcome:

int_G

Model Summary

R	R-sq	MSE	F	df1	df2	p
.3766	.1418	1.9306	8.0962	1.0000	49.0000	.0065

Model

	coeff	SE	t	p	LLCI	ULCI
constant	2.0070	.6362	3.1548	.0027	.7285	3.2854
Order	1.1126	.3910	2.8454	.0065	.3268	1.8984

Degrees of freedom for all conditional effects:

49

Condition 2 Outcome:

int_I

Model Summary

R	R-sq	MSE	F	df1	df2	p
.0281	.0008	2.1189	.0389	1.0000	49.0000	.8446

Model

	coeff	SE	t	p	LLCI	ULCI
constant	3.4829	.6665	5.2260	.0000	2.1436	4.8222
Order	-.0807	.4096	-.1971	.8446	-.9039	.7425

Degrees of freedom for all conditional effects:

49

Order positively predicts
interest in **class with group**
work

and does not significantly
predict interest in **class with**
individual work.

Using MEMORE for CASC WS data

```
MEMORE w = order /y = int_G int_I /model = 2 /plot = 1.
```

```
%memore(w=order,y = int_G int_I, model = 2, plot = 1, data =  
CompSci_WS) ;
```

Data for visualizing conditional effect of X on Y.

Paste text below into a SPSS syntax window and execute to produce plot.

```
DATA LIST FREE/order YdiffHAT int_GHAT int_IHAT.
```

```
BEGIN DATA.
```

1.0000	-.2826	3.1196	3.4022
2.0000	.9107	4.2321	3.3214

```
END DATA.
```

```
GRAPH/SCATTERPLOT = order WITH YdiffHAT.
```

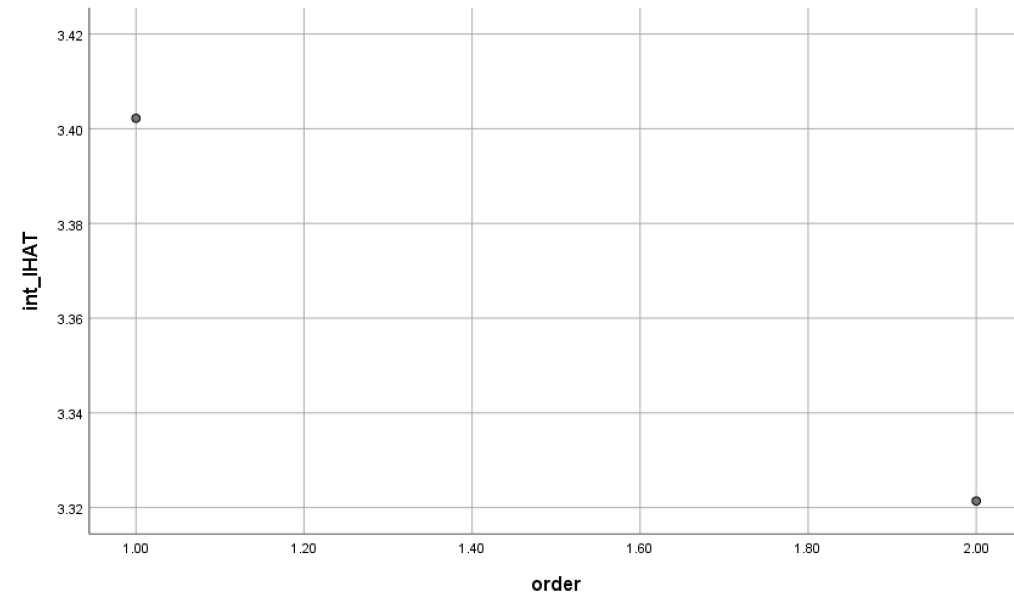
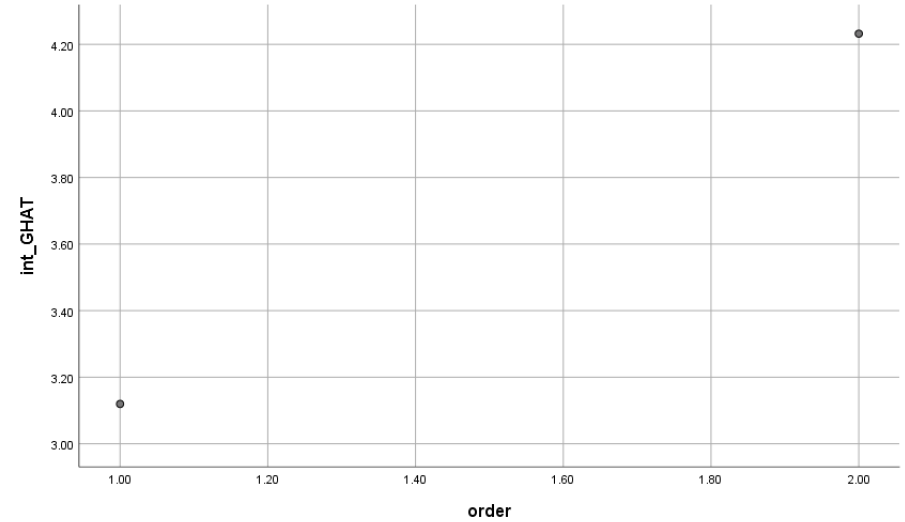
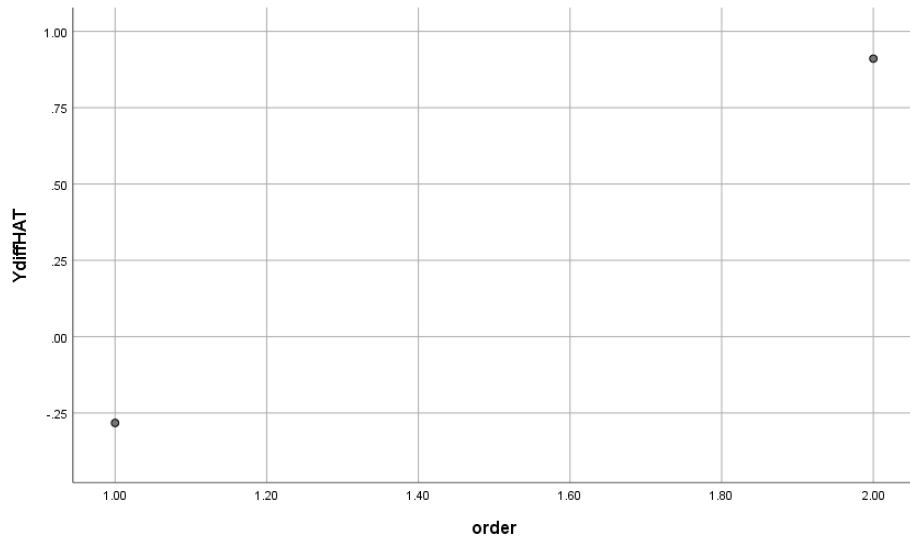
```
GRAPH/SCATTERPLOT = order WITH int_GHAT.
```

```
GRAPH/SCATTERPLOT = order WITH int_IHAT.
```

Code for plotting. You'll get three plots each with the moderator on the X axis and a different outcome on the Y axis.

- 1) Predicted Differences between Y's
- 2) Predicted Y from first condition
- 3) Predicted Y from second condition

SPSS Graphs



You can snaz these up in SPSS or export the data to something more fit for preparing graphs (e.g., R or Excel)

Using MEMORE for CASC WS data

```
MEMORE w = order /y = int_G int_I /model = 2 /plot = 1.
```

```
%memore(w=order,y = int_G int_I, model = 2, plot = 1, data =  
CompSci_WS);
```

Data for visualizing conditional effect of 'X' on Y.

	ORDER	Ydiff	INT_G	INT_I
	1.0000	-0.2826	3.1196	3.4022
	2.0000	0.9107	4.2321	3.3214

Only table with values is produced

```
data;  
input order Ydiff int_G int_I;  
datalines;  
1 -0.2826 3.1196 3.4022  
2 0.9107 4.2321 3.3214  
run;  
proc sgplot; reg x=order y=Ydiff; run;  
proc sgplot; reg x=order y=int_G; run;  
proc sgplot; reg x=order y=int_I; run;
```

Writing Up Results

Tips:

- Interpret the sign and the magnitude of the interaction coefficient with respect to X 's effect on Y (or M 's effect on Y ; or both).
- Provide probing results with interpretations
- Read the write ups of other's moderation analyses
- Provide a graphical representation of the effect of interest (like the ones we've done)

Does the effect of group work on interest in a computer science class depend on order of syllabus presentation?

Overall, the impact of including group work in a computer science class on interest in the class depends on the order that students read the syllabus ($b_1 = 1.19, p = .001$). Among those who read the individual work syllabus first, we observed a 1.19 unit larger difference between interest in group work and interest in individual work classes. Among those who read the group work syllabus first, they did not significantly differ on their interest in the two classes ($\theta_{X \rightarrow Y|W} = -.283, p = .48$). But among those who read the individual work syllabus first, they were significantly more interested in the group work class ($\theta_{X \rightarrow Y|W} = .9107, p = .0156$). Considering the interaction another way, this result shows that order predicts interest differently across the conditions. Those who read the individual work syllabus first were significantly higher on interest in the group work class than those who read the group work syllabus first ($\theta_{W \rightarrow Y|X} = 1.1126, p = .0065$); whereas, order did not significantly predict interest in the individual work class ($\theta_{W \rightarrow Y|X} = -.0807, p = .8446$). Overall, this suggests that there may be some unique aspect of reading about the individual work class first, and then the group work class which is driving differences in interest between the two conditions. It is worth considering whether it is ecologically valid to rely on order of presentation occurring in one way versus another, and leads to many limitations of the utility of introducing group work into computer science classes as an effective method for recruiting and retaining women.

Computer Science Within-Subjects Data Example

Montoya, A. K. (2013) Increasing Interest in Computer Science through Group Work: A Goal Congruity Approach (Undergraduate Thesis).

1. Data should be a two-condition within-subjects design with a person level covariate.

Research Question: Does the degree to which preference for group work predicts interest in computer science depend on whether or not the class has group work?

Or

Does effect of group work on interest in computer science classes depend on an individual's preference for group work?

CompSci_WS.sav

Subject	int_I	int_G	grppref
300	1.50	4.00	6.67
301	2.75	3.25	6.33
325	5.75	2.50	2.67
342	3.50	5.75	6.00
349	2.25	2.00	4.00
350	1.50	1.75	3.67
305	2.50	4.25	4.00
348	6.00	1.75	2.33
318	3.00	2.00	4.67
320	4.00	5.25	4.00
332	5.00	5.00	3.67
338	2.00	1.75	3.00
310	1.00	1.75	3.00
304	1.25	4.50	5.67
306	5.75	4.50	4.00
308	3.25	4.75	4.00
315	2.75	2.25	4.33
322	5.50	2.00	2.33
343	1.75	5.25	6.00
314	4.00	5.50	3.00
319	2.25	4.00	5.00

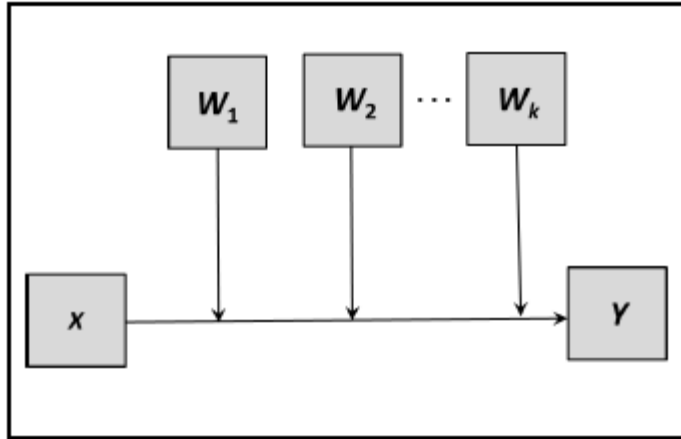
MEMORE

We can use MEMORE to estimate and probe this model.

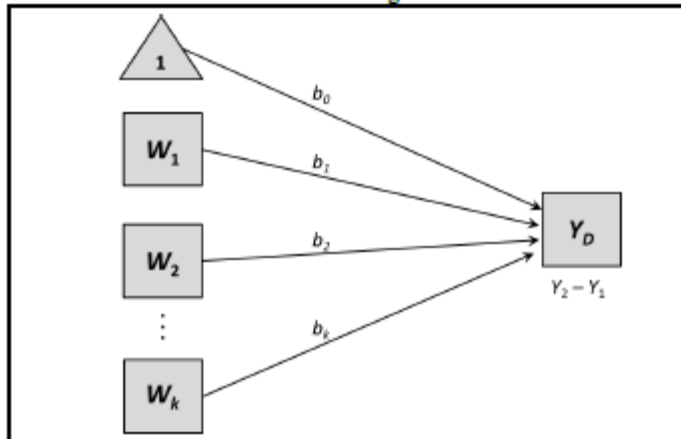
Model Templates for MEMORE V3.Beta

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Model 2 Additive Moderation
Conceptual Diagram



Statistical Diagram



Subscript k indicates moderators. Model can have up to 5 moderators

```
MEMORE w = grppref /y = int_G int_I  
/model = 2/jn = 1 /plot = 1.
```

```
%memore(w=grppref,y = int_G int_I, model =  
2, jn = 1, plot = 1, data = CompSci_WS);
```

- List moderator(s) in the w list
- List outcomes in the y list
- Can use `model 2` or `model 3` when you have 1 moderator there is no difference.
- JN option calls the Johnson-Neyman technique
- PLOT option calls a table of values for making a nice plot.

Using MEMORE for CASC WS data

```
MEMORE w = grppref /y = int_G int_I /model = 2/jn = 1 /plot = 1.
```

```
%memore(w=grppref,y = int_G int_I, model = 2, jn = 1, plot = 1,  
data = CompSci_WS);
```

```
***** MEMORE Procedure for SPSS Version 2.1 *****
```

```
Written by Amanda Montoya
```

```
Documentation available at akmontoya.com
```

```
*****
```

```
Model:
```

```
2
```

```
Variables:
```

```
Y = int_G int_I
```

```
W = grppref
```

```
Computed Variables:
```

```
Ydiff = int_G - int_I
```

```
Sample Size:
```

```
51
```

First part of output repeats what you told MEMORE to do. Always double check that this is correct!

I double checked to make sure the order of subtraction was the same as when we did this by hand.

Using MEMORE for CASC WS data

```
MEMORE w = grppref /y = int_G int_I /model = 2/jn = 1 /plot = 1.
```

```
%memore(w=grppref,y = int_G int_I, model = 2, jn = 1, plot = 1,  
data = CompSci_WS);
```

```
*****
```

```
Outcome: Ydiff = int_G - int_I
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.6741	.4544	2.2178	40.8067	1.0000	49.0000	.0000

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	-3.5500	.6485	-5.4742	.0000	-4.8532	-2.2468
grppref	.9936	.1555	6.3880	.0000	.6810	1.3062

```
Degrees of freedom for all regression coefficient estimates:
```

```
49
```

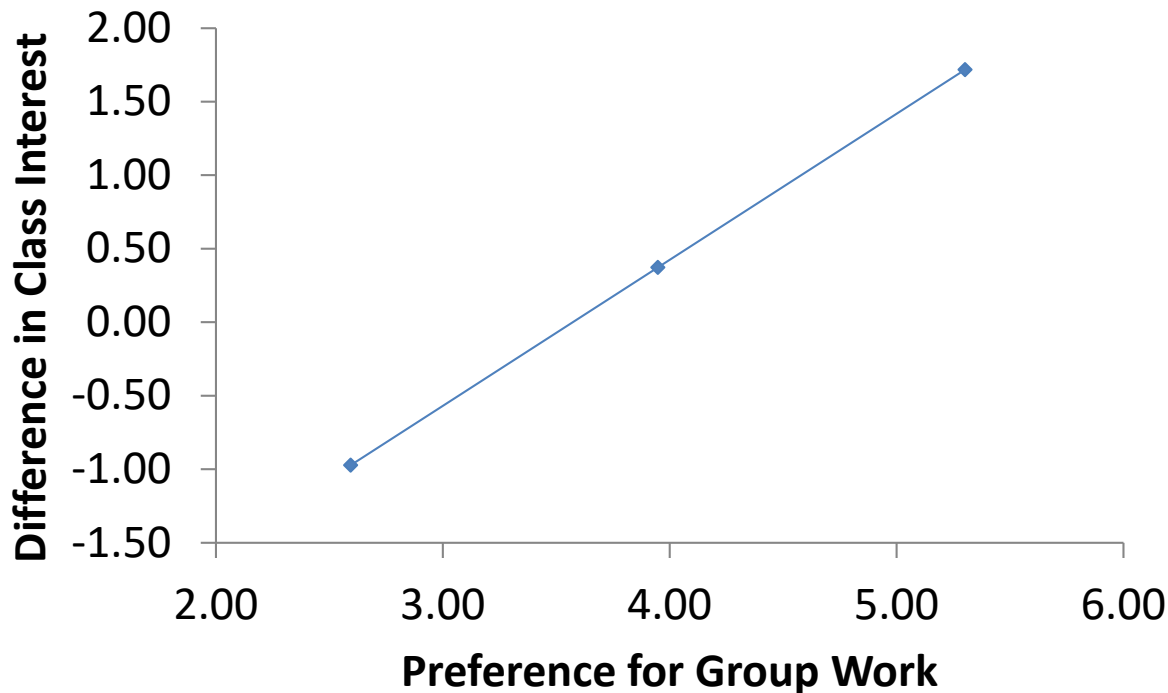
Strong evidence for moderation, where as preference for group work increases, the difference between interest in the two classes increases.

Probing an Effect of Condition on Outcome: The “Pick-a-Point” Approach

You must choose the points along the moderator to “probe” the effect of condition on Y .

Let’s look at an example with our computer science data:

$$Y_{Di} = -3.55 + .99W_i$$



W	$\theta_{X \rightarrow Y W}$	$s\theta_{X \rightarrow Y W}$	p
2.59	-0.97	0.30	0.00
3.95	0.37	0.21	0.08
5.30	1.72	0.30	0.00

Participants relatively low in preference for group work are more interested in the individual work class, and those high in preference for group work are more interested in the class with group work.

Using MEMORE for CASC WS data

```
MEMORE w = grppref /y = int_G int_I /model = 2/jn = 1 /plot = 1.
```

```
%memore(w=grppref,y = int_G int_I, model = 2, jn = 1, plot = 1,  
data = CompSci_WS);
```

Probing effect of condition on outcome at different values of the moderator

Conditional Effect of 'X' on Y at values of moderator(s)

grppref	Effect	SE	t	p	LLCI	ULCI
2.5938	-.9728	.2964	-3.2823	.0019	-1.5684	-.3772
3.9478	.3725	.2085	1.7865	.0802	-.0465	.7916
5.3019	1.7179	.2964	5.7963	.0000	1.1223	2.3135

Degrees of freedom for all conditional effects:

49

Values for quantitative moderators are the mean and plus/minus one SD from the mean.

This is the default. You can change this to the 10th, 25th, 50th, 75th, and 90th quantiles by adding `quantile =1` to the command line

The Johnson-Neyman Technique

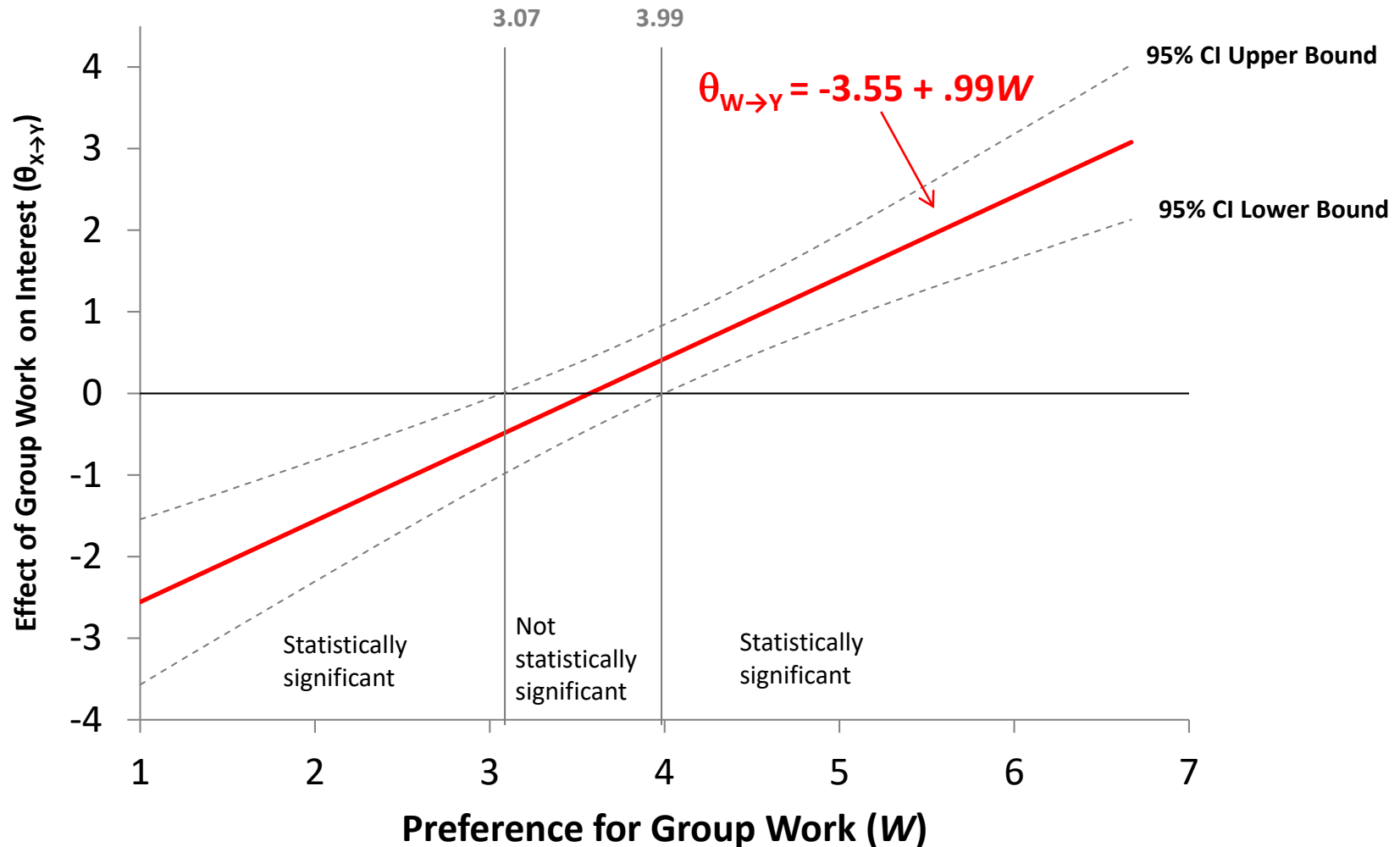
The Johnson-Neyman technique seeks to find the value or values of the moderator (W) within the data, if they exist, such that the p -value for the conditional effect of condition at that value or those values of W is exactly equal to some chosen level of significance α . Thus, no need to select values of W in advance.

To do so, we ask what value of W produces a ratio of $\theta_{x \rightarrow y}(W)$ to its standard error exactly equal to the critical t value (t_{crit}) required to reject the null hypothesis that $\theta_{x \rightarrow y}(W)$ is equal to zero at that value of W ?

$$t_{crit} = \frac{b_0 + b_1 W}{\sqrt{s_{b_0}^2 + 2W s_{b_0 b_1} + W^2 s_{b_1}^2}}$$

Isolating W yields to the solution in the form of a quadratic equation which always has two roots, though not always two that are interpretable.

A Plot of the “Region of Significance”



Using MEMORE for CASC WS data

```
MEMORE w = grppref /y = int_G int_I /model = 2/jn = 1 /plot = 1.
```

```
%memore(w=grppref,y = int_G int_I, model = 2, jn = 1, plot = 1,  
data = CompSci_WS);
```

```
***** JOHNSON-NEYMAN PROCEDURE *****
```

Moderator value(s) defining Johnson-Neyman significance region(s) and percent of
observed data above value:

Value	% Abv
3.0685	72.5490
3.9949	54.9020

Conditional Effect of 'X' on Y at values of moderator

grppref	Effect	SE	t	p	LLCI	ULCI
1.0000	-2.5564	.5037	-5.0752	.0000	-3.5687	-1.5442
1.2984	-2.2599	.4619	-4.8931	.0000	-3.1880	-1.3318
1.5968	-1.9634	.4210	-4.6641	.0000	-2.8094	-1.1174
1.8953	-1.6669	.3813	-4.3712	.0001	-2.4332	-.9006
2.1937	-1.3704	.3434	-3.9905	.0002	-2.0605	-.6803
2.4921	-1.0739	.3078	-3.4886	.0010	-1.6925	-.4553
2.7905	-.7774	.2755	-2.8218	.0069	-1.3310	-.2238
3.0685	-.5012	.2494	-2.0096	.0500	-1.0023	.0000
3.0889	-.4808	.2477	-1.9416	.0579	-.9785	.0168
3.3874	-.1843	.2260	-.8156	.4187	-.6385	.2699
3.6858	.1122	.2125	.5279	.5999	-.3148	.5392
3.9842	.4087	.2086	1.9591	.0558	-.0105	.8279
3.9949	.4193	.2087	2.0096	.0500	.0000	.8387
4.2826	.7052	.2149	3.2809	.0019	.2733	1.1371
4.5811	1.0017	.2306	4.3435	.0001	.5382	1.4652
4.8795	1.2982	.2539	5.1124	.0000	.7879	1.8085
5.1779	1.5947	.2830	5.6350	.0000	1.0260	2.1634
5.4763	1.8912	.3162	5.9804	.0000	1.2557	2.5267
5.7747	2.1877	.3525	6.2070	.0000	1.4794	2.8961
6.0732	2.4843	.3909	6.3560	.0000	1.6988	3.2697
6.3716	2.7808	.4308	6.4546	.0000	1.9150	3.6465
6.6700	3.0773	.4720	6.5200	.0000	2.1288	4.0258

Degrees of freedom for all conditional effects:

This will only print when we
include jn =1 in the command
line. JN technique does not
work for multiple moderators.

Using MEMORE for CASC WS data

```
MEMORE w = grppref /y = int_G int_I /model = 2/jn = 1 /plot = 1.
```

```
%memore(w=grppref,y = int_G int_I, model = 2, jn = 1, plot = 1,  
data = CompSci_WS);
```

Conditional Effect of Moderator(s) on Y in each Condition

Condition 1 Outcome:
int_G

Model Summary

R	R-sq	MSE	F	df1	df2	p
.4488	.2014	1.7964	12.3612	1.0000	49.0000	.0010

Model

	coeff	SE	t	p	LLCI	ULCI
constant	1.7874	.5836	3.0624	.0036	.6145	2.9603
grppref	.4922	.1400	3.5158	.0010	.2109	.7735

Degrees of freedom for all conditional effects:
49

Preference for group work
positively predicts interest in
class with group work

Condition 2 Outcome:
int_I

Model Summary

R	R-sq	MSE	F	df1	df2	p
.4710	.2218	1.6502	13.9671	1.0000	49.0000	.0005

Model

	coeff	SE	t	p	LLCI	ULCI
constant	5.3374	.5594	9.5415	.0000	4.2132	6.4615
grppref	-.5014	.1342	-3.7373	.0005	-.7710	-.2318

Degrees of freedom for all conditional effects:
49

and negatively predicts interest
in **class with individual work.**

Using MEMORE for CASC WS data

```
MEMORE w = grppref /y = int_G int_I /model = 2/jn = 1 /plot = 1.
```

```
%memore(w=grppref,y = int_G int_I, model = 2, jn = 1, plot = 1,  
data = CompSci_WS);
```

```
*****
```

Data for visualizing conditional effect of X on Y.

Paste text below into a SPSS syntax window and execute to produce plot.

```
DATA LIST FREE/grppref YdiffHAT int_GHAT int_IHAT.
```

```
BEGIN DATA.
```

2.5938	-.9728	3.0640	4.0368
3.9478	.3725	3.7304	3.3578
5.3019	1.7179	4.3968	2.6789

```
END DATA.
```

```
GRAPH/SCATTERPLOT = grppref WITH YdiffHAT.
```

```
GRAPH/SCATTERPLOT = grppref WITH int_GHAT.
```

```
GRAPH/SCATTERPLOT = grppref WITH int_IHAT.
```

Code for plotting. You'll get three plots each with the moderator on the X axis and a different outcome on the Y axis.

- 1) Predicted Differences between Y's
- 2) Predicted Y from first condition
- 3) Predicted Y from second condition

Using MEMORE for CASC WS data

```
MEMORE w = grppref /y = int_G int_I /model = 2/jn = 1 /plot = 1.
```

```
%memore(w=grppref,y = int_G int_I, model = 2, jn = 1, plot = 1,  
data = CompSci_WS);
```

Data for visualizing conditional effect of 'X' on Y.

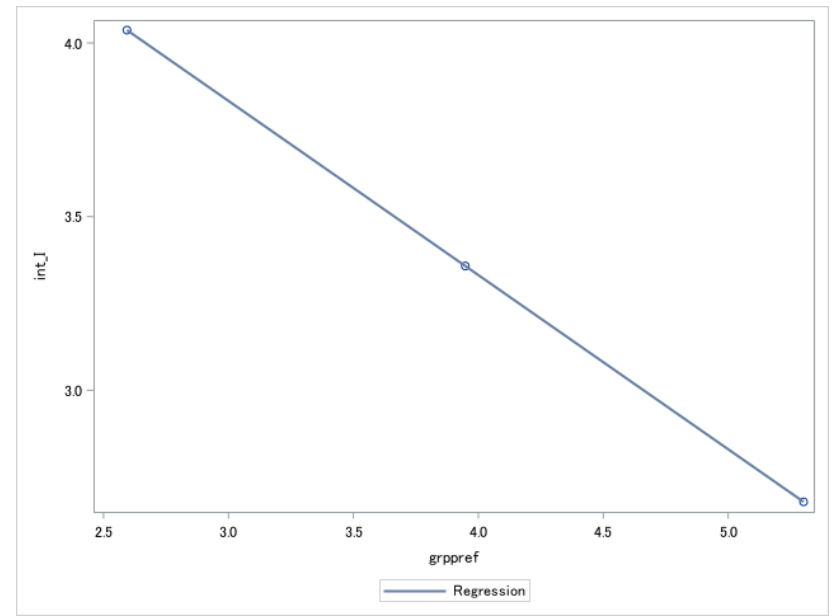
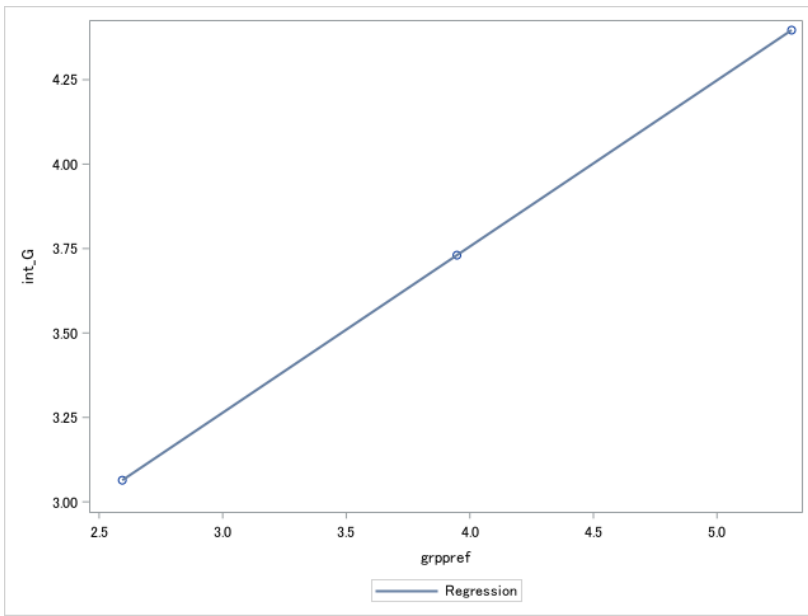
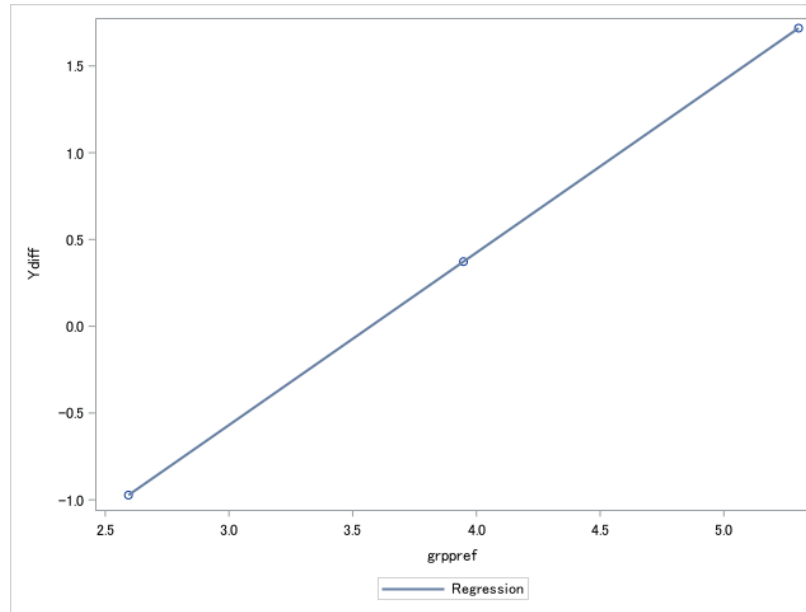
	GRPPREF	Ydiff	INT_G	INT_I
	2.5938	-0.9728	3.0640	4.0368
	3.9478	0.3725	3.7304	3.3578
	5.3019	1.7179	4.3968	2.6789

Only data table is generated, then write separate code to make the graphs

- 1) Predicted Differences between Y's
- 2) Predicted Y from first condition
- 3) Predicted Y from second condition

```
data;  
input grppref Ydiff int_G int_I;  
datalines;  
    2.5938 -0.9728 3.0640 4.0368  
    3.9478 0.3725 3.7304 3.3578  
    5.3019 1.7179 4.3968 2.6789  
  
run;  
proc sgplot; reg x=grppref y=Ydiff; run;  
proc sgplot; reg x=grppref y=int_G; run;  
proc sgplot; reg x=grppref y=int_I; run;
```

SAS Graphs



Writing up a Moderation Analysis

Tips:

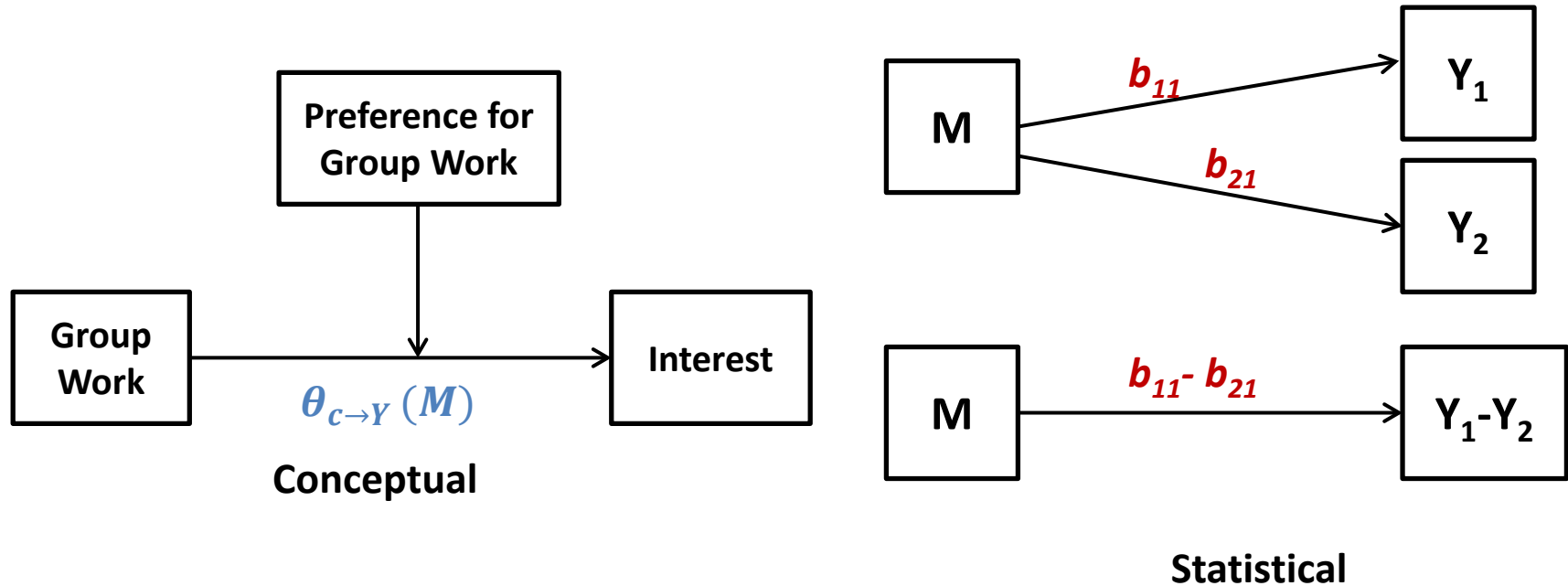
- Interpret the sign and the magnitude of the interaction coefficient with respect to X 's effect on Y (or M 's effect on Y ; or both).
- Provide probing results with interpretations
- Read the write ups of other's moderation analyses
- Provide a graphical representation of the effect of interest (like the ones we've done)

Does the effect of group work on interest in a computer science class depend on preference for group work?

Overall, the impact of including group work in a computer science class on interest in the class depends on an individual's general preference for group work ($b_1 = .49, p = .001$). As preference for group work increases relative interest in the class with group work compared to the class with individual work increases as well. (i.e. the group work class is more preferred as general preference for group work increases). Indeed we found that those who were relatively low in preference for group work preferred the individual work class over the class with group work ($\theta_{X \rightarrow Y}(M=2.59) = -.97, p = .002$). Whereas, those who were relatively moderate in preference for group work did not show a strong preference for one class over another, though they marginally preferred the class with group work ($\theta_{X \rightarrow Y}(M=3.97) = .37, p = .08$). Finally, those who showed a strong general preference for group work, unsurprisingly preferred the class with group work over the class with individual work ($\theta_{X \rightarrow Y}(M=5.30) = 1.72, p < .001$). The Johnson-Neyman procedure those whose preference for group work was less than 3.07 preferred the individual work class, and those who's preference for group work was greater than 3.99 preferred the group work class. Preference for group work was positively related to interest in the class with group work ($b = .49, p = .001$), and negatively related to interest in the class with individual work ($b = -0.50, p = .001$).

Visualizations

I recommend trying a number of different types of visualizations to decide what works best for your case.



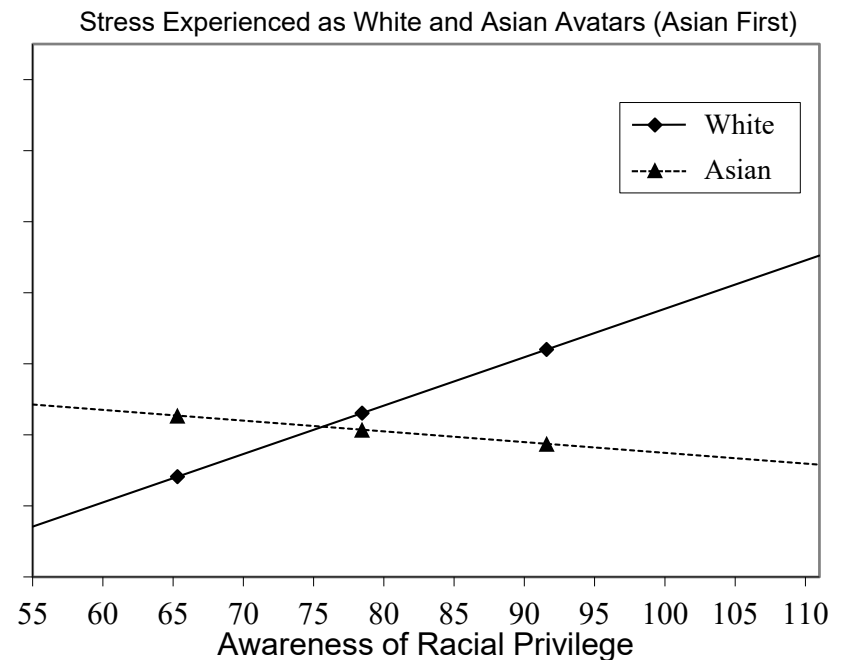
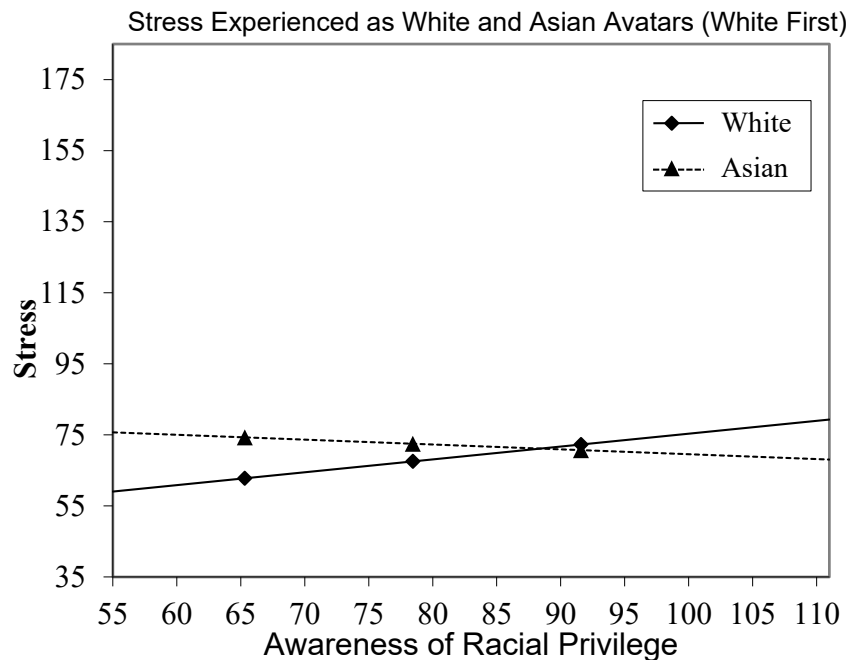
Tips:

- Try the different scales of the Y axis (difference vs. raw Y score with two lines for each condition)
- I do not like bar graphs with the effect of the moderator in each condition
- Provide path estimates on statistical diagram or in a table.

Visualizations: A Case Study

Tawa, J., Montoya, A. K.(2023). Racial embodiment in virtual environments for facilitating students' understanding of racism. *Technology, Mind, and Behavior*, 4(1), Special Collection: DOI: 10.1037/tmb0000024

White participants operated avatars of three difference races (White, Black, and Asian) and wrote heart monitors to measure their stress while operating each avatar. We found that individual's awareness of racial privilege moderated the effect of avatar race on stress, and that this effect depended on the order of operating the avatars.

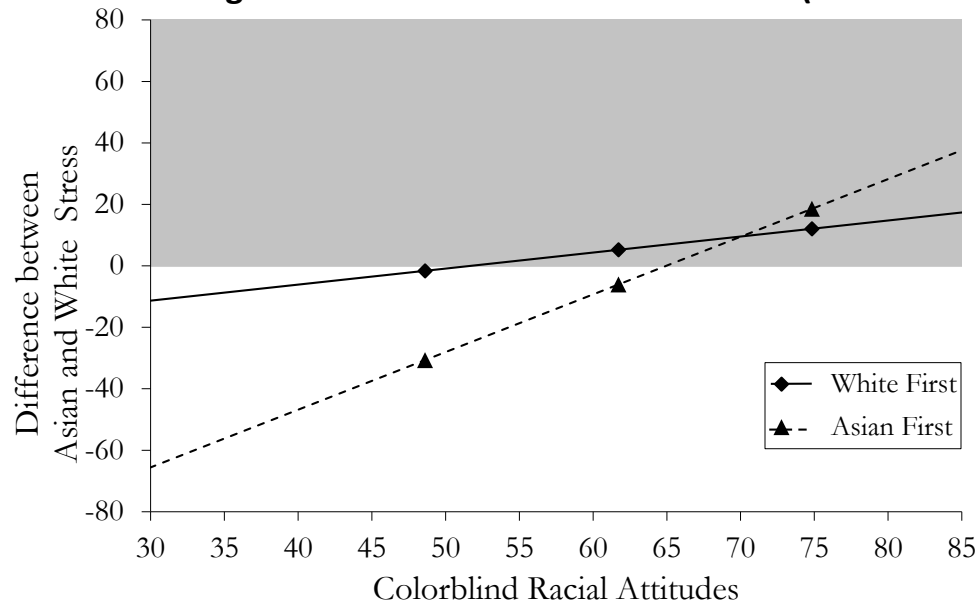


Visualizations: A Case Study

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White participants operated avatars of three difference races (White, Black, and Asian) and wrote heart monitors to measure their stress while operating each avatar. We found that individual's awareness of racial privilege moderated the effect of avatar race on stress, and that this effect depended on the order of operating the avatars.

Figure 3. Predicted difference in Stress (Asian Stress – White Stress), split by order.



Note. Scores above zero on the Y-axis represent greater predicted stress while piloting the Asian avatar than while piloting the White avatar. Points marked by shapes indicate predicted stress differences at the mean plus/minus one standard deviation on CBA.

Common Questions

- Can this method be used for more than two conditions?
YES! The same method for coming up with contrasts in Judd, Kenny, and McClelland (2001) describe a system for setting up contrasts among conditions can be used for moderation.
I recommend reading [Hayes & Montoya \(2017\)](#) on moderation analysis with a multicategorical IV if you want to try this out. I am happy to give instructions on how to get MEMORE to doing this.
ALTERNATIVES: Some of the other repeated-measures mediation options are more appropriate if you have more than two conditions (especially longitudinal), so take a look at those when thinking about these options.
- Can I use multiple moderators?
YES! MEMORE models 2 and 3 accept up to 5 moderators. (See Documentation for instructions).
- How do I control for covariates?
All of MEMORE's mediation analyses are within-person models, so you do not need to control for any between subjects variables such as age, gender, big-5. But you can include them as additional moderators (likely using model 2).

Power Analysis

Power analysis for within-subject moderation can be conducted with any tool that does power analysis for regression.

G*Power is a commonly used tool for power analysis.

The screenshot shows the G*Power 3.1.9.7 window with the following settings:

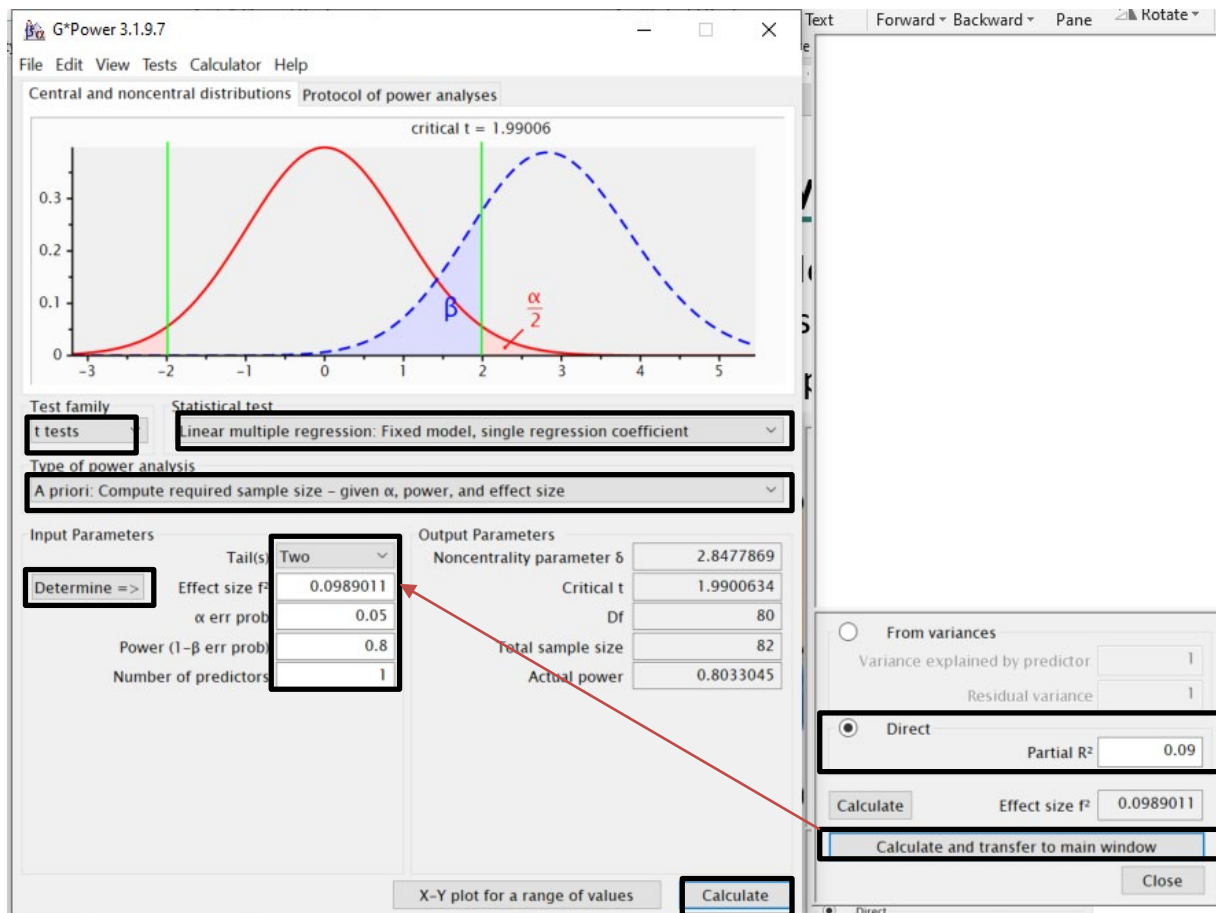
- Test family:** t tests
- Statistical test:** Linear multiple regression: Fixed model, single regression coefficient
- Type of power analysis:** A priori: Compute required sample size – given α , power, and effect size
- Input Parameters:**
 - Determine =>** (checked)
 - Tail(s):** Two
 - Effect size f:** 0.0989011
 - α err prob:** 0.05
 - Power (1- β err prob):** 0.8
 - Number of predictors:** 1

A red arrow points from the 'Calculate' button to the 'Effect size f' field. At the bottom, there is a button for 'X-Y plot for a range of values' and a 'Calculate' button.

Power Analysis

Power analysis for within-subject moderation can be conducted with any tool that does power analysis for regression.

[G*Power](#) is a commonly used tool for power analysis.



Recommended sample size for this case is 82

Generating Effect Size Estimates

Power analysis itself is often not difficult, but coming up with an effect size estimate is.

Possible approaches:

- Pilot studies
 - Often cannot produce a precise enough estimate to be useful or will be [biased](#)
- Effect sizes from prior studies, literature, or meta-analysis
 - *Beware publication bias*
- Smallest effect size of interest (SESOI)
 - [Lakens, 2017](#) & [Anvari & Lakens, 2021](#)

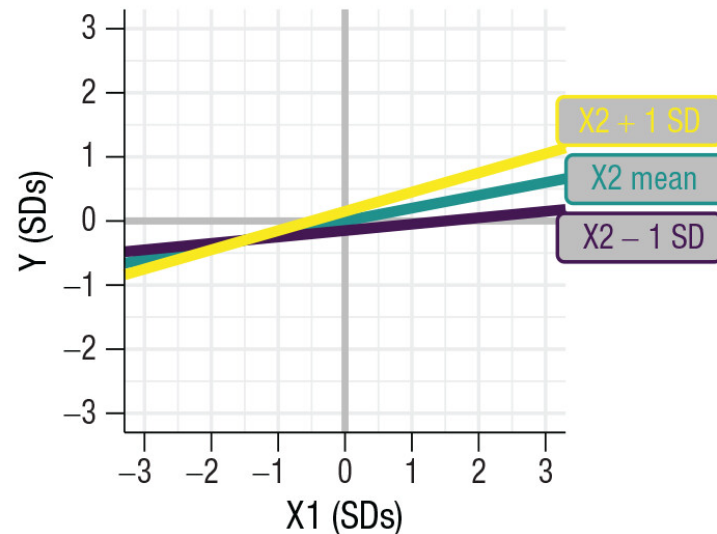
Moderation Specific Issues

Generating a predicted effect size for an interaction can be difficult and unintuitive, especially when it depends on many other effects.

a

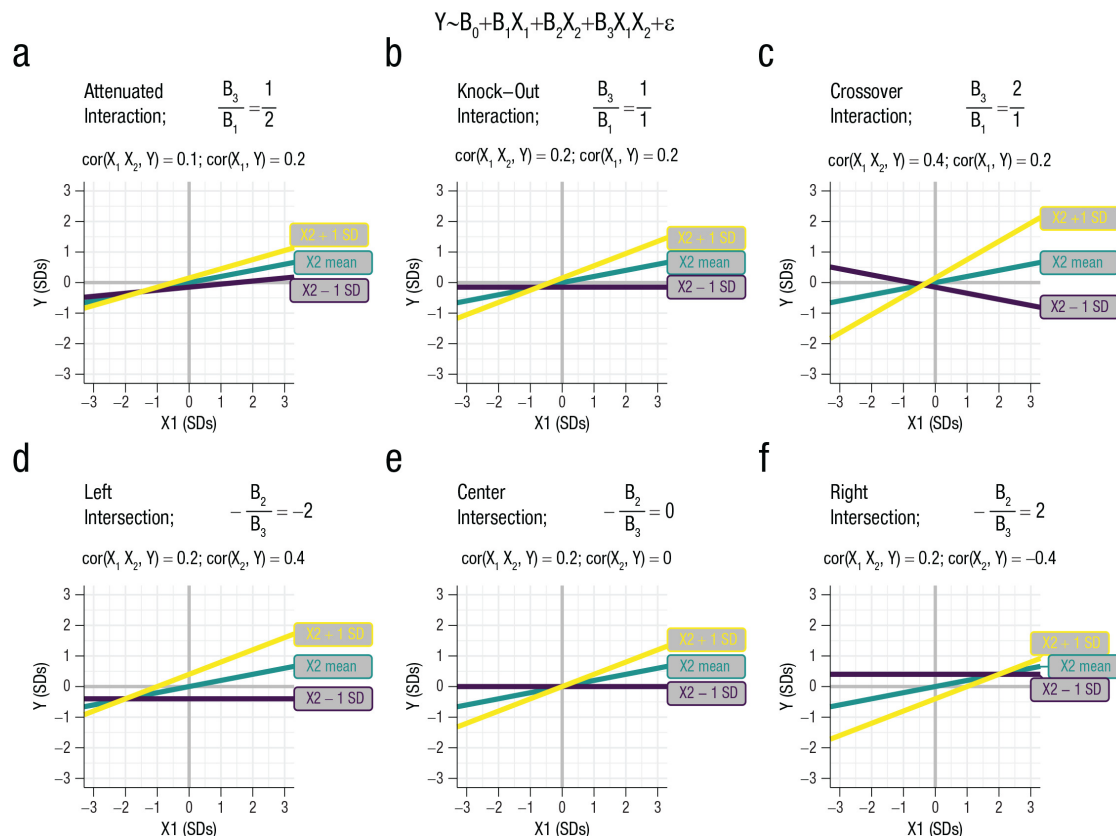
Attenuated
Interaction; $\frac{B_3}{B_1} = \frac{1}{2}$

$\text{cor}(X_1, X_2, Y) = 0.1$; $\text{cor}(X_1, Y) = 0.2$



Moderation Specific Issues

Generating a predicted effect size for an interaction can be difficult and unintuitive, especially when it depends on many other effects.



Preregistration

Preregistration: A process where you create a time-stamped, publicly accessible record of your plan for a specific study.

- ❑ Planned sample size
- ❑ α -level/CI-level for each test
- ❑ Role of different variables in the analysis (e.g., independent variable, moderator, outcome), and how they are computed
- ❑ Estimation Method
- ❑ Plans for probing: simple slopes, JN, probing at specific values of the moderator
- ❑ Tools and specifications

Finding Examples

DataStudio for finding examples

<https://lookerstudio.google.com/s/gFgefAkOjKA>

Publication Year Research Areas ModelNumb... (1) Xtype levelsX numYvars											
Journal Title		Covariates		Sample Size		Article Count 50					
Article Title (link)	Authors	Journal Title	Publication...	Research Areas	Study Num...	SampleSize	ModelNumb...	numXvars	Xtype	levelsX	numYvars
When should retail...	Jeong, H; Ye,...	JOURNAL O...	2021	Business & Economics	1	271	1 (Mediation)	1	Pre-post	2	1
What hinders resi...	Dong, XJ	TOURISM A...	2022	Social Sciences - Other...	2	130	1 (Mediation)	1	Experimental	2	1
The role of metad...	Hartley, S; P...	GROUP PRO...	2022	Psychology	3	239	1 (Mediation)	1	Experimental	2	2
The connotative ...	Motoki, K; P...	JOURNAL O...	2022	Business & Economics	1	154	1 (Mediation)	1	Experimental	2	3
The Self-Other Div...	Ring, C; Kav...	JOURNAL O...	2020	Social Sciences - Other...	1	100	1 (Mediation)	1	Experimental	2	1
The Impact of Mix...	Huang, XZ; Z...	INTERNATIO...	2022	Environmental Science...	2	434	1 (Mediation)	1	Experimental	2	1
The Hypoalgesic E...	Song, JS; Ka...	RESEARCH ...	2022	Social Sciences - Other...	1	40	1 (Mediation)	1	Experimental	2	1
Testing the effecti...	Brochu, PM	JOURNAL O...	2023	Psychology	1	45	1 (Mediation)	1	Pre-post	2	2
Suicidality and so...	Breitborde, ...	EARLY INTE...	2021	Psychiatry	1	38	1 (Mediation)	1	Pre-post	2	1
Putting the Me in ...	Hamilton, K...	NEW MEDIA...	2021	Communication	1	119	1 (Mediation)	1	Experimental	2	1
Positive reputatio...	Inoue, Y; Mif...	FRONTIERS ...	2023	Psychology	2	293	1 (Mediation)	1	Experimental	2	1
Perceptions of a P...	Pals, AM; Go...	JOURNAL O...	2022	Psychology; Family Stu...	1	52	1 (Mediation)	3	Experimental	2	1
Pandemic Pedago...	Armstrong, ...	SOUTHERN ...	2022	Communication	1	163	1 (Mediation)	1	Pre-post	2	4
Mind the ad: How ...	Kocak, A; Ro...	JOURNAL O...	2022	Psychology; Business ...	1	123	1 (Mediation)	1	Experimental	2	1
Mind the ad: How ...	Kocak, A; Ro...	JOURNAL O...	2022	Psychology; Business ...	2	151	1 (Mediation)	1	Experimental	2	1
Mediation and Mo...	Wong, CL; C...	JOURNAL O...	2020	Public, Environmental ...	1	1001	1 (Mediation)	1	Pre-post	2	1

Multiple Moderator Models

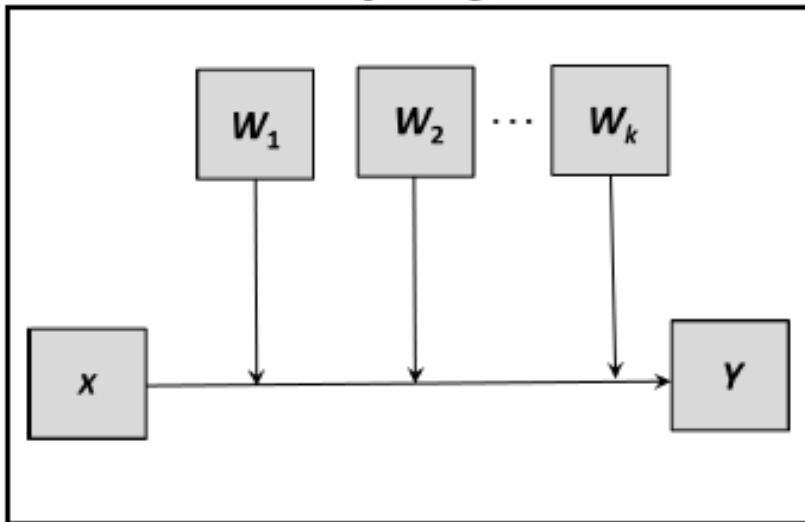
Model 2 vs. Model 3

When you have multiple moderators you are interested, consider whether you think those moderators will themselves interact or not.

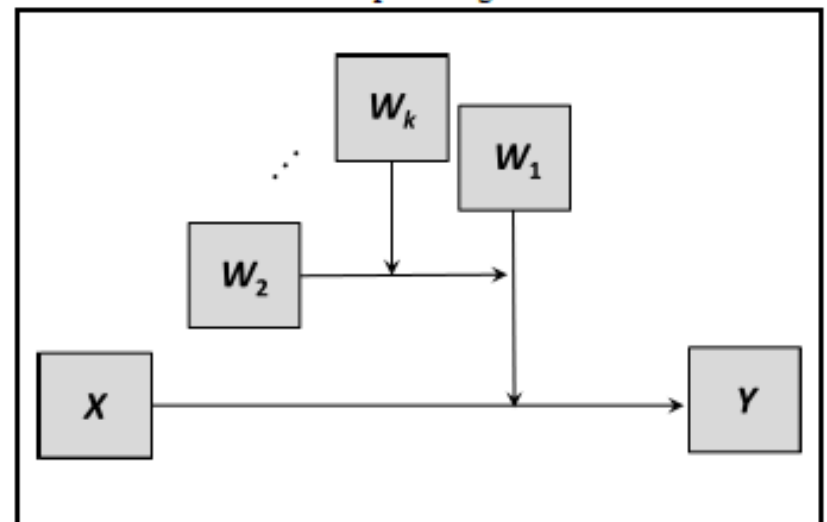
If you believe the moderators will interact **with each other** → Model 3

If you believe the moderators will **only interact with condition** → Model 2

Model 2 Additive Moderation
Conceptual Diagram



Model 3 Multiplicative Moderation
Conceptual Diagram



Multiple Moderator Models

```
MEMORE w = grppref order/y = int_G int_I /model = 2.
```

```
%memore(w=grppref order,y = int_G int_I, model = 2,  
data = CompSci_WS);
```

```
***** MEMORE Procedure for SPSS Version 3.0 *****
```

```
Written by Amanda Montoya
```

```
Documentation available at akmontoya.com
```

```
*****
```

```
Model:
```

```
2
```

```
Variables:
```

```
Y = int_G int_I
```

```
W1 = grppref
```

```
W2 = Order
```

```
Computed Variables:
```

```
Ydiff = int_G - int_I
```

```
Sample Size:
```

```
51
```

```
*****
```

```
Outcome: Ydiff = int_G - int_I
```

```
Model Summary
```

	R	R-sq	MSE	F	df1	df2	p
	.7113	.5059	2.0502	24.5734	2.0000	48.0000	.0000

```
Model
```

	coeff	SE	t	p	LLCI	ULCI
constant	-4.8074	.8294	-5.7269	.0000	-6.4952	-3.1196
grppref	.9562	.1505	6.3542	.0000	.6536	1.2588
Order	.9071	.4055	2.2372	.0300	.0918	1.7223

```
Degrees of freedom for all regression coefficient estimates:
```

```
48
```

Think of it like two two-way interactions:
Condition x Group Preference
Condition x Order

Multiple Moderator Models

```
MEMORE w = grppref order/y = int_G int_I /model = 3.
```

```
%memore(w=grppref order,y = int_G int_I, model = 3,  
        data = CompSci_WS);
```

Model:

3

Variables:

Y = int_G int_I

W1 = grppref

W2 = Order

Computed Variables:

Ydiff = int_G - int_I

Int1 = grppref x Order

Sample Size:

51

Outcome: Ydiff = int_G - int_I

Model Summary

R	R-sq	MSE	F	df1	df2	P
.7125	.5077	2.0862	16.1569	3.0000	47.0000	.0000

Model

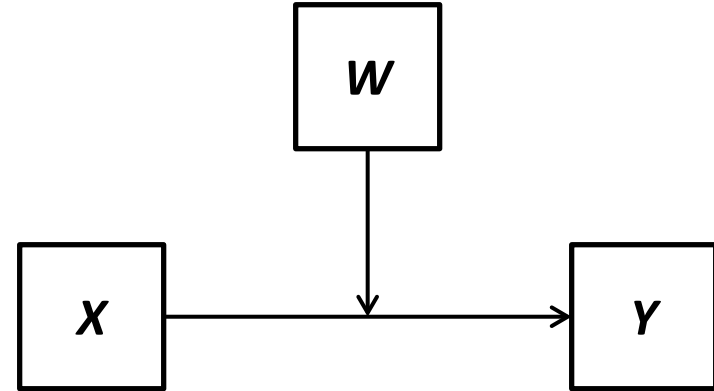
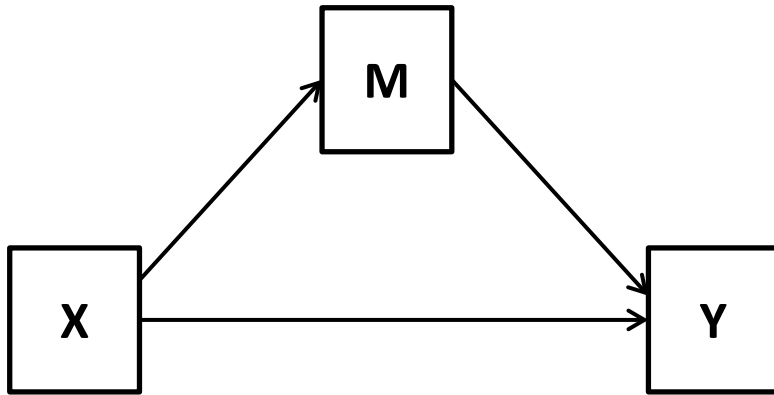
	coeff	SE	t	p	LLCI	ULCI
constant	-5.5239	1.9247	-2.8700	.0061	-9.3960	-1.6518
grppref	1.1401	.4690	2.4312	.0189	.1967	2.0836
Order	1.4057	1.2704	1.1065	.2742	-1.1501	3.9615
Int1	-.1263	.3048	-.4145	.6804	-.7395	.4868

Degrees of freedom for all regression coefficient estimates:

47

Think of it like three-way interaction,
and three two-way interactions:
Condition x Group Preference
Condition x Order
Group Preference x Order
Condition x Group Preference x Order

Combining mediation and moderation



Research questions:

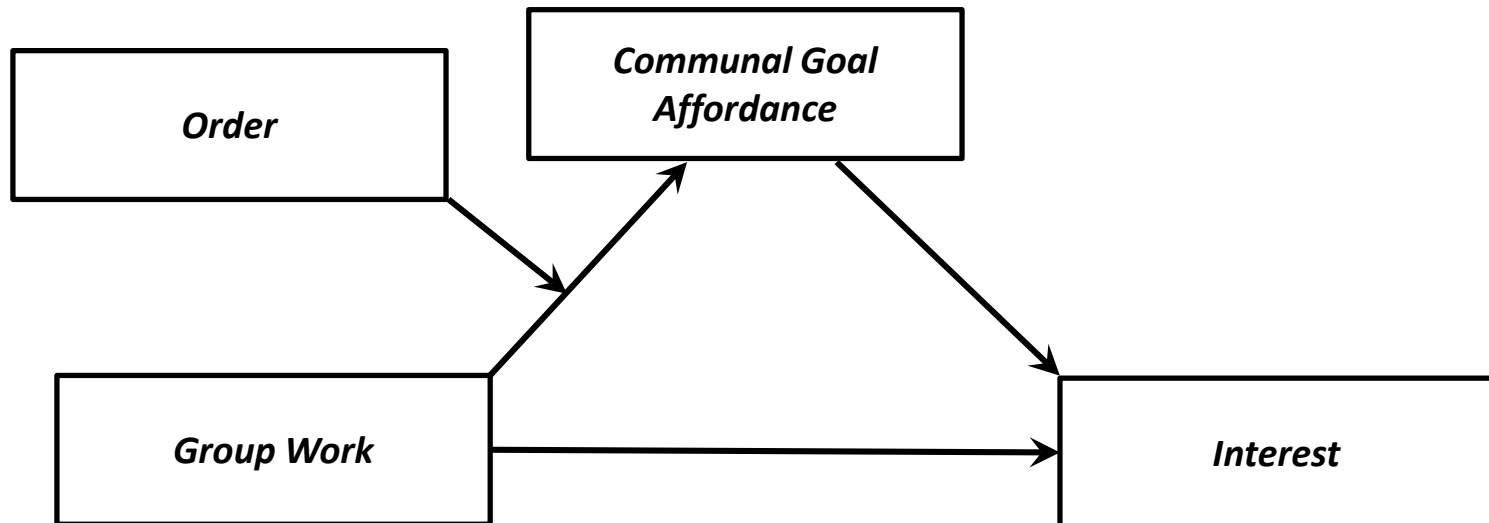
- Does the process through which X affects Y through M depend on W ?
- Are there certain groups where X affects Y through M and certain groups where this process does not occur?

Conditional process analysis allows a mediated process to be moderated. Now the indirect effect can be defined as a *function of the moderator*.

CPA in Two-instance repeated-measures designs

Extending the path analytic from Montoya & Hayes (2017) we can now allow for moderation of a mediated pathway.

First stage moderated mediation allows W to moderate the path between the within-subjects factor and the mediator.



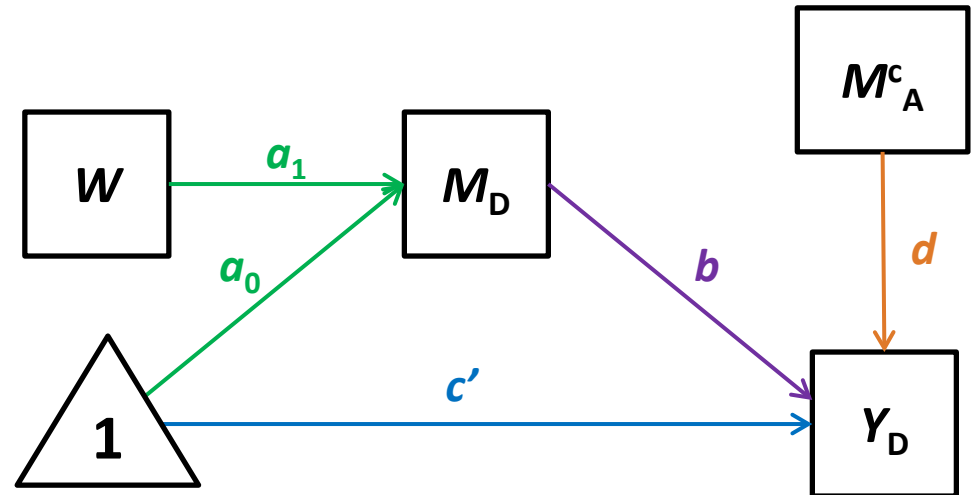
Equations and Path Diagram

First stage moderated mediation allows W to moderate the path between the within-subjects factor and the mediator.

$$M_{2i} - M_{1i} = a_0 + a_1 W_i + \epsilon_{Mi}$$

$$\theta_{X \rightarrow M}(W) = a_0 + a_1 W_i$$

$$Y_{Di} = c' + b M_{Di} + d M_{Ai}^c + \epsilon_{Yi}$$



What is the indirect effect?

$$\theta_{X \rightarrow M}(W) \times b = (a_0 + a_1 W)b = a_0 b + a_1 b W$$

Indirect effect is a *function* of the moderator

Inference

$$\theta_{X \rightarrow M}(W) \times b = (a_0 + a_1 W)b = a_0 b + a_1 b W$$

Conditional Indirect Effects

Select a value of W , plug that into the equation for the indirect effect, and use bootstrapping to make inference about the indirect effect at that value

Does the indirect effect *depend* on the moderator?

If $a_1 b = 0$ then the indirect effect *does not* depend on W

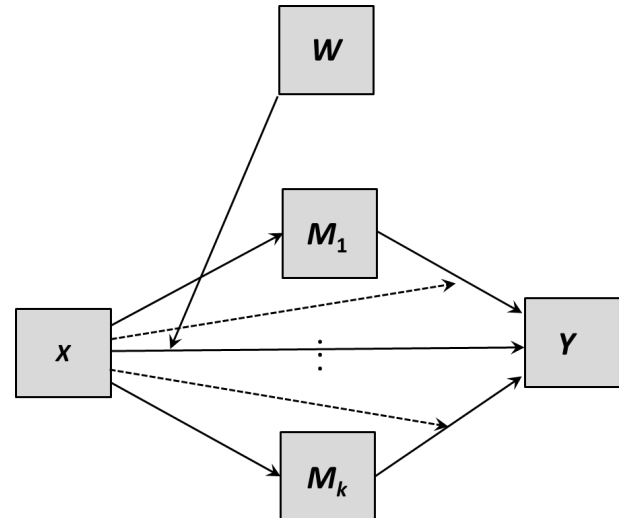
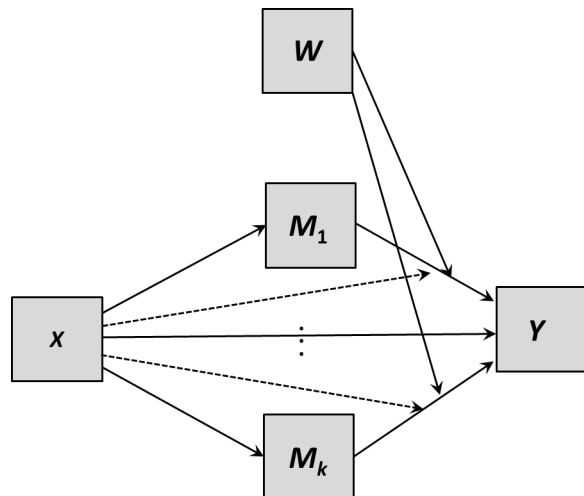
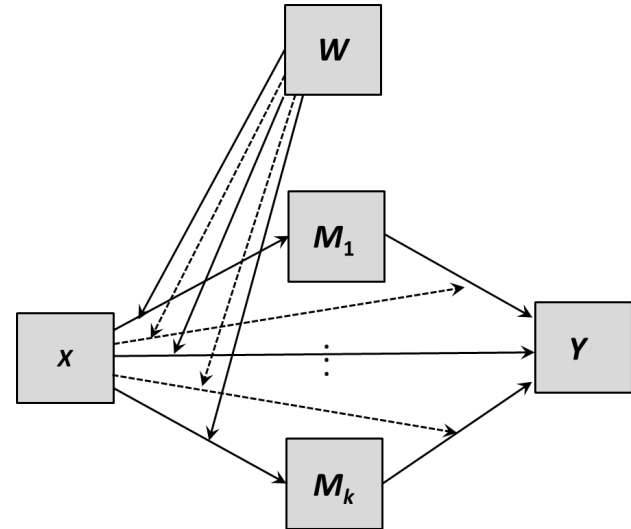
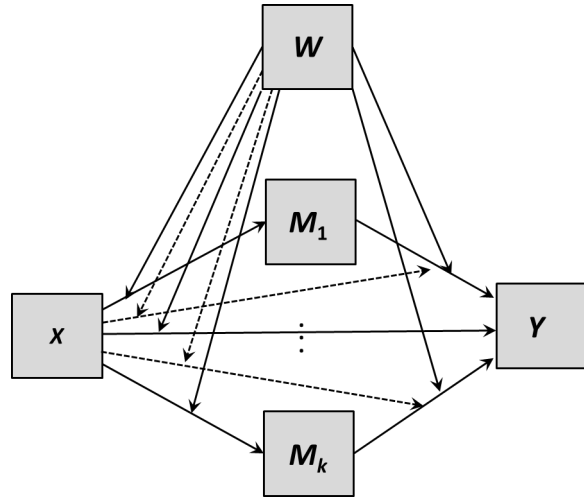
$$\theta_{X \rightarrow M}(W) \times b = a_0 b + 0 * W = a_0 b$$

$a_1 b$ can be called the **index of moderated mediation**

A test on the index will indicate if the indirect effect depends on W . We can do this formal test using bootstrapping.

MEMORE V3: Models 4 - 18

The latest version of MEMORE has expanded to models with a single moderator on any combination of paths in the mediation.



MEMORE

```
MEMORE m = comm_G comm_I /w = order/y = int_G int_I /model = 15.
```

```
%memore(m=comm_G comm_I, w=order,y=int_G int_I,model=15,  
        data = CompSci_WS);
```

```
***** MEMORE Procedure for SPSS Version 3.0 *****
```

```
Written by Amanda Montoya
```

```
Documentation available at akmontoya.com
```

```
*****
```

```
Model:
```

```
15
```

```
Variables:
```

```
Y = int_G    int_I
```

```
W = Order
```

```
M = comm_G   comm_I
```

```
Computed Variables:
```

```
Ydiff =      int_G    -      int_I
```

```
Mdiff =      comm_G    -      comm_I
```

```
Mavg = (      comm_G    +      comm_I    )      /2      Centered
```

```
Sample Size:
```

```
51
```

Moderated Mediation with MEMORE

```
MEMORE m = comm_G comm_I /w = order/y = int_G int_I /model = 15.
```

```
%memore(m=comm_G comm_I, w=order,y=int_G int_I,model=15,  
data = CompSci_WS);
```

Outcome: Ydiff = int_G - int_I

Model Summary

R	R-sq	MSE	F	df1	df2	p
.3005	.0903	3.6978	4.8629	1.0000	49.0000	.0322

Model

	Effect	SE	t	p	LLCI	ULCI
constant	-1.4759	.8804	-1.6764	.1000	-3.2452	.2934
W	1.1933	.5411	2.2052	.0322	.1058	2.2808

Degrees of freedom for all regression coefficient estimates:

49

Conditional Effect of Focal Predictor on Outcome at values of Moderator(s)

Focal: 'X' (X)

Outcome: Ydiff (Y)

Mod: Order (W)

Order	Effect	SE	t	p	LLCI	ULCI
1.0000	-.2826	.4010	-.7048	.4843	-1.0884	.5232
2.0000	.9107	.3634	2.5061	.0156	.1804	1.6410

Values for dichotomous moderators are the two values of the moderator.

Degrees of freedom for all conditional effects:

49

Moderated Mediation with MEMORE

```
MEMORE m = comm_G comm_I /w = order/y = int_G int_I /model = 15.
```

```
%memore(m=comm_G comm_I, w=order,y=int_G int_I,model=15,  
data = CompSci_WS);
```

```
Outcome: Mdiff = comm_G - comm_I
```

```
Model Summary
```

R	R-sq	MSE	F	df1	df2	p
.3393	.1151	2.8567	6.3752	1.0000	49.0000	.0149

```
Model
```

	Effect	SE	t	p	LLCI	ULCI
constant	.4339	.7738	.5606	.5776	-1.1213	1.9890
W	1.2009	.4756	2.5249	.0149	.2451	2.1568

```
Degrees of freedom for all regression coefficient estimates:
```

```
49
```

```
Conditional Effect of Focal Predictor on Outcome at values of Moderator(s)
```

```
Focal: 'X' (X)
```

```
Outcome: Mdiff (M)
```

```
Mod: Order (W)
```

Order	Effect	SE	t	p	LLCI	ULCI
1.0000	1.6348	.3524	4.6387	.0000	.9266	2.3430
2.0000	2.8357	.3194	8.8779	.0000	2.1938	3.4776

```
Values for dichotomous moderators are the two values of the moderator.
```

```
Degrees of freedom for all conditional effects:
```

```
49
```

Moderated Mediation with MEMORE

```
MEMORE m = comm_G comm_I /w = order/y = int_G int_I /model = 15.
```

```
%memore(m=comm_G comm_I, w=order,y=int_G int_I,model=15,  
data = CompSci_WS);
```

Outcome: Ydiff = int_G - int_I

Model Summary

R	R-sq	MSE	F	df1	df2	p
.5639	.3180	2.8299	11.1909	2.0000	48.0000	.0001

Model

	coeff	SE	t	p	LLCI	ULCI
constant	-.9814	.3884	-2.5269	.0149	-1.7623	-.2005
Mdiff	.5902	.1346	4.3845	.0001	.3195	.8608
Mavg	-.5505	.4328	-1.2718	.2096	-1.4208	.3198

Degrees of freedom for all regression coefficient estimates:

48

Moderated Mediation with MEMORE

```
MEMORE m = comm_G comm_I /w = order/y = int_G int_I /model = 15.
```

```
%memore(m=comm_G comm_I, w=order,y=int_G int_I,model=15,  
data = CompSci_WS);
```

***** CONDITIONAL TOTAL, DIRECT, AND INDIRECT EFFECTS *****

Conditional Total Effect of X on Y at values of the Moderator(s)

Order	Effect	SE	t	df	p	LLCI	ULCI
1.0000	-.2826	.4010	-.7048	49.0000	.4843	-1.0884	.5232
2.0000	.9107	.3634	2.5061	49.0000	.0156	.1804	1.6410

Values for dichotomous moderators are the two values of the moderator.

Direct effect of X on Y

Effect	SE	t	df	p	LLCI	ULCI
-.9814	.3884	-2.5269	48.0000	.0149	-1.7623	-.2005

Conditional Indirect Effect of X on Y through Mediator at values of the Moderator

Ind: Indl
Med: Mdiff (M)

Order	Effect	BootSE	BootLLCI	BootULCI
1.0000	.9648	.2835	.4278	1.5511
2.0000	1.6736	.4349	.7777	2.4990

Values for dichotomous moderators are the two values of the moderator.

Indirect Key

Indl 'X' -> Mdiff -> Ydiff

Indirect effect is
significant and
positive in both
order conditions

Moderated Mediation with MEMORE

```
MEMORE m = comm_G comm_I /w = order/y = int_G int_I /model = 15.
```

```
%memore(m=comm_G comm_I, w=order,y=int_G int_I,model=15,  
        data = CompSci_WS);
```

```
***** INDICES OF MODERATION *****
```

Test of Moderation of the Total Effect

	Effect	SE	t	df	p	LLCI	ULCI
W	1.1933	.5411	2.2052	49.0000	.0322	.1058	2.2808

Index of Moderated Mediation for each Indirect Effect.

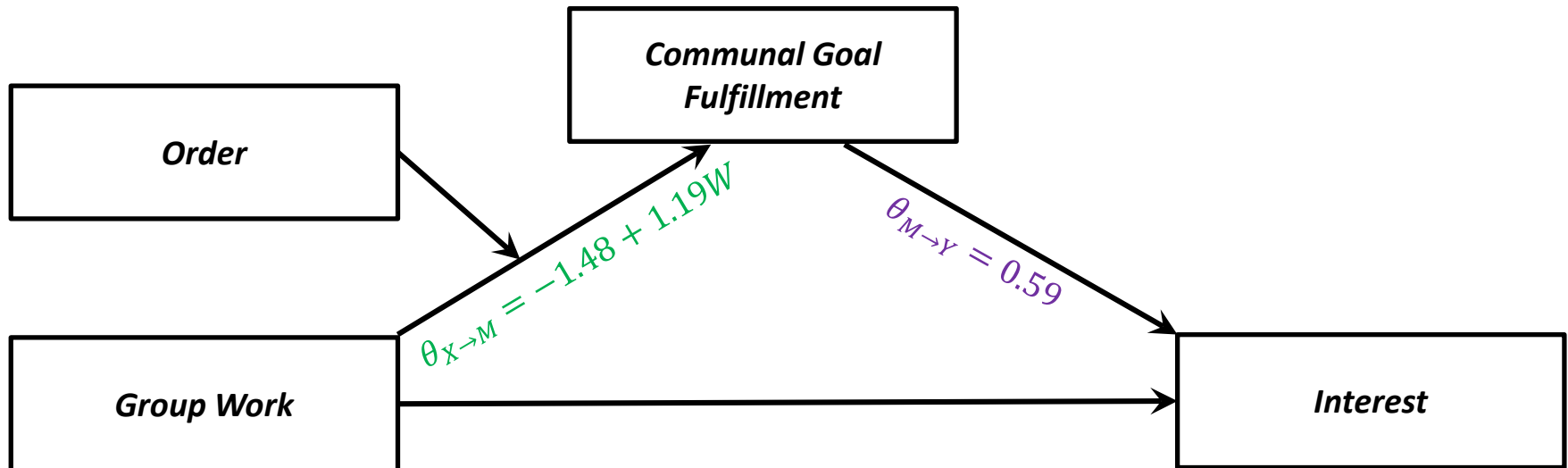
	Effect	BootSE	BootLLCI	BootULCI
Ind1	.7088	.3629	.1035	1.5004

The indirect effect is statistically larger when participants read about the individual work class first.

Computer Science Study

```
MEMORE m = comm_G comm_I /w = order/y = int_G int_I /model = 15.
```

```
%memore(m=comm_G comm_I, w=order,y=int_G int_I,model=15,  
data = CompSci_WS);
```



The indirect effect for both orders was such that **the group work class increased interest through communal goal fulfillment** (Group-first: 0.96 [0.43, 1.55], Individual-First: 1.67 [1.78, 2.50]).

The *index of moderated mediation* was significantly different from zero (0.71 [0.10, 1.50]), meaning the **indirect effect through drawing on the communal goals was stronger for those who read the individual work syllabus first, compared to reading group work first.**

Wrapping Up

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Upcoming versions of MEMORE expanding functionality

- More moderated mediation
- Dichotomous outcomes
- More than 2 observations

Feel free to contact me with questions or consultation for data with other structures, extensions. This is frequently how new methods get developed!

Mediation for Between Subject Data: [PROCESS](#) (Available for SPSS, SAS, and R)

Mediation for Dyadic Data: [MEDYAD](#) (Available for SPSS, SAS, and R)

Mediation for Multilevel Models: [MLMED](#) (Available for SPSS)

[Github.com/akmontoya/SHwithin](https://github.com/akmontoya/SHwithin)